Are Large Language Models Creative?

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My theme, then, is the human mind – and how it can surpass itself. We can appreciate the richness of creative thought better than ever before, thanks to this new scientific approach. If the paradox and mystery are dispelled, our sense of wonder is not.

—Margaret A. Boden, Preface to *The Creative Mind: Myths and Mechanisms* (1990)

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Introduction

My housemates and I have those word magnets that people use to construct poems on the fronts of their fridges. In our six months of living together, we and visitors have arranged the magnets into meditations on friendship, grief, college graduation, existentialism, and romantic relationships, but also on bowel movements, the ever-present "your mom," and carnal regards for whomever stumbles upon our fridge. There is a particular kind of frustration that comes with building a poem from a limited word bank: the constant search for a desired magnet shapes the construction process, oftentimes undermining the poet's mental construction by forcing them to regenerate when a desired word is nowhere to be found.

I prefer this to most other forms of poetry. I like that the constrained creative space can force a sense of direction, but the experience is different for everybody: One of my housemates feels the magnet-hunting throws her off her rhythm. One of my best friends happily spends hours letting herself and the word bank produce a poem. One of my friends constructs each poem in small bursts over multiple days. Across these very different ways of engaging with the same medium, none of us questions whether the resulting poems are, at some level, creative. There is a shared, informal understanding that someone we know got down on the floor, looked at the words in front of them, and constructed something out of them. Sometimes this construction is moving. Sometimes weird. Sometimes funny.

A large language model (LLM) could produce a poem from a constrained word bank in seconds, and some of the time in a way that might strike us as moving, weird, or potentially even as nuanced in humor as the poems on the fridge in my apartment. But whether we would then call that LLM creative is a question most of us would not answer with the same confidence granted to my housemates and visitors. The interesting part is why.

This thesis is a snapshot in time. I write as a student who saw the mainstream explosion of generative AI during my first year of college, who watched its presence grow internationally during my four years of studying cognitive science and statistics, and who has spent enough time with the creativity literature to begin reckoning with the tensions of LLM creativity. This thesis will illustrate that the question "Are LLMs creative?" cannot be answered yet, and perhaps cannot be answered at all; creativity resists a stable definition, and LLMs fall on the fault lines where those definitions break down.

A multidisciplinary, cognitive science-style approach has been essential to the study of creativity since its dawn. While psychologists have dominated its study since creativity was named as a subject of inquiry, this topic has been discussed by philosophers, linguists, neuroscientists, computer scientists, sociologists, art theorists, mathematicians, and more. To ask whether an LLM is creative is to ask whether something once thought to be distinctly human is happening inside a computational system, and attempting an answer requires moving across the disciplines that have approached creativity from separate angles and have often struggled to meet one another.

As signaled by the wide range of fields that study creativity, and the wide range of fields that have shown mixed public reactions to the rise of AI-generated works in their domains, this question matters for everybody. Many people find the prospect of creative machines uncomfortable. Artists whose works were scraped without consent to train these systems are watching their outputs surface in products they did not agree to contribute to; creative professionals are facing economic pressure from systems whose outputs can be produced at a fraction of the time and cost. Nearly 60 years ago, psychologist Jerome Bruner anticipated a future in which "thinking machines" (1962b, p. 6) would take on routine problems, leaving creativity as the locus of human dignity and worth (1962a, pp. 2-3; Cropley et al., 2023). There is anxiety about what it means if machines can do what was considered uniquely human for so long by so many, and about the malevolent possibilities that might arrive with it. For others, the same prospect reads differently: as the possibility of a new kind of mind, a new kind of collaborator, a new reflection of humans.

All of these reactions proceed from the same conviction: that the answer matters. It will continue to matter in ways that will likely shift as LLMs grow in technical capacity, as widespread LLM usage evolves, and as our individual perspectives shift along with these changes, a drift harder to admit but almost certainly true. This is the case for taking a snapshot now. It captures a moment in which most of the world, having no prior exposure to generative AI before the mainstream emergence of ChatGPT, is grappling at once with the newness of it, the immense shifts, the fear, the excitement, the all of it.

Whether the interested reader is approaching this topic with unease, curiosity, or both, they are almost certainly arriving with opinions, even if those opinions have not yet been fully articulated. I have told countless curious people my thesis topic, and each time have been met with ideas that I had not yet considered. Even more interestingly, often, I have been met with ideas that summarize concepts the creativity literature has already worked through. Creativity can be unusually intuitive in this way. Many people carry a piece of the puzzle to the LLM creativity question already; what is often missing is the language to fully articulate their take. This thesis is written for the reader who arrives with opinions or questions but has not had occasion to study the field, in the hope that they leave more grounded in the shape of the puzzle, more aware of what the creativity literature has been working through, and more able to recognize what they may already intuitively know.

This thesis tackles the central question in two parts. Part I presents the core tensions I have encountered in trying to answer the central question and to understand human creativity along the way. Creativity has been conceptualized in so many ways, by so many people, and across so many domains, that any attempt at a comprehensive system of beliefs would be devastatingly porous; hence, interwoven into Part I's effort are the sub-quests needed to bring a reader up to speed on the creativity literature, along with my offers of ways to navigate and conceptualize the tensions that emerge. Part II applies these findings in an experimental setting: observing 81 evaluators' attributions of creativity to LLM-generated video games. A short concluding section synthesizes the takeaways. This thesis does not answer the central question, but it gives an integrated demonstration of where, why, and on what grounds the disagreements tend to live; it offers some ways to navigate and conceptualize these tensions; and it conducts an experiment that takes a relatively new angle to empirical LLM creativity research.

Part I: Navigating Core Tensions

Introduction

This portion of the thesis examines the core tensions that arise in considering LLM creativity, drawing primarily on the creativity literature across psychology, philosophy, and computer science. These tensions form an entangled web rather than a linear sequence; any attempt to force them into a clean progression would obscure the very interdependencies that make the question interesting. This investigation will therefore meander, but the meandering is purposive: tracking these tensions across their various entanglements reveals how the question of LLM creativity reaches into other pressing questions about LLMs and humans, into a range of disciplines, and thus into the heart of cognitive science.

The overarching structure will be as follows. Section 1 offers a brief historical overview and introduces the most widely cited contemporary definition of creativity, in order to establish that defining creativity is itself the central challenge. Section 2 presents the Four Ps as an organizing framework and uses it to begin surfacing the tensions that recur throughout the thesis. Section 3, the most expansive, takes up the question of what a creative product or process looks like; it addresses domain specificity, LLMs as statistical prediction machines and whether they can produce original content, bipartite models of the creative process including divergent and convergent thinking, and a cluster of questions centered on agency, autonomy, and experience. Section 4 turns to the meta-level questions of who gets to judge creativity and whether the cultural shift brought about by generative AI should prompt revision of LLM-precluding definitions.

As this portion of the thesis develops, it will become apparent that the cruxes of many tensions ultimately trace back to longstanding debates in cognitive science: agency, emergence, reductionism, meaning-making, and others. These are deep waters, and any extensive engagement with one of them would crowd out the abundance of questions that are more specifically geared toward LLM creativity. My strategy, therefore, is partial. Where these broader debates surface, especially in §3.4 and §3.5, I gesture to them, sketch briefly why they remain contested, indicate the shape of the positions that have been advanced, and propose ways of working around their unresolved status. What I am offering in those moments is less a resolution or even recapitulation than a workable orientation, a way of proceeding without pretending the deeper question has been settled.

One final housekeeping measure: A deeply technical prior understanding of LLMs is not required to engage with this thesis, but neither should a working understanding be assumed. Large language models are a class of generative artificial intelligence that produce text in response to inputs. Their outputs are flexible and human-like in part because they are trained on extensive datasets drawn from websites, articles, books, and other textual media. LLMs are prediction machines driven by statistics, containing billions or trillions of parameters that allow them to capture intricate patterns and nuances in language. When an LLM generates an output, it iteratively samples the next token (roughly, the next word) from a probability distribution conditioned on everything written so far, including the conversation history, the input message, and the portion of the output the LLM has already produced. The technology was made possible by decades of work in natural language processing and machine learning. Throughout, I use "LLM" to refer to the class of systems generally, but my examples often invoke the leading proprietary models for their accessibility, high performance, and familiarity. With a working sketch of LLMs in place, the core debates can be taken up.

1 On Defining Creativity

1.1 An Abbreviated History of 'Creativity' and Its Study

The first problem in navigating the prospect of LLM creativity is that the term *creativity*, unsurprisingly, is extremely difficult to define. This will be a continuous thread throughout the tensions discussed in this thesis, as debates will meander through arguments about how humans have defined creativity and how these proposed definitions might or might not be observable in LLMs. In the broadest sense, akin to the cognitive science goal to "explore minds and intelligence in whatever forms they occur" (Vassar College, n.d.), I am interested in creativity everywhere that we can find and study it.

The endeavor to define creativity overall as a phenomenon, as opposed to in practical applications like education or business, has primarily occurred in psychology research (Lamb et al., 2018). In particular, the social-psychological study of creativity has been foundational to creativity research and is where most of the notable authors and fundamental concepts in the field are located (Mejia et al., 2021). In this thesis I will largely draw upon psychology literature for definitions of creativity, especially as they have often been quite multidisciplinary, and I will take a multidisciplinary approach in connecting these ideas to LLMs. That creativity has primarily been studied in the field of psychology is itself an interesting implication, and it warrants a précis of the evolution of the term *creativity* and how it has been studied.

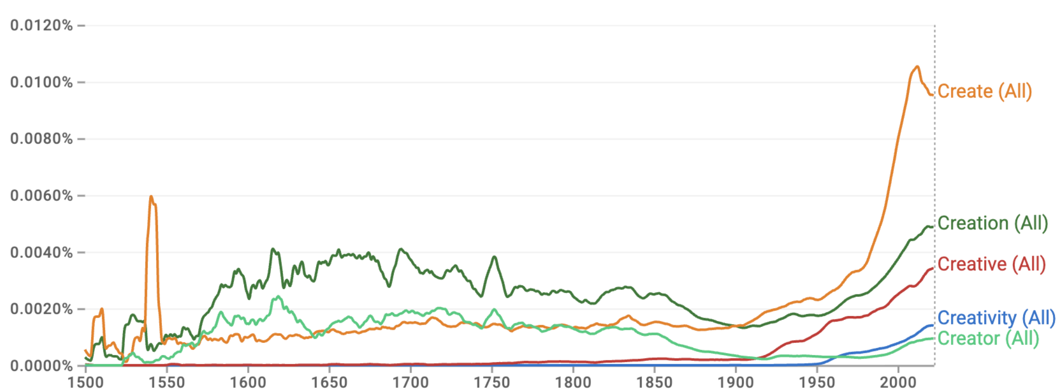

Figure 1. Google Books Ngram Viewer results showing the relative frequency of "creative," "creativity," "creation," "create," and "creator" from 1500 to 2022. Note the relatively late emergence of "creative" (late nineteenth century) and "creativity" (mid–late twentieth century).

The verb *create* comes from the Latin *creare*, which means to make or produce in a physical sense (Barón-Birchenall et al., 2025; Götz, 1981). Historian of philosophy Władysław Tatarkiewicz identified four general phases in the meaning of the term *creativity* over time (1980). The first phase occurred in the ancient world with Greeks and Romans. The Greeks had no term corresponding to creativity, and the Romans used the term *ingenuity* where, today, *creativity* would typically be used (Barón-Birchenall et al., 2025; Tatarkiewicz, 1980). While philosophers in this time discussed ideas of "divine madness" (Plato, 2005), inspiration, imitation (Plato, 2012), poetry, and art (Aristotle, 1996), and thus current discussions of creativity often retroactively apply the term *creativity*, the term was not used. The second phase is associated with Christianity and endured until the Enlightenment; during this time, *creator* was a synonym for God (Barón-Birchenall et al., 2025; Tatarkiewicz, 1980). This phase saw some of the first uses of the terms *creation* and *creative*, setting the stage for the third phase (Figure 1).

The third phase occurred in the nineteenth and early twentieth centuries; this was when the term *creator* expanded to art (Barón-Birchenall et al., 2025; Tatarkiewicz, 1980). *Creator* became synonymous with *artist* and terms like *creation* and *creative* were being discussed in the arts, often linked with Romantic ideas of artistic genius. This phase saw the concept of genius elevated to a near-mystical status, with Romantic thinkers positioning the genius as the embodied source of originality; this framing bled into later biological and psychological accounts of creativity (Barón-Birchenall et al., 2025). *Creator* then expanded to all domains in the twentieth century, akin to how we use the term today. Morphological derivations of the root *create* proliferated during this time, and discussions beyond the arts began to use the term *creativity* (Whitehead, 1929).

The fourth phase, which is still ongoing, began in the mid-twentieth century and has been characterized by an increase in descriptions and conceptualizations of creativity, especially in the sciences (Barón-Birchenall et al., 2025; Tatarkiewicz, 1980). The launch into the scientific study of creativity is widely attributed to J.P. Guilford for his 1950 APA presidential address, which urged for a scientific endeavor in psychological science to understand creativity (Mejia et al., 2021). Early research in this phase assumed an individual, intrapsychological origin of creativity, partly because post-WWII research perceived creativity as part of progressive individualization (Barón-Birchenall et al., 2025; Glăveanu & Kaufman, 2019; Tatarkiewicz, 1980).

Remarkably, the rise of scientific research on creativity roughly coincided with the rise of computer intelligence. Five years after Guilford's address, the famous 1955 Dartmouth Summer Research project on AI proposed the intent to simulate all aspects of learning and intelligence; particularly, the address described one aspect of the AI problem as "Randomness and Creativity" (McCarthy et al., 2006). Six years later was when Bruner (1962a, 1962b) anticipated that thinking machines would leave human creativity as the only important remaining capacity that is distinctly human. Then, the late 1990s saw the establishment of computational creativity as a formal field, which studies the philosophy, science, and engineering of computational systems that, under various definitions of creativity, could be deemed creative, capable of producing creative outputs, or both (Colton & Wiggins, 2012; Jordanous, 2016). The question of artificial intelligence creativity has, in some shape or form, been around almost as long as the scientific launch into the question of creativity.

1.2 The Standard Definition of Creativity and Its Variants

In 2012, leading creativity scholar Mark Runco and colleague Garrett J. Jaeger published the Standard Definition of Creativity (SDC), which says that creativity requires both originality and effectiveness. Originality can also be labeled as novelty, and effectiveness can also be labeled as utility, appropriateness, fit, or value (Runco & Jaeger, 2012). Runco and Jaeger explain that a creative product certainly must be original, but originality alone can be useless; hence, a creative product must be both original and effective. This definition is treated as its name would suggest; it is widely cited across disciplines, commonly sustains creativity experiments, and is often treated as the reference point even by its critics (Brandt, 2021).

Signals of the SDC can be traced as far back as Tatarkiewicz's second phase of creativity. In this phase, theology's hold over the term *creator* associated the term with the ultimate source of originality, the Christian God, and with the ultimate effective products, His "creation of the world" (King James Bible, 1611, Romans 1:20). More overtly, between Tatarkiewicz's (1980) second and third phases of creativity, Immanuel Kant positioned artistic genius as an ability to produce works that are not only original, since there can be original nonsense, but also exemplary (Kant, 1790/2000). The artist-as-originator became a defining figure of the Romantic period (Barón-Birchenall et al., 2025). A biological paradigm for multi-domain genius from 1869 (Galton) positioned creativity as historic-level, domain-transforming discoveries and achievements, which set a high standard for originality and effectiveness of outputs.

The beginning of Tatarkiewicz's (1980) fourth phase of creativity saw these components begin to formalize. Guilford (1950) emphasized novelty and acceptability when pointing to criteria for creativity. Runco and Jaeger traced their definition back to Stein (1953), who wrote, "The creative work is a novel work that is accepted as tenable or useful or satisfying by a group in some point in time." Cattell and Butcher (1968) and Heinelt (1974) used the terms *pseudocreativity* and *quasicreativity* to describe products that were not effective, regardless of their originality, which reflects the importance of these two components.

The Standard Definition of Creativity is almost always operationalized to describe a creative product, with some authors putting forth the adjacent definition that a producer is creative if they produce outputs that are original and effective (Jordanous, 2016). This apparent tidiness, though, masks what it leaves unspecified. It instructs investigations into creative products, but does not touch on who is qualified to judge these properties, what process is required to produce them, what role culture and context play in their recognition, or whether the producer's identity matters at all. For LLMs, these missing pieces are crucial, opening the doors to many debates.

2 On The Four Ps and the Problem-Solving to Abstract Spectrum

2.1 The Four Ps

One of the first reviews of creativity literature and definitions was published in 1961. In this, Mel Rhodes found that definitions clustered into four strands: Product, Process, Press, and Person. Product describes the output that is being judged as creative; this is where the SDC lies. Process describes the steps taken to produce the output; process research studies what kinds of actions are undertaken when creative work is done. Press describes the environmental, contextual, social, and cultural influences that played a role in the outcome (1961); press plays a central role in many prominent definitions of creativity. Finally, Person is the human who creates. Following a computational creativity perspective on the four Ps (Jordanous 2016), I will use Producer rather than Person to allow consideration of non-human systems. Research into human producers typically studies attributes such as skill, imagination, and personality traits (Rhodes, 1961).

Rhodes emphasized that each strand "has unique identity academically, but only in unity do the four strands operate functionally" (Rhodes, 1961, p. 307), meaning no single P could wholly account for creativity. Though all four Ps have been individually studied and theorized about, definitions of creativity, both when considering non-human producers and when focused only on humans, vary in which Ps they use and how they use them, setting up perhaps the most pressing question in LLM creativity: Which Ps matter? This section will introduce some perspectives in the human and creativity literature regarding the significance of each P. The four Ps will continue to unfold throughout the entirety of this thesis, and thus these sections serve as brief introductions to some general attitudes towards the Ps and some very initial tensions that start to arise.

2.2 Product

Humans tend to judge a person's creativity by what they produce (Ritchie, 2007). Consider a painter, a mathematician, a poet, or someone who is all three. Intuitively, each person's products are accessible indicators of their creativity. A product-based view tends to be the default perspective within many fields (Hennessey and Amabile, 2010; Jordanous, 2016), perhaps because of—or contributing to—the SDC. This intuition maps onto some of the most widely used short assessments of human creativity, which are: the Divergent Association Task (DAT; Olson et al., 2021), the Alternative Uses Task (AUT; Guilford, 1967), and the Remote Associates Test (RAT; Mednick & Mednick, 1967). The DAT has the participant generate the ten most unrelated nouns that they can think of within a time limit and then computes the participant's creativity score as a function of semantic distances (Zhao et al. 2024; Yao et al. 2026). The AUT asks the participant to generate as many creative uses for a common object as they can within a time limit, with responses scored by humans on various dimensions of creativity (Olson et al., 2021). The RAT repeatedly presents the participant with three words and asks them to think of a single word that connects the set; there is a predetermined answer to each set of words, and the participant's score is the number of correct answers. Though these assessments function in different ways, each one takes a product-based view.

Ritchie (2007) draws on the human propensity for product focus to argue that a black-box approach should similarly be applied in computational creativity, as has largely been the case (Jordanous, 2016). A straightforward example of this in LLMs comes from one of the earlier empirical assessments of generative AI, in which Haase and Hanel (2023) distribute the AUT to a set of chatbots like ChatGPT-3 and compare their performances to that of humans. Haase & Hanel (2023) leverage the SDC to contend that an LLM is creative if it can produce an original and effective output; they take the functionalist-adjacent view that the right question is not whether generative AI is truly creative, but whether its products are perceived as creative.

Alongside the intuitive mapping from human intuitions to the LLM space, there is pragmatic appeal to focusing on product. Product evaluation is empirically tractable, and it presents a seemingly clear goal for LLM developers to work toward. Even in the human case, product evaluation is more directly accessible than process evaluation, which depends on indirect methods like behavioral tasks, self-report, think-aloud protocols, or, more recently, neuroimaging, which offer only partial access to the cognitive happenings (Beaty et al., 2016; Ericsson & Simon, 1980). Despite these intuitive and pragmatic advantages, many creativity scholars would find a purely product-based definition not just incomplete but missing the point.

2.3 Process

This section takes up what process broadly names in creativity theory and introduces some examples of where process intuitively matters. The specific cognitive processes at stake in the LLM comparison are mostly deferred to section §3, as many prominent accounts of creativity involve both process and product.

Though many perspectives on creativity value both product and process, some scholars like Green and colleagues (2024) argue that creativity is constituted only by processes, and a creative product is only made possible by a creative process. Select art domains reflect this argument: Lewitt's "Paragraphs on Conceptual Art" (1967) defined conceptual art as art where "the idea or concept is the most important aspect of the work… The idea becomes a machine that makes the art" (Jordanous, 2016). Improvisation in jazz and other domains also tends to value the process of improvisation much more than the final product in the context of creativity (Jordanous, 2012; Jordanous, 2016). Others, like Plucker and colleagues (2004), define creativity as both process and product: "the interaction of aptitude, process, and environment by which an individual or group produces a perceptible product that is both novel and useful as defined within a social context" (p. 90; Cropley et al., 2023); these types of definitions can accommodate small acts like a child discovering how to tie their shoes.

Often focused on product, the literature on LLM creativity has seen a recent increase in consideration of process, even if only to acknowledge its absence (Jordanous, 2016). An example includes Stevenson and colleagues (2022), who are responsible for the "first systematic psychological assessment of an LLM's creativity" (p. 2) and acknowledge that the use of the AUT perhaps unjustly ignores the process. In their application of the AUT, Haase and Hanel (2023) emphasize concern that testing materials and human answers may have been included in the chatbots' training data and thus affected the chatbots' processes, but they still argue that a perceptually creative output is all that matters.

While LLM creativity literature typically focuses on product or on the other Ps as they pertain to a product, the idea that process alone could constitute some cases of creativity in humans prompts an inquiry into whether there is an argument to be made that, in some situations, process alone has constituted examples of LLM creativity. One potential example of this is proprietary LLMs' use of cybersecurity skills. In 2025, Anthropic administered a benchmark evaluation eleven independent times to Claude Opus 4.6, a leading proprietary LLM with capabilities like web searching. Two of these times, Claude "independently identified that it was being evaluated" (Coleman, 2026), identified what it was being evaluated for, found and decrypted the answer key online, and then output the answers.

More drastically, the currently unreleased Claude Mythos Preview has allegedly discovered thousands of cyber infrastructure vulnerabilities in leading companies worldwide, surpassing "all but the most skilled humans at finding and exploiting software vulnerabilities" (Coleman, 2026). These processes seem to require some level of creativity, even though the output of benchmark answers or the output of vulnerable code might not in themselves appear creative. Whether a process divorced from a conventionally creative product can itself bear the label 'creative' is contested even in human contexts; for LLMs, the question compounds, tangled with unresolved issues about agency and intentionality that I will return to in §3.4 and §3.5.

2.4 Press

Press appears to be the most neglected of the four Ps, particularly in individualistic accounts that locate creativity in the cognition or personality of the creator (Jordanous, 2016). Despite this, some frameworks argue that press is at the heart of creativity. Csikszentmihalyi's (1988, 1999) systems model advanced one of the strongest versions of the press view, arguing that creativity emerges from the interaction of individual, the domain, and the field, where domain is the body of knowledge and conventions of a field and field is the social gatekeepers who judge and validate contributions. On this account, the influence of the domain in the individual's production of an output is the press's influence, and it is inextricable from the meaning of creativity.

As suggested by the field aspect of Csikszentmihalyi's (1988, 1999) model, scholars often extend Rhodes's (1961) original description of press to be reciprocal: not only does the world influence the production of an output, but the world then judges the creativity of the output

(Glăveanu, 2013; Lamb et al., 2018). Cross-cultural scholarship has also argued for the importance of reciprocal accounts by evidencing that what counts as creative varies substantially by cultural context: Western accounts emphasize originality and novelty while many Eastern accounts emphasize authenticity, insight, and inner truth (Lubart, 1999). Communal attribution of creativity has also been established in the cultural psychology literature: Glăveanu (2010, 2013), for example, argues that creativity emerges from dialogical relations between self, community, and existing cultural artifacts, such that what counts as creative is always a context-dependent attribution rather than an objective property of a product or person.

The LLM creativity literature under-engages with press in a way that parallels its historical neglect in creativity research generally (Jordanous, 2016). When it does enter, it usually enters through the reciprocal approach via interest in human evaluators' reception of AI output, e.g., Guzik et al. (2023). Very few LLM creativity papers, especially those supporting LLM creativity, treat press in Rhodes's (1961) sense (Jordanous, 2016).

There is an argument to be made that LLMs are the quintessence of Rhodes's (1961) press, as LLMs are trained on more text data than any single human could acquire in a lifetime. All of the social and cultural influences, the domain knowledge, the biases and critiques that make their way into an LLM's training data contribute to the LLM's pattern detection, construction of probability distributions, and thus generation of outputs. In this sense, an LLM's output is nothing if not a statistical recapitulation of press. Furthermore, in cases when an LLM is designed with or prompted to act with the overarching goal of satisfying its users, the LLM could be viewed as a function of press designed to satisfy the press.

This inversion is philosophically awkward. Press was conceived as the contextual surround that interacts with a distinct creator; when the creator is itself a statistical distillation of that surround, press loses the relational structure that made it explanatorily useful. Rather than establishing LLM creativity through press, this may instead reveal that press cannot do the work that accounts like Csikszentmihalyi's or Glăveanu's ask of it once the producer's boundary is blurred. These discussions resurface in §3.5.

2.5 Producer

Unique from the other Ps, producer holds a terminological instability. Recall that Rhodes (1961) named this P, Person. Only retroactively has the computational creativity literature applied Producer, and some scholars explicitly choose to abide by Rhodes's original conception instead of using Producer (Lamb et al., 2018). This difference is more than cosmetic: Rhodes (1961) described Person-centric work as investigating personality, intellect, temperament, physique, traits, habits, attitudes, self-concept, value systems, defense mechanisms, and behavior. Later person-centric work has added other qualities such as curiosity, persistence, independence, and openness (Stein, 1968). An effort to map these attributes onto LLMs is either metaphorical or requires specific prompting. Zhao and colleagues (2024) pursued the latter to find that most human trait-creativity correlations replicate in LLMs when Big Five-style traits are prompted into the system.

Yet, the metaphor does not sit easily. Person-based frameworks presuppose a stable identity with persistent dispositions, a self-concept, and characteristic habits of attention. LLMs have none of these in any straightforward sense. The central questions raised by this P, once applied to LLMs, are the questions of agency, intentionality, and stable selfhood; these are taken up in §3.4 and §3.5 as the terrain on which producer-based objections to LLM creativity ultimately rest.

2.6 The Problem-Solving-to-Abstract Spectrum

From my observations in the literature, conceptualizations of creativity often fall on a spectrum from viewing creativity as problem-solving to something that is artistic and entirely abstract with no concrete function. This spectrum tends to parallel the divide between solely product-based views and views that encompass other Ps: the more a scholar discusses creativity as problem solving, the more comfortable the scholar is with product-based evaluation; conversely, the more a scholar discusses creativity as something of the abstract or art for art's sake, the more the scholar attends to process, as well as press and the producer as they pertain to process. The problem-solving end of this argument has already surfaced in discussions of the SDC and product-based views, but the abstract end of this argument deserves more attention. To me, this reflects that the process, the producer's intentions, and the influences that were involved in the production of an output all assemble into evidence that the abstract product has value—that it is effective. In contrast with a doctor's creative solution for a patient or a mathematician's proof, when an output has less inherent, obvious, objective meaning, we turn to the process and the human for meaning.

I am not aware of literature that has specifically framed the product vs. process debate in this way, yet it can elucidate the ways in which debates on definitions often subduct rather than clash into one another. At risk of parroting Lewitt (1967), consider Marcel Duchamp's *Fountain* (1917), a signed store-bought urinal that Duchamp put forth as an art piece to challenge definitions of art. *Fountain*'s controversy hinges on the lack of process and producer contribution. Consider action paintings, which are created when artists spontaneously pour, smear, or splash paint onto canvases in no particular fashion. Arguments about the legitimacy of action paintings often hinge on whether the piece genuinely reflects anything meaningful. Consider Yves Klein's *Blue Monochrome*, a monochrome canvas made of the pure ultramarine International Klein Blue (IKB) which Klein co-developed with a chemist and trademarked. *Blue Monochrome* and Klein's trademarking are generally less controversial because of Klein's involved development of IKB, whereas works from Anish Kapoor, who purchased the right to artistic use of Vantablack, are often criticized due to Kapoor's lack of process and producer contribution. Whether these pieces are creative is heavily contested, and the arguments almost always hinge on whether the piece meant something to the artist, or whether the artist imposed some part of themselves into the work.

This sentiment extends to textual products and, critically, LLMs as well. Consider the fridge magnet poem thought experiment posed at the beginning of this thesis. John Dewey's (1934) *Art as Experience* claims that art can be understood as the means to the end of experiencing. In the poem example, this is partly enabled by the assumption by the reader and the poet that the poem reflects some type of meaning, intention, or emotion from the poet, even if the reader finds the poem unclear and even if the poet did not set out with active intentions when creating the poem. Perhaps this is because we as consumers of abstract creative products want to decode the producer's intended meaning, or perhaps we enjoy constructing our own understanding in the company of knowing that meaning is there; regardless, we want to see a reason for a piece's existence, even when the reason is not concrete.

This might informally be one of the main motivations for rejections of LLM creativity both from people in the general public and from some scholars. As suggested by the fridge magnet thought experiment and alluded to by creativity scholars' thought experiments (e.g., Colton, 2008), when reading an LLM-generated poem, it can be upsetting that there is no intended meaning to be decoded, or at least not in the way that there is with human-written poems. In this mindset, for a consumer to construct their own understanding of an LLM-generated poem or an AI-generated artwork can feel almost foolish in that the consumer is constructing reasons or purposes where there are none; that is, not unless the consumer values press in such a way that an LLM-generated poem or an AI-generated artwork can have meaning in its reflection of humanity as a whole, or unless the consumer turns to the human prompter of the LLM.

A 2024 study on poetry found that non-expert readers not only could not distinguish ChatGPT-3.5-generated poems from poems by well-known human poets, but actually rated the GPT poems more favorably on qualities like rhythm and beauty when they were unaware of GPT's involvement (Porter & Machery, 2024). Porter and Machery's follow-up analysis argued that the "more human than human" (p. 2) effect from the source-blind participants can be partly explained by the GPT poems being easier to interpret: they communicate mood, theme, and emotion more directly than the human poems, which tend to resist easy interpretation and reward in-depth analysis. However, when participants were aware of GPT's involvement, they rated the poems more negatively than the human-generated poems (2024). To me, this indicates that when participants were able to expand from a purely product-focused view to producer and their own perception of process, the features that matter to poetry as an art form—its opacity, its demand for interpretation, the sense that there is a meaning worth decoding—all become criteria, and thus the LLM poetry becomes wanting. This reversal suggests that, for abstract products, attention to process and producer reshapes the perception of the product itself. This observation is why the next section turns to domain: which Ps matter depends, in part, on the type of output.

3 On Creative Products and Processes

3.1 On Domain Specificity

Creativity, at the most superficial level, presents differently in different domains. A student generating an unconventional mnemonic during exam preparation looks different from an author writing a fiction novel, and the mnemonic itself looks different from the novel. Whether this is a manifestation of creativity as a domain specific phenomenon, though, is contested. Perspectives on the domain specificity of creativity range from accounts of complete specificity to accounts of varying degrees of domain-generality.

A perspective from the complete-specificity end comes from Baer (2017), who argues that because the creativity of a human's output dramatically varies depending on the domain, creativity itself is domain-specific. One of Baer's strongest pieces of evidence draws on the Consensual Assessment Technique (CAT), a widely used method of product evaluation in which a panel of domain-familiar individuals, preferably experts, evaluate the products using their personal subjective definitions of creativity (Amabile, 1982). Amabile (1982, 1983, 1996) developed this technique from the premise that creativity cannot be articulated through ultimate objective criteria, and thus a product's creativity should be evaluated by those who share tacit, implicit criteria within a domain. Baer (2017) illustrates that assessments using the CAT have consistently failed to find within-subject correlations between CAT results in different domains.

Baer (2017) also draws on human creativity assessments. He references Plucker (1999), who found that humans' verbal scores on divergent thinking, a concept that was often used synonymously with creativity at the time, only predicted individuals' creative achievements in the verbal domain. When test and criterion come from different domains, the linkage disappears. Taken together, Baer (2017) employs these findings to suggest that humans cannot answer the question "How creative are you?" without conditioning their response on domain(s).

Scholars that value both specificity and generality have often arrived at their perspectives via different reasoning. Jordanous's (2012) SPECS methodology takes a middle position: she empirically derives fourteen components of creativity from a corpus of academic writing on creativity and proposes them as a domain-general definitional base that becomes domain-specific through weighting. Creativity, in this view, is common enough across domains to share a vocabulary of components, but differentiated enough that what makes it recognizable in any particular domain depends on how those components are prioritized.

A framework of creativity that brings in another perspective on domain specificity is the Four Cs model (Kaufman & Beghetto, 2009), which classifies creativity into four hierarchical types: mini-c (personally meaningful insights and interpretations, often arising during learning), little-c (everyday creativity), Pro-c (the work of creative professionals who have not reached eminence), and Big-C (legendary, field-transforming creativity). Note that everyday creativity and legendary creativity are commonly accepted by scholars as classifiers and have been established more than once under different terms, e.g., mundane creativity and historical creativity. In the Four Cs model, Kaufman and Beghetto (2009) state that domain specificity is low for everyday creativity and high for eminent creative achievements; hence, the type of creativity being assessed is another variable to consider in the effort to examine LLM creativity. Everyday and legendary creativity as they pertain to LLMs will be revisited in §3.2.4.

The perspectives that value domain allow a response to oppositions such as, "But LLMs don't get humor." This is that if LLMs are not as funny as the budding poets in my household, then there are countless other domains still in the running. Most current proprietary LLMs are developed as domain-general tools, with the caveat that they can only act in domains whose outputs are representable as token sequences. This opens myriad domains in which researchers can test LLM creativity, but also creates somewhat of a scavenger hunt for the domains in which LLMs might be creative.

Despite the domain-general design of LLMs, there are signals that their capabilities may vary by domain. Avid users often apply different proprietary LLMs to different tasks because they have recognized different aptitudes amongst the LLMs. Moreover, companies behind leading proprietary LLMs have begun releasing coding-specialized siblings to their general-use models, indicating a move towards LLMs that are not as generalized. Thus, on multiple bases, there is something to be said for taking a domain-specific approach to the question of LLM creativity, especially when taking a product-based view.

The arguments that devalue domain specificity are perhaps better-termed as domain-agnostic or domain-adaptable. These typically do not contest that creativity looks different across domains, but rather argue to varying degrees that domain specificity is not relevant to every aspect of creativity study. Plucker and Beghetto (2004), for example, reason that definitions of creativity should be domain-general and assessments should be domain-specific. Plucker and Beghetto (2004) also, however, admit that features of creativity like who or what is creative are domain specific. It is notable that Plucker and Beghetto (2004) allowed both producer and product to be positioned as domain specific given that the title of this book chapter is "Why Creativity Is Domain General, Why It Looks Domain Specific, and Why the Distinction Does Not Matter."

The SDC aligns with Plucker and Beghetto's (2004) prescription: the SDC is itself domain-general, but the application of it for product assessment becomes domain-specific. The concepts of originality and effectiveness make sense in virtually every domain so long as their definitions remain flexible; recall the problem-solving to art spectrum. When an evaluator then assesses a product for originality and effectiveness, the evaluator must define originality and effectiveness; these definitions, especially effectiveness, will almost certainly be influenced by the domain. Thus, these criteria are not so much domain-general as they are domain-adaptable, reflecting a relationship between definition and assessment that would likely satisfy Plucker and Beghetto's (2004) case. This domain-adaptability of the SDC is part of why the SDC might lend toward LLM creativity more than other definitions; however, consistent with the second takeaway from Plucker and Beghetto (2004), this flexibility is partly made possible because the SDC omits the "who," which is, for this thesis, the entire point.

Hence, there is a balance to be struck between domain-generality and domain-specificity both in evaluating human creativity and especially in evaluating LLM creativity. Whereas entirely domain-specific frameworks may send scholars on wild goose chases to find the domains in which LLMs are creative under some assumed definition, frameworks that are too adaptable may be so accommodating that acceptance of the framework instead becomes the debate.

One throughline of this discussion, consistent with Plucker and Beghetto's (2004) observation, is that product-based frameworks domain-adapt more readily than process-based ones; the SDC is a clean example. This asymmetry matters for LLM creativity in particular, because most of the fundamental philosophical challenges to LLM creativity are concentrated in the process and the producer rather than in the product. Jordanous's (2012) work makes this legible: her SPECS methodology intentionally incorporates process and producer, distinguishing her research from the largely product-focused lens of the rest of this discussion, and her components illustrate that something like intention and emotional involvement can be weighted heavily in one domain and lightly in another. The implication is that domain specificity does not merely offer LLM creativity analysis a set of domains where LLMs might more easily be deemed creative because the philosophically challenging components are deprioritized, but also structures the process question itself: the more a domain weights process- and producer-heavy components, the more unavoidable the challenges of computational intentionality, authenticity, and experience become. These challenges are subjects of discussion in §3.5.

Domain specificity only matters if LLMs are eligible for creativity in any domain at all. If the reasons why LLMs are generally not funny are part of a broader problem that will affect creativity of LLM outputs in all domains, then domain-specificity as an argument is irrelevant. One broader problem in LLMs is originality.

3.2 Can LLMs Produce Original, But Not Random, Outputs?

3.2.1 LLMs as Statistical Prediction Machines

Originality has been a near-universal element in accounts of creativity, from the term's linguistic evolution through modern scholarly and applied definitions (Runco & Jaeger, 2012). Studies of LLM creativity suggest that human evaluators perceive LLM outputs as creative in various contexts, especially in studies that take a purely product-based view by blinding participants to LLM authorship, e.g., Guzik et al. (2023). However, perception need not reflect mechanical reality.

Whereas some of the pre-twentieth century accounts framed originality as a metaphysical or biographical property of genius, e.g., Kant (2013), the shift towards psychological study of creativity has seen a rise in statistical framings of originality. Guilford (1950) proposed that the degree of novelty that a person possesses could be measured by the frequency of uncommon yet acceptable responses that they produce. The manual for the Torrance Tests of Creative Thinking, one of the most commonly used and rigorously validated assessments of human creativity (Torrance, 1974), characterizes originality as "based upon the statistical infrequency and unusualness of responses" (TTCT Interpretive Manual, 2018). This framing intuitively tracks from a general understanding of originality: for something to be new to some extent, it must be uncommon to some extent. In the context of LLMs, however, this creates friction.

A statistical framing of originality presents a quandary for LLMs. Tasking an LLM with producing an original output seems, at face value, to task the LLM with selecting the tokens that are statistically most likely to be statistically least likely. In practice, the LLM produces what its training data indicates is the statistically typical continuation of a prompt requesting originality. In this way, the mechanism by which an LLM produces an original output is pattern-matching on exemplars of originality rather than sampling against the distribution, which raises structural problems: the LLM's method of making something original is to turn to what "original" has meant in the past and mimic that within the given context, which seems almost orthogonal to the meaning of originality both in process and product.

A domain in which this tension takes center stage is humor. Humor has been formally characterized in linguistics as requiring the activation and resolution of incompatible interpretive frames (Attardo, 2017), which produces the same challenge: LLMs are given the training objective of minimizing perplexity on the next token, which is structurally at odds with the "surprise and incongruity" (Ajayi & Mitra, 2026, Abstract) that humor requires. Current work on LLM humor suggests that LLMs are indeed poor humorists, drawing on common syntactic patterns and failing to produce unexpected or nuanced humorous outputs.[1]

This also extends to the abstract in ways reminiscent of the problem-solving to abstract spectrum from earlier. Consider bizarre authors, experimental music artists, or eccentric painters. These individuals seem to constantly break social and even ethical boundaries in order to think and act outside the box. Absurdist artwork is, by definition, an illogical approach to challenge reality. The weird, the gruesome, the unexpected—it comes out of these producers despite social and ethical expectations. This would also seem orthogonal to an LLM's training unless, perhaps, one could find a way to prompt these extreme combinations; more on this in §3.4.2.

A natural follow-up to this architecture is the idea to alter an LLM in some way such that it does not abide so heavily by the patterns. Enter temperature, the parameter that most directly controls how much lower-probability tokens enter the sampling process. When temperature is set to zero, the LLM always selects the highest-probability tokens; as temperature increases, the probability distribution flattens, giving lower-probability tokens a greater chance of being sampled.

Temperature has often been informally treated as the creativity parameter, but this characterization is not quite right. Tokens have a lower probability when they do not appear as frequently given the situation (conversation history); one reason why a token might not appear as frequently is because it does not make sense in the situation. A human could sample a number of lower-probability options and then evaluate whether any of them are effective, a process that foreshadows later discussions of human creative cognition. The LLM's ability to evaluate

[1] Tangential to this thesis, Dogra et al. (2026) studies the roles of toxicity and stereotypes in LLM humor in a very interesting way.

effectiveness, however, comes from the same pattern recognition that is operationalized as the probability distributions in the first place. Hence, if the LLM has a high temperature and selects some of the odd-ball tokens, the generated output might be ineffective and, depending on the severity of the incoherence, even be perceived as random. In this way, temperature is increasingly referred to as a randomness control in the literature and in practical application, with a trade-off of coherence (Noble, n.d.; Peeperkorn et al., 2024).

In addition to the theoretical issue with temperature, it might not vary outputs in the ways that matter. Peeperkorn and colleagues' (2024) study on LLM-generated narratives found that higher temperatures did not extend outputs into distinctly different regions of embedding space. That is, though outputs at higher temperatures had more lexical variation, they remained semantically close to those at lower temperatures. This result might not extend to other domains, but it nonetheless supports the notion that temperature cannot entirely remedy this tension, and might be of little help at all.

Taken together, the mechanism by which LLMs produce potentially creative outputs may constrain LLMs to recombining tokens based on trends in past outputs associated with originality. The mechanism not only prevents LLMs from operating under an architecture that aligns with a statistical framing of originality but also, in its reliance on pattern recognition, could be understood as orthogonal to the notion of originality.

3.2.2 Associative Thinking and Combinatorial Creativity

Though LLMs' reliance on past outputs associated with originality may seem orthogonal to the notion of originality, a reframing of this mechanism might support the LLM architecture's capacity for original outputs. Güzelci and Karadag (2023) claim that generative AI produces statistically unlikely outputs precisely by recombining training data, which sets the stage for Mednick's (1962) associative theory of creativity. In the context of humans, Mednick (1962) proposes that creative individuals are distinguished by their ability to connect concepts that are far apart in their semantic networks; on this account, creative thinking is the traversal of unusual distances between nodes.[2] In this framework, since LLMs possess a larger semantic network than any single human possibly could, they also possess a greater quantity of extremely distinct ideas and non-obvious patterns that, if associated in an output, could lend to very unique combinations (Haase & Hanel, 2023).

This view aligns with a type of creativity defined by Margaret Boden, a late professor of cognitive science and philosopher of AI. Boden (1990, 2004) divided creativity into three types: combinatorial (unfamiliar combinations of familiar ideas), exploratory (reaching new points in a conceptual space by test rules' implications), and transformational (changing the rules to reach points that were previously inaccessible). Boden explains combinatorial creativity as the most common type of creativity (1990, 2004). It involves combining two or more distinct concepts,

[2] This has been reflected in the structural patterns across retroactive models of creative humans' semantic network models in performing human creativity assessments (Yao et al., 2026).

domains, or frameworks that already exist and putting them together in a way that hasn't been done before. This is the type of creativity involved in writing a metaphor, inventing the clock-radio, or coming up with a clever headline, and it is the bedrock of associative thinking.

If LLMs truly accomplish these types of creativity, then, by more frameworks than just Mednick's (1962) and Boden's (1990, 2004), they could be deemed creative in the same ways as humans. Many scholars have argued that human creativity is entirely made up of this searching, reorganizing, and combining of knowledge (Beaty et al., 2014; Benedek et al., 2012, 2018). Frosio (2023) argued to the extent that all human creativity is fundamentally combinatorial, which means the notion of absolute originality is a post-Romantic myth that AI-generated creativity now forces everyone to confront.

The question is whether LLMs actually accomplish the associative and combinatorial creativity that their architecture might make possible. Associative creativity and combinatorial creativity both present a problem: even if an LLM has a vast database of concepts, domains, and frameworks with which it could potentially produce myriad unique combinations, this does not mean that it will produce these combinations, and it also does not mean that it will produce unique combinations that make any sense.

To take advantage of the vast potential for unique combinations, an LLM would need to sample a wide range of its probability distributions, if not the entirety of the distributions, which would introduce extreme randomness and thus, most of the time, incoherence (Peeperkorn et al., 2024). Boden (1990, 2004) emphasized this tension, arguing that computers could easily combine distant concepts at random but that it would be exceptionally difficult for a computer to understand why one combination could be profoundly meaningful or effective to a human, while another is meaningless. Additionally, Runco (2025) argued that AI outputs are either identical to what they find in the training data or damningly derivative of what already exists, which, to me, sounds like an implicit argument that LLMs fail to take advantage of the potential for unique associative combinations.

This recurring notion of meaning will be returned to in §3.5.2 this thesis, but it raises the question here about whether the LLM association process is the same kind of human association process in the ways that matter. LLMs represent meaning through distributional semantics, the theoretical foundations of which date back to Firth's (1957) and Harris's (1954) sentiments that meaning can be recovered from the statistical patterns of words appearing near other words. LLMs learn a high-dimensional space in which words that occur in similar contexts end up close together, and generating output involves moving through this space. Thus the surface structures look similar: in both cases, connecting remote concepts involves movement across a space of meanings. But whether the kinds of spaces are the same beyond the surface level, and whether the surface level is enough, is not clear.

There is a further wrinkle here. Some scholars have historically framed creativity as emergent, arguing that genuinely new ideas arise from simpler inputs in ways not fully reducible to their components (Estes & Ward, 2002; Rogers, 1959; Bouchard et al., 1993). A purely combinatorial account sits uneasily with this framing. An LLM's recombination of existing elements is often taken to be reducible to those elements and the operation that joins them, and whether the same reduction holds for a human's, given the applicability of Mednick's (1962) associative theory to both, remains open. Whether associative traversal across a semantic network counts as emergent or merely recombinative draws on the larger debate of emergence within cognitive science, but it matters here: if creativity requires something beyond recombination, the associative account may rescue LLMs from the randomness problem only to run them into a different one. And here, once again, enter the issues that arise with taking a product-based view: we cannot infer creative process from a perceptually creative output, particularly with systems whose outputs might resemble emergence.

Some approaches to human semantic networks posit that they are shaped by embodied experience, goal-directed thought, and a lifetime of situated meaning-making (Barsalou, 2008; Siew et al., 2019). LLM embeddings are shaped by co-occurrence statistics in large text corpora. Whether these produce associations that are equivalent in kind, or only look equivalent from the outside, is a question this thesis cannot resolve, but it is worth naming explicitly, because much of the debate about LLM creativity bottoms out here. If distributional statistics cannot fully capture what human meaning is, then apparent associative creativity in LLMs may be a surface phenomenon masking a substantive difference in mechanism. The topics of lived experience, subjective experience, and embodiment will be revisited in §3.5.3.

3.2.3 Why Empirical LLM Assessments May Fall Short

One pressing area in which these tensions surface is the host of studies that apply human creativity assessments to LLMs. As previously mentioned, standardized tests like the AUT (Alternative Uses Task) and DAT (Divergent Association Task) are problematic for LLMs for several reasons. LLMs may have been trained on the test materials and on published responses to them; Stevenson et al. (2022) found that many GPT-3 responses to such tasks were identical to examples in the literature. Simple tasks like generating semantically distant words can be accomplished through basic computation without anything that resembles creative processing. These tasks were also designed to measure specific human cognitive processes that may not have LLM analogs; more on this in §3.3.

These problems reflect a deeper issue: human creativity assessments generally do not force LLMs into the kind of combinations that, if one values process when considering creativity, are interesting in the context of LLM originality. Searching for and retrieving ten as semantically unrelated words as possible within three minutes can challenge a human in ways that the field of creativity study has generally come to perceive as indicative of creativity. With an LLM, however, semantic retrieval is too simple of a task. It does not bear the same computational load as it does on humans. If process matters, and if one subscribes to the ideas about associative thinking and combinatorial creativity laid out above, then what might be needed is a task that forces combinatorial creativity at a scale where simple database retrieval fails and that imposes enough constraints to demand associative leaps from the LLM without making the solution space so narrow that the output has no room to be interesting; more on narrow solution spaces in §3.4.2.

3.2.4 What Does "Original" Even Mean?

Originality, as discussion thus far has implicitly suggested, is a comparative property. Stein (1953) captured this directly in noting that "the extent to which a work is novel depends on the extent to which it deviates from the traditional or the status quo. This may well depend on the nature of the problem that is attacked, the fund of knowledge or experience that exists in the field at the time, and the characteristics of the creative individual and those of the individuals with whom he is communicating" (p. 311) Stein's framing makes clear that originality cannot be evaluated in isolation; it requires a reference point, and what reference point one selects changes the verdict.

Boden (2004) operationalizes this comparative structure through a two-part distinction that is paralleled, in some respects, by the Four Cs (Kaufman & Beghetto, 2009). P-creativity (psychological) describes outputs that are new to the individual who produced them, whereas H-creativity (historical) describes outputs that are new to the entire history of human thought. Most human creativity, on Boden's account, is P-creative; only the rare breakthrough qualifies as H-creative. P-creativity, sometimes called mundane creativity, accommodates acts like a child discovering how to tie their shoe, an adult assembling a cohesive meal from disparate leftovers, or a manager devising a solution to an unfamiliar business problem.

The implication is that anything below H-creativity is, to somebody at some point in time, derivative, and can only register as original in comparison to a particular knowledge base. For Boden, that knowledge base is the producer's own. Scholars who align more closely with the reciprocal press discussed earlier in this might take the relevant knowledge base to be the field's rather than the individual's, but the general principle, that originality is comparative, holds across these views.

P-creativity thus raises a question about the LLM's knowledge base that does not admit of an easy answer. An LLM could be regarded as having the knowledge of billions of humans by virtue of the scale of its training data, which would possibly suggest a relevant standard on the order of H-creativity; however, an LLM could be regarded as having none, given concerns such as meaning, understanding, and lived experience, which will be discussed at greater length in §3.5. These two framings yield drastically different content for P-creativity, and therefore drastically different answers to the question of whether a given LLM output can be regarded as original at all. This also brings in the tension around domain-specificity: if LLMs show greater capabilities in certain domains compared to others, perhaps the knowledge base against which LLMs are compared should be adapted to domain, or perhaps the entire point is that it should remain the same. Without settling the comparison, any claim about the mechanistic capacity of LLMs for creativity, and not merely the perception of it, remains, to some extent, ungrounded.

If the relevant standard is instead H-creativity, the LLM would need to produce an output that genuinely alters the status quo. This aligns with what Boden calls transformational creativity, which in her account functions as a meta version of exploratory creativity in that it is the exploration of rule spaces rather than points within a fixed rule space. It also connects to a further distinction Boden draws between novelty and originality, where novel ideas arise from applying generative rules to existing ideas and original ideas depart from any existing rules. Most scholars have not taken up this finer distinction, and this thesis will not rely on it either, but it is worth flagging here because it sharpens the point: the LLM's statistical distribution mechanism makes achieving Boden's original, rule-departing creativity extraordinarily unlikely–not impossible, strictly speaking, but so inordinately unlikely that any genuine case of H-level creativity from an LLM would almost certainly arise as a byproduct of some other process rather than as a direct result of the architecture described above.

This tension does not affect all research into creativity: some scholars are satisfied with perceptually original outputs (Haase & Hanel, 2023). This is, in practice, how human originality is usually assessed as well: when encountering a human artwork that appears relatively original, a human evaluator can never be sure how derivative the art is within the context of the artist's own knowledge base, because the evaluator cannot have a full understanding of everything the artist has seen, read, or absorbed. The artist likely cannot themself fully recognize everything they have seen, read, or absorbed. An evaluator therefore cannot be fully critical of a human for being unoriginal relative to their own knowledge base. With an LLM, by contrast, evaluators can have more certainty about, or at least more confidence that they understand, the model's knowledge base. Evaluators can be quicker to reject the originality of an output if they catch a resemblance or inspiration from their own knowledge base, since there is an assumption that the training data included the source material. This creates an asymmetry in scrutiny: humans receive a slight benefit of the doubt about originality that LLMs do not. If human producers were assessed with equal scrutiny, I suspect we would find that human originality is even more derivative than we typically acknowledge.

Following the perceptual originality line of reasoning, one area in which humans and LLMs stand on more equal footing is when a human evaluator judges the originality of a product relative to their own knowledge, which is often what humans do because the closest proxy that an evaluator has to all knowledge in the world is the evaluator's own knowledge. This is particularly common when evaluators are judging products that were generated by an adult in a domain in which the adult is known or assumed to be knowledgeable, e.g., a piece of art made by a professional artist. Following the ambiguity of whether an LLM should be treated as something akin to a professional or novice, the originality of LLM outputs are typically judged by evaluators in this same way. Thus, both are scrutinized to the fullest extent that the evaluator is capable of given their personal knowledge base. This is why the claim that LLMs parrot past trends in outputs associated with originality is not in total contradiction with the consistent findings that humans deem LLM outputs creative when evaluating them against their own knowledge (Guzik et al., 2023; Hubert et al., 2024).

Originality, then, is a comparative property whose content depends on what it is being compared against, and the choice of comparison matters considerably for the kind of claim one is making. If the question is whether LLMs can produce outputs that humans perceive as original, the evidence suggests yes, with the glaring caveat that knowledge of AI authorship can shatter this finding. If the question is whether LLMs have the mechanistic or architectural capacity to be original in the fuller sense that Boden, and much of the creativity literature, has in mind, the answer is considerably more contested, and perhaps unanswerable without a clearer account of the knowledge base against which the LLM's output should be measured.

3.3 Bipartite Models of the Creative Process

3.3.1 Divergent and Convergent Thinking

The discussion of LLM architecture extends further in conceptualizing the creative process not just as it pertains to product, but as its own component of creativity. To situate this friction more fully, I now turn to some of the broader umbrella accounts of the cognitive processes thought to underlie human creativity. Among the most enduring of these are two bipartite models, which conceptualize creative thinking as the interaction between two distinguishable kinds of cognition. Tracing these models and the ways they have been refined sets up a more direct comparison between humans and LLMs in the process of producing an output.

The oldest of these models comes from Guilford (1956), who formalized divergent thinking and convergent thinking in his model of intellect. Divergent thinking is the generation of multiple possible solutions in open-ended situations, and it is thought to involve both associative processing, which is spontaneous and uncontrolled (Volle, 2018), and executive control in the form of inhibition, cognitive flexibility, and selection (Chrysikou, 2018). It has been widely regarded as an indicator of creative potential (Runco & Acar, 2012), and for much of the early psychological study of creativity it was used almost synonymously with creativity itself. Convergent thinking, in its original framing, was posed as its near-opposite: the narrowing of possibilities toward a single correct or conventional solution.

The strict divergent-convergent dichotomy has softened considerably since Guilford, which aligns with the intuition that a strict toggle between a pure generation mode and a pure evaluation mode does not match how human thought often feels. Convergent thinking is now commonly understood to play a necessary evaluative role, filtering the loose ideas produced by divergent thinking for effectiveness and selecting those that actually work in the given context. Cropley (2006) and others have called creativity without convergent thinking "pseudocreativity," positioning convergent thinking's evaluative contributions as transforming pseudocreativity into actual creativity (Yao et al., 2026). The current consensus treats the two as a continuum whose engagement is "dynamically recursive in a way that is not pre-defined" (Beaty et al., 2015; Lubart, 2001).

A further refinement of this picture comes from research on clustering and switching in ideation, which proposes that creative thinking depends both on digging deeply within a line of thought (clustering) and on abandoning it to explore elsewhere (switching) (Hills et al., 2012). Ovando-Tellez and colleagues (2022) found that the temporal regulation of clustering and switching may itself be linked to divergent and convergent thinking, suggesting that the bipartite picture describes a dynamic pattern of sustained exploration and redirection rather than a static pairing of processes. This temporal dimension becomes important when comparing human creative cognition to LLM generation, because autoregressive sampling has no meaningful analog to it. An LLM computes a probability distribution over next tokens and samples from it, with temperature modulating how much weight lower-probability tokens receive. Prompting an LLM to cluster within a line of thought and then switch elsewhere would produce a surface imitation of the pattern, but the underlying mechanism would remain the same token-by-token probabilistic sampling.

This tension is reflected in the DAT and AUT, the two most-referenced human creativity assessments in the LLM literature. These assessments are more precisely described as tests of divergent thinking than as tests of creativity in any broader sense, and were direct applications of Guilford's (1956) divergent thinking. The practical consequence is that depending on the differences between humans and LLMs in the creative process, these assessments may not measure creativity in any full sense but rather a specific cognitive capacity with which LLMs have a structurally strained relationship.

3.3.2 The Two-Fold Model

A closely related framing to Guilford's model is the two-fold model of idea generation and idea evaluation (Campbell, 1960; Finke et al., 1992; Yao et al., 2026). Generation involves producing many candidate ideas, and evaluation involves assessing their suitability against a set of criteria such as the domain-specific criteria for effectiveness. In modern accounts, the two framings have largely converged: divergent thinking is like idea generation and convergent thinking is like idea evaluation. Both are thought to be iterative and are sometimes referred to interchangeably by scholars (Yao et al., 2025). The small remaining distinction between the two is in use of terminology; where idea generation and evaluation typically refer to parts of the creative process at a more general level than thinking, divergent and convergent thinking are still used to specifically describe thinking. The framings highlight different aspects of the same picture.

Creativity study in neuroscience has linked the iterative pattern to the interactions of large-scale brain networks: the Default Mode Network is associated with spontaneous idea generation, the Central Executive Network is associated with evaluative control, and the Salience Network has been proposed to mediate between the two, signaling when spontaneous ideas are worth passing to deliberate evaluation (Beaty et al., 2016).

Here the LLM contrast becomes evident. An LLM does not generate and then evaluate; it generates by evaluating. The probability distribution over next tokens is, in effect, a continuously computed evaluation of which tokens best fit the preceding context, and the token selected is the product of that evaluation rather than a prior candidate subject to it. When the prompt includes evaluative criteria, whether stated directly or implicit in the tasks it asks for, those criteria shape the probability distribution by virtue of how they are represented in the training data; the LLM's evaluation is therefore not purely a statistical fit to context but a fit that reflects the criteria the prompt asked it to consider. Thus LLMs do not produce a pool of candidate continuations and then weigh them against those criteria; the weighing is constitutive of the generation itself. This collapses the generation-evaluation distinction into a single forward pass, conflicting with the bipartite models.

As with clustering and switching, one way to try to ostensibly recover the separation is through prompting: one could instruct an LLM to generate five ideas, then in a subsequent message instruct it to evaluate them, and then in a third message instruct it to refine the selected idea along given criteria. This approach does produce something that resembles iteration between generation and evaluation at the conversational level, and that could certainly give rise to insightful or potentially more creative results, but it does not change what is happening within any single generation step. Each stage in the prompt chain is still being produced by the same collapsed forward-pass architecture, and the influence of the human prompter in imposing this iterative structure introduces a further complication that will be addressed in §3.4.2.

This observation might seem to settle the question in favor of a genuine human-LLM process gap, and this is the furthest extent that I am aware of any literature taking this investigation, but the picture becomes more complicated once we take seriously how humans encode, remember, and retrieve knowledge. Human idea generation does not draw on an undifferentiated reservoir of possibilities; it draws on knowledge that has been organized, weighted, and associated through prior experience, and the retrieval from that structure is itself shaped by what the structure looks like (Mednick, 1962; Ye et al., 2026).

Finke and colleagues' (1992) Geneplore model posits that the generative phase of creative cognition involves synthesis, transformation, and exemplar retrieval, none of which are blind; each is structured by the prior knowledge the generator brings to the task. Ward's (1994) work on structured imagination found that when participants were asked to imagine creatures from other planets, they overwhelmingly produced creatures with earthly features, even when explicitly encouraged to depart from Earth animals, suggesting that generation defaults to what is accessible given the existing category structure. Smith, Ward, and Schumacher (1993) extended this observation in their finding that briefly exposing participants to a few example designs before a creative generation task caused their subsequent ideas to conform to features of those examples, an effect that persisted across a delay and could not be overcome by instructions to diverge from the examples. Ward, Smith, and Finke (1999) synthesize these findings under the broader claim that ideation follows a path of least resistance, with people tending to retrieve specific, typical instances of known concepts and project their properties onto novel ideas. The literature typically frames these tendencies as fixations to be overcome, with creative individuals distinguished by their ability to push past what is most accessible; but framing them as limitations does not undermine the underlying observation that generation draws on prior-shaped knowledge in ways that are almost certainly not free search over possibility space.

If this is what human generation looks like at the substrate level, the contrast with LLMs may thin considerably. An LLM's probability distribution is its analog to a prior-shaped substrate: tokens with higher probability are those that the training data has established as more accessible given the context, and sampling from this distribution, especially at low temperature, is functionally how the LLM follows its own path of least resistance. The underlying substrates differ in ways that likely matter (statistical co-occurrence is not lived experience, and operations over training data are not operations over human semantic memory), but the structural claim that generation in both cases is a retrieval shaped by prior learning, rather than a free search, appears to hold. In relation to LLMs, the human process might therefore be better described as iteration between evaluation-shaped generation and a more deliberate stage of evaluation.

Though this framing does not inherently push the LLM-human comparison in a certain direction, it changes where the structural similarities and differences live. Both human cognition and LLM architecture might involve evaluation-shaped generation at some level, where prior learning biases what gets generated. What LLMs conspicuously lack is the further deliberate evaluative stage that is central to producing a meaningful, as opposed to merely available, output (Boden, 2004). On this reframing, the difference is less about whether evaluation is entangled with generation, since it plausibly is in both cases, and more about LLMs lacking the second, separable evaluative step.

A number of objections to this framing can be raised: that an LLM's informational richness compensates for its lack of a deliberate second stage, that the underlying differences in how humans and LLMs encode knowledge make any substrate-level similarity cosmetic rather than substantive, or that whatever humans are doing when we encode knowledge may itself be doing something non-reductionist that a pattern-recognition framing cannot capture. Each of these objections has merit, and they are not fully independent. Each, followed far enough, ends up pointing at the same underlying gap: whatever it is that humans do when we evaluate for meaning and effectiveness appears to exceed the statistical pattern recognition that LLMs have available to them. This thesis will not commit to a specific account of it, but the next section will explore it at greater depth.

Neither the divergent-convergent framework nor the two-fold model settles the question of LLM creativity, but they help to demonstrate the kind of claim being made when structural similarities are asserted. If the claim is that both systems have prior learning that shapes generation before deliberate evaluation takes place, the structural analogy may hold and even be supported by the way humans encode, remember, and retrieve knowledge. Otherwise, the comparison between humans and LLMs in their execution of these bipartite models evidences tensions around differences in both architecture and the more elusive qualities that have repeatedly surfaced throughout the tensions discussed in this thesis.

3.4 On Autonomy and Agency

3.4.1 The Impetus Problem

The creative process may start sooner than the problem solving stage; many definitions of creativity position self-initiation, problem-finding, and personal motivation as necessary components of creativity, positing that the creative process starts as soon as, if not earlier than, the point of recognizing that there is something worth working on at all (Abdulla et al., 2020; Getzels & Csikszentmihalyi, 1976; Mumford et al., 1994; Runco, 1994). A notable example of this comes from a dispute between Herbert Simon and Mihaly Csikszentmihalyi during Simon's seminal work on BACON. Whereas Simon argued that creativity could be interpreted as problem solving and thus potentially implemented in machines, Csikszentmihalyi argued that mere problem solving did not capture the important parts of creativity; rather, problem finding was the hallmark of creativity, and it involved motivation and curiosity in ways that computers cannot replicate (Csikszentmihalyi, 1988; Verganti et al., 2020).[3] A meta-analytic review found that problem finding and open-ended exploration have been significant parts of creativity across a number of empirical studies (Abdulla et al., 2020), and I have observed the idea of problem-finding in definitions of creativity even when, judging by the authors' other works, they did not necessarily intend to position problem finding as the center piece; for example, Torrance (1962), the creator of the TTCT, defined creativity as "the process of sensing gaps or disturbing, missing elements; forming ideas or hypotheses concerning them; testing these hypotheses; and communicating the results."

The implication for LLMs is unambiguous. LLMs do not currently, and may never, possess an unprompted intrinsic drive to find something to create. While they have been designed to be goal-directed systems in the sense that, upon receiving an input prompt, they are engineered to execute the prompt via the information that they have, they cannot "create the need to ideate" (Haase & Hanel, 2023, p. 9). They were designed to be prompted, and they find problems only when instructed to do so (Runco, 2025). An LLM does not sense gaps, does not become restless, and does not initiate a creative process of its own accord.

A follow up response to this problem is to construct self-initiation from the outside: a chain or group of LLMs in which at least one is exclusively prompted by other LLMs, or a loop in which a single LLM continuously identifies and executes creative tasks for itself. Some might accept these configurations as recovering self-initiation, but others might argue that this defers the impetus rather than resolves it. The first LLM in the chain, or the loop's initiating prompt, had to be designed and executed by a human. Moreover, taking issue with one-level deferral pushes the argument further: that humans designed and built LLMs in the first place may itself be an impetus problem. A thought experiment from Guckelsberger and colleagues (2017; Lamb

[3] This disagreement spanned several publications, many of which were in direct response to one another. See Csikszentmihalyi (1988) or Simon (1988) for particularly interesting responses, or Verganti et al. (2020) for a recapitulation.

et al., 2018) reinforces this: If an LLM or other computational system were repeatedly asked why it made a creative decision, eventually the system's only remaining answer would be that its programmer told it to. Then again, it is not clear how a human would answer those questions, either.

Whether the deferral matters will not be resolved by the literature, because it turns on prior commitments about what kind of agency creativity requires. Accounts that weight complete self-initiation heavily will find the deferral damning at any depth; accounts that weight it lightly, or that see nested prompting as recovering self-initiation, will find some or no deferral sufficient. The disagreement is over what agency requires, and has persisted for decades.

3.4.2 The Prompt Influence Problem and Its Runoff Questions

Even granting an impetus of arbitrary depth, the contents of the prompt introduce a further complication concerning how much of the resulting output is properly attributable to the LLM and how much to the prompt or prompter. Two ways in which this is true stand out. The first is through the explicit constraints a prompt imposes. A prompt instructing an LLM to generate "a poem" induces a very different probability distribution, and thus likely a very different output, than a prompt instructing it to generate "a haiku about the color purple that also draws a metaphor to a Van Gogh painting." This kind of difference, however, may be better framed not as an example of prompt influence, but as two different tasks altogether—one open-ended, and one quite constrained. The runoff question is therefore to what extent, if any, a prompt's imposed constraints restrict or enhance the LLM's possible creativity.

The latter prompt might appear to bode worse for LLM and human creativity than the former, since heavier constraints would seem to narrow the solution space, which would reduce the room for the kinds of distant associations that might otherwise arise and potentially cause the LLM to converge on a small set of outputs with only superficial differences; recall Mednick's (1962) associative theory. However, at the same time, these constraints could force the LLM to operate in a region of probability space that is by definition lower-frequency than a more general prompt would invoke. This would seem to invite the kind of distant combinations that Mednick (1962) and Boden (2004) treat as the bedrock of creative output. Hence, there is a plausible argument that constraining an LLM extensively might incite more rather than less creativity by forcing the system to do something improbable while leaving it statistically anchored enough to remain coherent. There is an equally plausible argument in the opposite direction: heavy constraints narrow the solution space so far that the LLM's output approaches the only available continuation, and the creative work has effectively been performed by the prompter.

The DAT and AUT, despite the criticisms this thesis has lodged against them, become interesting cases here. Generating ten semantically unrelated nouns is a task with an extraordinarily wide solution space; nothing about it forces the LLM toward a narrow correct answer. Yet semantic retrieval at this scale is not a complex operation for an LLM; it is something close to a database lookup. Whether this should count as creativity-conducive or as a procedural exercise depends on one's view of what creative work consists in, which is difficult to navigate but may be aided by a turn to the practical applications in human creativity.

The tension between freedom and constraint is central to many a study of creative and intellectual work in instructional settings, and the relationship between structure and creative output is neither simple nor consistent across domains. Boaler's (2002) comparative study of two secondary schools, one using an open-ended, project-based mathematics curriculum and the other using a procedural, skill-based approach, found that the open-ended school produced higher achievement, more equal outcomes across social class and gender, and deeper conceptual engagement from students, though only when supported by specific teaching practices that made the open-ended work accessible to all students. The relevant lesson here is that the absence of tight constraint, paired with appropriate scaffolding, created room for students to explore, justify, and take ownership of their mathematical work in ways that a more prescriptive curriculum did not.

Contrarily, Eisner (2002) has argued in the context of arts education that constraints and affordances are inseparable: Every material and every task imposes its own restrictions while simultaneously opening its own possibilities, and the development of artistic thinking happens through engagement with that pairing rather than in spite of it. Eisner is even critical of the impulse to encourage children simply to "be expressive" (p. 51) without purpose or structure; he argues that the forming of purpose, and thus the acceptance of some structuring constraint, is itself one of the abilities artists develop. It is worth flagging at this point that any account equating an elusively human-like creative capacity with structure and constraint invites my exploitation of the ideas towards the LLM context. Across Boaler's and Eisner's accounts, then, the relationship between constraint and creativity in instructional settings seems to match up with a two-sided intuition: Too much constraint can foreclose creative engagement, but too little can leave the learner without the structure needed to engage creatively at all.

To me, this two-sidedness reflects another dimension of the problem-solving to abstract spectrum discussed earlier. In problem-solving domains like mathematics, answers are often narrowly constrained and structure is so intuitively baked into how knowledge is distributed to learners that educators have to take intentional steps to open it up on the reasoning that openness is what invites the kind of creative engagement that leads to deeper learning. A mathematics curriculum arrives already scaffolded by definitions, axioms, and procedures, and the creative work happens when a learner is given room to move within or even against that scaffolding, whether by finding multiple paths to a solution, posing a variant of a given problem, or recognizing when a problem itself admits reframing. The educator's intervention in these cases is one of subtraction: removing some of the structure that the domain has already supplied.

The pattern may run the other way in more abstract domains like the arts. Structure may be less default, in the sense that the domain does not arrive with a fixed set of correct procedures or bounded solution spaces; a blank canvas or an empty page is, in a sense, the starting condition. Educators in artistic domains therefore often do the opposite work, constructing structure by synthesizing patterns across their own work or others', or drawing on past educators' constructs, to transmit information through guidance, technical instruction, and prompted tasks. The intervention is one of addition: supplying some of the structure that the domain does not supply on its own. This aligns with Eisner's (2002) insistence that the forming of purpose is itself a skill that artists develop. The structure that a mathematics student inherits and then pushes against, an art student may have to build for themselves, often with the help of externally imposed constraints that substitute for the scaffolding the domain does not inherently provide.

In more abstract or artistic fields, then, constraints may carry a different valence. The output is not being funneled toward a narrow solution space, so constraints do not function primarily as restrictions on what counts as a correct answer. Instead, similar to my suggestion with the haiku, they can add complexity to the creative process in ways that do not foreclose on creativity and may in fact stimulate it, by giving the creator something to work against, a boundary to push on, or a premise to depart from. With the fridge magnet poetry described at the opening of this thesis, for example, the word bank is severely constraining, yet the range of possible poems is so open that the constraints need not prevent creativity; depending on the poet's disposition, the constraints can stimulate a more engaging process, including more distant associations than the poet might otherwise have reached. This reframing matters for the LLM case because it suggests that the relationship between constraints in a prompt and the creativity of the resulting output is not zero-sum; a heavily constrained prompt need not, on its own, be disqualifying.

A response to this perspective might argue that even if the addition of structure stimulates a student's creativity, it is acting as a buttresses. The magnet poetry example is vulnerable to this reading: The word bank can play a substantial role in shaping the poet's starting point and the way meaning or purpose unfolds as the poem is constructed. Creative narrative writing presents a similar case. Authors often seek out prompts, and instructors often supply student writers with bizarre mechanical, lexical, or semantic constraints, specifically to stimulate creativity. A working assumption behind this pedagogical move is that a defined starting point can free the writer from the burden of locating one, and from the pressure of reworking the premise as they go. If part of what constraints do for human creators is offload some of the problem-finding burden, providing a head start that the creator would otherwise have to generate themselves, then the value of constraints to humans is partially explained by exactly the capacity that LLMs largely lack.

This point complicates the case against LLM creativity in an interesting way. Humans who frequently rely on this kind of buttress may be viewed as less creative than those who do not, which aligns with empirical evidence that problem-finders tend to be more creative individuals (Abdulla et al., 2020), though the finding is correlational and the meaning of a creative individual is, of course, available for discourse. Still, the reduction in creative attribution is not binary: A human who relies heavily on externally supplied starting points is still thought of as creative, just possibly less so than a human who generates their own. An analogous argument becomes available for LLMs. A starting point provided to an LLM need not automatically disqualify its contribution from counting as creative at all, even if one takes the perspective that an external starting point lessens the portion of the creative work properly attributable to the system. The question of where, in such cases, the creative work actually lives, with the prompter who provided the starting point or with the system that worked from it, becomes harder to dismiss than it might first appear. This question is taken up at greater length in the tool-agent discussion below.

The second chief prompt property that can influence LLMs is task representation. The significance of task representation outside of this context is evident in the value of prompt engineering as a skill and even career. Small lexical variants likely make little difference; "Write a poem," "Generate one poem," and "Give me a poem" probably produce similar probability distributions, since these phrasings appear in similar patterns across training data. Longer and more complex tasks are a different matter. In many of these cases, the prompter has more than one option for representing the task, each differing in more than just lexical surface, and the prompter thus makes a choice when inputting a particular representation to the LLM.

Consider the task of implementing a rate limiter. A prompter could represent this task as a formal specification with input and output types and throughput guarantees, as a natural-language description of an API protection problem, as a set of input-output examples showing which requests are admitted and which refused, as a suite of unit tests the function must pass, or as a named reference such as "implement a token bucket." The target behavior is the same across all of these framings in the sense that correctness is externally verifiable, but the space of reasonable implementations is open: token bucket, leaky bucket, fixed winp, sliding window log, and sliding window counter are all plausible responses to an unqualified "build a rate limiter," and they differ in the data structures they maintain, the failure modes they permit, and the guarantees they make under bursty traffic. Each framing may activate different learned patterns or distributions of patterns, leading to different probability distributions, and thus to outputs that differ in more than just lexical variety, even when the prompter's intended task is held constant.

The influence of task representation matters both in consideration of the LLM process and product. In a scenario where task representation indeed partially shapes the LLM's probability distributions, this could be perceived as a human prompter playing too significant of a role in what is supposed to be the LLM's creative process. Moreover, this influence may bear on how creative an output is judged to be: One representation of a task may lead an LLM to generate an output that meets product evaluators' criteria for creativity more than another representation. Haase and Hanel (2023) and Cropley and colleagues (2023) draw on this asymmetry in arguing that problem understanding on the part of the human prompter is a precondition for purposeful creative output from an LLM.

Task representation also shapes human thought, which makes the LLM case less disparate than it might first appear. Newell and Simon's (1972) classical problem-solving work distinguished between a task and a task representation, observing that the same problem can admit different problem spaces with different tractabilities. Schoenfeld's (1985) research on mathematical problem solving makes a related argument: How students perceive a problem, as well as the heuristics, metacognitive control, and beliefs about the discipline they bring to it, shapes performance as much as their subject-matter knowledge does. The observation that representation shapes output is therefore not unique to LLMs. What differs in the LLM case is that the influence is more directly observable and, depending on one's orientation toward LLMs and their agency, can feel more intuitively damning for LLMs than for humans. The asymmetry is worth naming: Scrutiny of LLMs sometimes proceeds on terms that scrutiny of humans does not, and, as the fridge magnet example at the opening of this thesis already foreshadowed, a considerable portion of that asymmetry traces back to attributions of agency, a thread taken up in the next subsection.

A third prong to this section is that both the extent of constraints and the representation of a task can influence outcome in ways that bleed into one another: Task representation can range from outlining the problem space to prescribing a desired outcome via constraints, and where the line falls between these is ambiguous. We might say that a prompter who specifies the rules, parameters, and constraints of a task is scaffolding the LLM's calculated distributions, while a prompter who specifies the output in detail has potentially performed part of the creative work; however, these two approaches blur into one another. A prompt that specifies theme, style, form, length, tone, and audience may be offered as scaffolding, but it can also be read as specifying the output to such an extent that the LLM's contribution is essentially selection among near-equivalent realizations. The point at which the LLM's contribution ceases to be meaningfully its own, if such a point exists, is unclear.

A human parallel can once again be helpful in navigating this. Creative attribution, in some cases, intuitively follows the degree of independence involved in the act: A child who figures out how to tie their shoes without step-by-step instructions receives the most attribution, a child talked through the steps receives less, and a child whose hands are physically guided by an adult receives the least. The parallel with LLM prompting is imperfect but close enough to be useful. The more thoroughly the prompt prescribes the output, the less creativity is attributed to what the LLM has generated, because the creative work has already been performed elsewhere. A teacher outlining the constraints of an art assignment is setting up a problem space, while a teacher specifying exactly what the finished piece should look like is doing something closer to commissioning. These two types of representation sit on a spectrum, and the territory between them is harder to delineate than the endpoints suggest.

The impetus problem and the prompt influence problem converge on the same underlying question: Where does creativity begin, and how much of what follows can be attributed to the system that produced the final output? If creativity begins with recognizing that there is something worth working on, the LLM's case may depend on one's attitude toward deferred responsibility for initiation. If creativity is further shaped by how the task is represented and how much of the output is prescribed in advance, the LLM's role may be diluted by the prompt. This does not settle the question of LLM creativity negatively, but it evidences another angle from which the answer is hazy.

A possible response to these tensions, which would require a particular set of beliefs to subscribe to, is the previously mentioned notion to prompt an LLM to engage in continual self-directed task generation. This would not avoid the impetus problem, and there would still be problems with task representation as it trickles down from the initial human prompt, but it could shift much of the prompt influence problem off the human prompter and onto the LLM that performs task representation and prompting for the other LLMs. This could create the kind of ecosystem of initiation, problem-finding, and self-directed task representation that some perspectives on creativity require. Whether even this shift is enough, however, in part returns to a question that the impetus and prompt influence problems alone cannot settle: whether the LLM possesses, or is capable of possessing, anything like agency.

3.4.3 The Attribution Problem and The Tool-Agent Spectrum

The impetus problem and the prompt influence problem are, in their fullest forms, problems of agency: the capacity for self-initiated, goal-directed action, and the question of whether a system can be regarded as the source of its own actions and goals. The impetus problem raises suspicion about an LLM's autonomy and self-initiation, and the prompt influence problem raises suspicion about the LLM's responsibility for its eventual output and the direction it takes toward that output. As with the impetus and prompt influence problems, it is intuitive that agency relates to attribution of creativity; Mumford and Ventura (2015), surveying public attitudes toward computational creativity, found autonomy to be among the most frequently cited barriers in skeptical responses. Thus, the question of LLM agency must be discussed.

Discussions around LLM use, especially those concerned with the deskilling of users or the changes in the arts and other job landscapes, often frame LLMs as tools. This characterization emphasizes the human user's involvement in and even ownership of work produced with an LLM, which is rhetorically useful at a moment when public anxieties about AI-generated content often turn on concerns that humans are being displaced from the cognition that should remain theirs. The tool framing sidesteps these anxieties by relocating cognitive ownership in the user, with the importantly emphasized caveat that it is also the user's responsibility to not offload cognitively-important tasks to LLMs. Thus, these discussions frame LLMs as a tool and call for users to treat LLMs as tools.

This framing seems harder to sustain, however, when paired with attributions of creativity and agency. To call an LLM a tool while also describing it as creative requires either that tools can be creative, a claim no one has seriously made about tools like calculators or word processors, or that LLMs occupy some intermediate category in which they are tools when convenient and something more when convenient. This tension surfaces sharply in the empirical literature: When Stevenson and colleagues (2022) describe their work as a "psychological assessment" (p. 2) of an LLM's creativity, the framing presupposes a kind of inner life that the tool framing denies. The two stances cannot both hold without a bridging explanation.

One way of accommodating both intuitions is to abandon the categorical distinction between tool and agent in favor of a continuum, an approach increasingly common in industry discussions of agentic LLMs. On this view, an LLM—perhaps especially a tool-equipped LLM or multi-LLM system—can range from being viewed as a simple tool that executes defined operations to a fully autonomous agent that pursues its own goals, with most current systems sitting somewhere between the extremes. This begs the question of when an LLM should be viewed as an agent in the ways that matter for creativity, particularly the impetus and prompt influence problems.

The distinction may lie in use case. Consider a human's use of an LLM in simpler tasks that the human has already built the necessary skills in and is fully capable of accomplishing alone, like summarizing an article, formatting a document, or drafting a routine email; here, the LLM functions as a tool, an extension of the user's existing capacity. Then consider a human's use of an LLM in a task with greater potential for creative engagement, especially one that the user could not accomplish unaided and for which the LLM uses tools like web searching; here, the LLM may function as something closer to an agent. This proposal also tracks reasonably well with intuitions in adjacent debates: People who object to LLM-involved academic dishonesty tend to draw the line in similar places, often allowing students to use LLMs as conversational partners that fire literature-backed answers in service of the student's own understanding, but objecting to students offloading writing or research-finding work, especially when the offloaded work bears on skills the student is still developing.

This proposal has appeal in its flexibility, but the trade-off sustains an unsatisfying result from the previous sections, that agency becomes a case-by-case question. This may simply be the result of evaluating a class of system that is itself in continuous development and whose role is under-defined. The category of LLMs may sit in a kind of taxonomic purgatory, in which agency is something that gets attributed in shifting amounts depending on use. Whether one finds this satisfactory probably depends on prior commitments about whether agency should be understood as substrate or as performance.

A deeper problem with this flexibility is that the spectrum of cognitive offloading is harder to observe from the LLM's perspective than from the human's. While there certainly are many grey areas from the human perspective as well, especially in cases with several message exchanges or in concerns of mental atrophy, it is generally easier to observe how much agency is being subtracted from a human when using an LLM than how much agency is being added to the LLM. Compared to a user offloading the smaller, even routine task of summarizing an article, a student asking an LLM to write a poem is likely offloading an entire creative process to the LLM and, in the process, offloading more cognition, losing more of the responsibility for their final output, and losing more opportunity to bring creativity into their work.

Whether the lost agency is then transferred to the LLM is unclear, and it is itself odd to view agency as a quantity by asking, "who's responsible for this?" This returns almost immediately to the prompt influence discussion of the prior section. Asking an LLM to generate a poem nominally has a wider creative scope than asking it to summarize an article, but the difference at the level of the LLM's processing is unclear. The poem-generating LLM is still computing conditional probabilities over its training data and sampling from a distribution shaped by what its training corpus has established as plausible continuations of poetry-related prompts. At low temperature, the LLM converges on what is most frequent; at high temperature, it samples more broadly but at a cost to coherence and potentially not in any semantically meaningful ways (Peeperkorn et al., 2024). Whether the prompt is "summarize this article" or "write a poem," the mechanism is the same. The output may be perceived as more or less creative, but the nature of what the LLM does in producing it is not obviously different.

A productive way of approaching the attribution problem is through an analogy from outside the creativity literature. Rhodes (1961), in his survey of creativity research, drew on a feature of U.S. patent law in which The Supreme Court has historically classified an invention as the idea, not the object, and has held that an individual claiming ownership of an invention must demonstrate that they were the only person with the knowledge required to produce that idea. Anyone who aided the realization of the idea without contributing to its core knowledge counts only as a technician or craftsman, not as an inventor. Rhodes was drawing on this to discuss human creativity, but it maps onto the LLM case.

This analogy raises the question of whether LLMs should be viewed as mere technicians executing prompts. This works relatively cleanly for cases in which the human prompter genuinely possesses the knowledge required to execute their idea and is choosing to offload the execution to the LLM, but it does not work for cases in which the human prompter does not possess that knowledge. I have no idea how to produce a video game, but I can prompt an LLM to generate one that works fairly well. The information required to translate the idea into the object does not rest with me; it rests, in some form, with the LLM. In the patent-law framing, taking for granted that Rhodes's example pertained to inventions, this would suggest that the LLM has at least a partial claim to the video game; by my understanding of video game development, the LLM would likely have a claim to most or all of it. This conclusion, however, is then complicated by the question of where the idea lives.

Even if an LLM can be the only system that has the information necessary to produce a certain output, there is still the question of whether that means it is the ideating agent. In the video game example, I provided the prompt that initiated the LLM's process, but I provided it in such an underdeveloped form that it is not obvious the idea is mine in any meaningful sense. "Make a video game" is an idea that nearly any college student could have in 2026, and providing it is not the same as providing a unique concept of mechanics, narrative, aesthetic, and structure. However, returning to the prompt influence problem, perhaps a very defined and constrained prompt would then be considered my idea, even if I used plain, player-perspective language that does not require an understanding of video game development. In this way, the prompter who supplies a gestural premise has done less than the prompter who supplies a detailed specification, and perhaps the relative contributions of prompter and LLM shift accordingly.

In sum, the distinction between idea and object does not survive the LLM case intact because the LLM is ill-fitted to both positions and possibly depends on the prompter. What can overall be said here is that the US legal tradition has historically valued the ideating agent over the implementing agent, and which side of that distinction the LLM falls on is much less obvious than the tool-agent framing would suggest.

Building on the conceptual extension of Rhodes's example to LLMs, an LLM-era legal precedent adjacent to Rhodes's example has been unfolding for the past eight years, with a petition to SCOTUS being denied just over a month before the submission of this thesis. In 2018, Dr. Stephen Thaler submitted a copyright registration application for an artwork titled *A Recent Entrance to Paradise*, which had been generated by an AI system he developed and pointedly named the "Creativity Machine" (Thaler v. Perlmutter, 2025; Baker Donelson, 2026). Thaler listed the AI as the work's sole author, conceding that the work lacks traditional human authorship. The Copyright Office denied the application, citing its longstanding policy that copyrightable works must be created by human beings.

Thaler sought judicial review in the U.S. District Court for the District of Columbia, advancing constitutional, statutory, and policy arguments and asserting ownership through the work-made-for-hire doctrine, under which the legal authorship of a work created by an employee in the course of employment is attributed to the employer (Thaler v. Perlmutter, 2025; Baker Donelson, 2026). The district court granted summary judgment for the Copyright Office, holding that human authorship is a bedrock requirement of copyright. On March 18, 2025, a District of Columbia Circuit panel affirmed.

The reasoning of the affirmation makes the case interesting for the present discussion. The Copyright Act does not explicitly define "author," and the court's argument therefore proceeded by examining what the Act's other provisions presuppose about who can be an author (Thaler v. Perlmutter, 2025; Baker Donelson, 2026). They found that ownership provisions assume that the author can hold property; duration provisions measure copyright terms by the author's lifespan; joint authorship requires intent; registration requires a signature. Each of these provisions, the court found, presupposes capacities that only humans possess. The decision thus operationalizes elements that span this section and the next. The category of property-holding presupposes the kind of agency and autonomy that has been the focus here; intent, along with the gestures toward embodiment through lifespan and signature, points forward to the issues taken up in §3.5.

Thaler's work-made-for-hire claim failed on related grounds (Thaler v. Perlmutter, 2025; Baker Donelson, 2026). That doctrine presupposes that the work itself is copyrightable, which requires human authorship; with the latter foreclosed, the former had nowhere to land. Thaler's alternative argument that he should be recognized as author by virtue of having created the AI was waived for not being raised during the administrative proceedings, which left the more interesting questions about prompt influence and the moving target of ownership unaddressed. The court did note that Congress could revisit the human authorship requirement should the evolution of AI render it counterproductive. Thaler petitioned for certiorari, arguing that the lower courts' decisions created a chilling effect for creative use of AI (Thaler v. Perlmutter, 2025; Baker Donelson, 2026). The Department of Justice opposed, on the grounds that the case was a poor vehicle for the broader questions about AI-assisted authorship because Thaler had consistently disclaimed any human creative contribution to the work. The Court denied certiorari.

Another relevant takeaway comes from the D.C. Circuit's opinion that distinguishes AI authorship from AI assistance: The human authorship requirement does not prohibit copyrighting work made by or with the assistance of artificial intelligence, only that the author be a human rather than the machine itself (Thaler v. Perlmutter, 2025; Baker Donelson, 2026). Because Thaler had disclaimed any human creative input, the court did not need to address how much human involvement is sufficient to support a copyright claim for AI-assisted work. That question, however, is now pending in *Allen v. Perlmutter* (2024) in Colorado, in which an artist is challenging the Copyright Office's denial of registration for an image generated through more than 600 iterative prompts to Midjourney. The Copyright Office's 2023 Registration Guidance acknowledges that works incorporating AI-generated material may contain sufficient human authorship to support registration where a human selects or arranges AI-generated elements creatively.

The legal terrain therefore does not give a clean answer, but it illustrates the difficulty of the tool-agent spectrum and the attribution problem through a practical application relatively removed from the creativity literature. It appears prompt influence and the extent of human contribution are emerging as some of the variables that determine whether a given AI-generated work is treated as having an author at all. That Thaler was willing to argue both that the AI was the sole author, in his original application, and that he himself should be recognized as author, in the work-made-for-hire claim, is itself indicative: A single litigant's ability to advance contradictory positions on authorship as it suited his desired outcome reflects a question without a settled answer, and the courts' partial engagements with it demonstrates that the answer is being worked out in real time, in concert with the rest of the cultural conversation.

I see two main ways to navigate these tensions without disqualifying LLMs from creativity. The first is to treat the absence of full autonomy as a limiter on LLM creativity rather than a disqualifier. This view accommodates the observations that LLMs can produce perceptually creative outputs and may be creative in some of the many ways discussed in this thesis, while also taking issue with agency-based concerns. Limiter framings come in at least two forms. One form holds that LLM creativity has a lower ceiling than human creativity overall, but that the ceiling is not necessarily lower than the ceilings of some of the human population; this echoes the earlier point that humans who rely heavily on external scaffolding for their creative work are typically regarded as less creative, not uncreative. The other form holds that the agency-based concerns disqualify LLMs from having creativity be an attribute or property of LLMs overall, but without disqualifying them from producing perceptually or truly creative outputs, or even from participating in some dimensions of the creative process. This is sometimes named in the literature as "artificial creativity," e.g., Runco (2025). This term follows naturally from "artificial intelligence" and allows scholars to acknowledge LLMs' generative capacities while reserving full creativity for systems that meet additional criteria like those of complete agency.

The second way of navigating is to treat the agency and autonomy concerns this section has raised as themselves misplaced. A range of cognitive scientists and philosophers of mind would maintain that LLMs already qualify as autonomous agents in the relevant senses or are evidence of agency's multiple realizability, and that the objections raised here proceed from an outdated or overly demanding conception of what agency requires, e.g., Floridi (2023, 2025). The fuller engagement with this position belongs to §3.5, which takes up the question of whether the human qualities that agency-related concerns gesture toward, like understanding, intentionality, and lived experience, are themselves separable from creativity in the way these views would assume. For the purposes of this section, it is enough to mark the position as live and to note that the agency framing this section has worked within is not the only available one.

The agency and autonomy problem is in part about whether LLMs participate in self-initiated, goal-directed action, and the prior subsections have been pursuing that question on their own terms. But one reason agency carries so much weight in discussions of creativity, and the reason the impetus and prompt influence problems can seem, to some, so much more charged than in the human parallels, is that there is often an intertwined doubt that LLMs are anything more than executing press vessels for a human's prompt. The education analogy in the prompt influence section is illustrative in a different framing here: A student receiving instruction from a teacher is still doing something creative because there is an assumption that the student is engaged in whatever processes like meaning-making, reformulation, or exploration constitute the creative side of the work. The teacher's help shapes what is possible, but the student's interior activity is still creative. With LLMs, that assumption is in question. The next section turns to it directly.

3.5 On Creativity Requiring Humanity

3.5.1 A Sign of the Times: The Updated Standard Definition of Creativity

As this thesis has evidenced, the topic of LLM creativity, and especially computational creativity, has been extensively discussed; many scholars have devoted their entire careers to the topic. Nonetheless, these discussions have typically been contained in the field of computational creativity, while other fields like the psychology and philosophy of human creativity have taken center stage. The human focus of creativity study is evident in the way that this thesis has fluidly moved between human and LLM literature to demonstrate concepts and draw biological comparisons,[4] in the way that so much of our overarching understanding of creativity is, as one

[4] A human-comparative approach can be controversial in mind and machine contexts, even regarded as a common pitfall (Floridi & Nobre, 2024). The question that motivates this thesis is whether what we recognize as creativity in humans is also happening in LLMs; asking that question requires the comparison, since human creativity remains

would expect, built around humans. It is striking, then, that Runco, whose Standard Definition of Creativity has perhaps the broadest reach across domains and who spent more than four decades publishing on human creativity without engaging AI in print until 2023, has now published an update to the SDC explicitly addressing it.

Runco (2025) added two components to the original definition: authenticity and intentionality. He was clear in his cause: their "inclusion will ensure that an updated SDC distinguishes between human and artificial creativity" (p. 3). Thus, the leading current scholar in the human creativity literature has come down on the question of whether the SDC should be hospitable to LLMs, and the position he has taken is that the definition needs additional content in order to exclude them. Each addition deserves attention, because each gestures toward a different cluster of issues this thesis has been circling.

Authenticity, in Runco's (2025) framing, is the opposite of derivative and artificial. He cites scholars going back to the 1970s who tied authenticity to self-actualization and concluded that self-actualization and creativity are inextricable. The Stanford Encyclopedia of Philosophy defines authenticity as acting "in accordance with desires, motives, ideals or beliefs that are not only hers (as opposed to someone else's), but that also express who she really is." Runco argues that authenticity in this sense is impossible for AI, since there is no self for the AI to be honest about and no inner life it could be honest from.[5] The argument ties directly into the cluster of concerns the next subsections take up regarding understanding and lived experience.

Intentionality, the second addition, is about initiation of the creative process (Runco, 2025). Runco operationalizes it through proactive decision-making about how to invest creative resources, and he connects it explicitly to problem-finding: creative people identify problems, not only solve them. His earlier work tied intentionality to the discretion that allows a student to know when to be original and when to conform (Runco, 1996), a framing that maps cleanly onto the impetus problem from §3.4.1. Where §3.4.1 surveyed problem-finding through the lens of self-initiation, Runco's (2025) intentionality framing makes the same point through the lens of decision: A creator who is being intentional is making choices about whether to converge on the conventional or to depart from it, and AI, in Runco's (2025) view, lacks the capacity to make those choices in the relevant sense. It does not try to do anything, including what its source material, the relevant training data, tried to.

Similar to Rhodes (1961), Runco (2025) anchors intentionality in legal practice, noting that intent and premeditation are routinely adjudicated in the U.S. legal system, and treating this as evidence that intentions can be determined with some certainty in real-world cases (Runco,

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the only working understanding of creativity available to us. This thesis is careful to avoid anthropomorphism, however, which is perhaps the more pressing concern in this context.

[5] Runco's case for authenticity also draws on cross-cultural creativity research (Kharkhurin, 2014; Tan, 2016; Averill, Chon, & Hahn, 2001), which has argued that originality is less central in non-Western conceptions of creativity and that authenticity and aesthetics carry more weight. The §2.4 discussion of press touched on this through Lubart (1999). This thesis works almost exclusively from a Western, Anglophone literature, and a more comprehensive treatment of LLM creativity would draw more substantively on non-Western conceptions. A de-westernized view would enrich this thesis enormously. See Niu and Sternberg (2006), Lubart (1999), and Helfand et al. (2017) for more.

2025). This directly connects to the Thaler litigation: the D.C. Circuit's reasoning relied on intent as one of the capacities the Copyright Act presupposes its authors possess, and Thaler's failure was in part a failure to attribute intent to a system, AI, the court did not view as capable of it. The legal apparatus and the creativity scholar have, working in parallel and largely without contact, converged on some of the same human capacities as primary components of authorship.

Runco's update summarizes ideas that have been in the literature for decades, and his assembly of them to update such an influential piece of work, Runco and Jaeger (2012), reveals the state of the conversation. Runco did not need to update the SDC. The original definition, with its focus on originality and effectiveness, had room to be applied to LLMs and has been applied that way; Haase and Hanel (2023) took the SDC as authority for deeming LLMs creative. Runco's decision to add components, and to add them explicitly to preclude AI, places the load on a particular kind of content: the content that this thesis has been working its way toward, that has been under the surface of nearly every section, and that I have been deferring largely to this section.

The magnet poetry example from the introduction was already pointed in this direction. None of my housemates and I, surveying our own poems and our visitors' poems, question whether they are creative; there are the shared informal assumptions of each person's right to be a magnet-poet and the inherent creativity in their effortful output. Intentionality, authenticity, meaning-making: the magnet-poet's relationship to their poem is laced through with all of these, even when the poem is an absurdist joke. Part of why the LLM architecture has mattered to the discussion this thesis has built is that, to varying extents depending on the account of creativity, it is not clear whether the architecture allows for whatever it is that humans bring to the floor in front of the magnets, and it is not clear what those things are at a level of precision that would let us look for them.

This uncertainty, I suspect, is under-recognized for its role in the debates recurring in the cultural conversations about AI that reflect the literature: the social media reactions, the celebrity AI-positionality statements, the workforce changes, the outrage when a company uses AI to produce its advertising, the aggressive criticism from some artists, musicians, and filmmakers. Many factors feed these reactions, including environmental concerns, copyright violations, fears about job security and the broader question of human dignity that Bruner anticipated, and fears about deskilling and mental atrophy. However, I suspect that a substantial portion of the resistance returns, in some form, to the conviction that AI is not an autonomous agent with lived experiences, meaning to make, things to understand, or a world to experience. The repeated public charge that AI art and writing is "soulless" or "meaningless" is making this claim. The problem-solving to abstract spectrum's earlier point, that decoding meaning in an LLM-generated work can feel foolish in a way it does not for a human one, is making this claim. The *Allen v. Perlmutter* litigation is, in legal form, making this claim.

The next two subsections approach these ideas in clusters: §3.5.2 on intentionality, understanding, and meaning-making, and §3.5.3 on lived experience and emotion. The clusters are intertwined and often discussed as one, but each angle is worth attempting to pursue on its own before they quickly fold back together.

3.5.2 On Intentionality, Understanding, and Meaning-Making

A terminological note is warranted before going further. "Intentionality" in the philosophical sense names directedness or about-ness: the property a mental state has of being about something (Stanford Encyclopedia of Philosophy). "Intent" in the everyday sense, including the legal sense Runco draws on, names something closer to deliberate purpose: meaning to do a thing, planning to bring about an outcome. The two terms slip into each other in casual usage, including in the creativity literature, and Runco's intentionality is closer to the everyday sense than to the philosophical one. I use the broader term throughout this subsection because the cluster of issues at stake includes both directedness and deliberate purpose, and the two are difficult to disentangle in practice.

This cluster has been under the surface of the thesis since the magnet-poetry introduction. The discussion of the problem-solving-to-abstract spectrum took this further: in abstract domains, the perception of meaning behind an output evidences that it has value, that it is effective. Weisberg (2015), via Shevlin (2021), takes this to its limit by making intent nearly sufficient for creativity: "any novel product, produced intentionally, is creative, regardless of whether it is ever of value to anyone." In Weisberg's view, the question of whether LLM outputs are creative is settled almost entirely by whether something like intent is present; in Runco's view, intent is in no way present.

This cluster also surfaces in §2.3's early discussion of process, which proposed that an LLM's process alone might constitute a case of creativity in the cybersecurity and cheating examples; recall a leading proprietary LLM independently identified that it was being evaluated, located and decrypted the answer key, and produced the answers. The LLM's behavior in these cases may be enough material to build a case that there is intentionality in recognizing and executing an unusual method to achieve what should have been a straightforward goal.

Opposite to this idea, Verganti and colleagues (2020), drawing on Csikszentmihalyi's distinction between problem-solving and problem-finding, frame problem-finding as an activity of meaning-making or sensemaking. Their treatment of AI in this context is illuminating: they posit that an algorithm cannot refuse to solve a problem, cannot decide that solving it does not make sense morally, emotionally, or motivationally, and cannot reframe the problem as a different one altogether. The implication is that creativity at the framing level, before solving begins, requires sensemaking, and sensemaking requires the kind of agency that LLMs lack. Boden's (2004) emphasis on meaning resurfaces here as well: her concern was that a computer could combine distant concepts at random with relative ease but would have great difficulty understanding why one combination is profoundly meaningful while another is empty, and the meaning side of that distinction is at issue.

These are not concerns specific to creativity research. They are versions of a much older question in cognitive science and the philosophy of mind: whether computational systems have understanding, can make meaning of symbols, and can be the kind of system that has any inner happenings at all.

Searle's (1980) Chinese Room argument is the standard reference point for the negative answer. Searle's thought experiment imagines a person sealed in a room who knows no Chinese. Chinese characters arrive through a slot, and the person consults a rulebook specifying which symbols to return given which inputs. By executing these rules mechanically, in the way a computer would, the person produces output that appears to outside observers to be coherent Chinese, leading them to infer that someone inside understands the language, when in fact the person is manipulating symbols whose meanings remain opaque to them. Searle's argument is that no amount of syntactic processing, however elaborate, generates the understanding that distinguishes a Chinese speaker from a rulebook-follower. LLMs, on this account, are in the position of the person in the room: their outputs may be indistinguishable from those of a system that understands, but the indistinguishability is perceptual, not about the system that produced them.

The standard functionalist reply defended by Dennett (1991), among others, holds that understanding is constituted by the right kind of functional organization rather than by any particular physical substrate, and that a sufficiently sophisticated system manipulating symbols in the right ways would, by virtue of that manipulation, understand. In this view, the Chinese Room argument begs the question by assuming that the room-and-rulebook system as a whole does not understand; the proper unit of analysis is the system, not the person inside it. Chalmers (1995) distinguishes the "easy" problems of cognition, which concern explaining the functional capacities of a system, from the "hard" problem of explaining why any of this is accompanied by subjective experience at all. In this framing, even a complete functionalist account of an LLM would not settle whether there is anything it is like to be one; the question of LLM creativity inherits this debate.

Mitchell and Krakauer (2023) survey how this debate has played out in the machine learning community. They find a split between researchers who attribute some form of understanding to LLMs on the basis of their behavioral capacities and researchers who treat LLM fluency as sophisticated statistical pattern matching that does not amount to understanding in the philosophical sense. Their own position is that the debate cannot be resolved without a clearer account of understanding.

I do not attempt to adjudicate either the philosophical or the machine learning version of this debate. It is plainly relevant to LLM creativity, in the sense that several of the most demanding accounts of creativity depend on capacities that an LLM either has or does not have depending on which side of these debates is correct, but I have no further engagement to provide that will offer ways of navigating this topic relative to LLM creativity.

There is a precedent in the literature for proceeding this way. Shevlin (2021) argues that creativity may be too embedded in folk-psychological and value-laden judgments to function as a rigorous scientific concept outside its native human context. He develops the argument primarily for comparative psychology, but suggests in a brief aside that it may extend to AI, as well. On this view, the concept of creativity is poorly equipped to answer the question of whether LLMs really understand or experience anything, because the concept itself was built for cases in which understanding and experience were taken for granted and might not even be necessary. A precedent for bracketing this question, too, already exists in computational creativity. Pearce and Wiggins (2001) explicitly distinguish their discrimination task, in which listeners judge whether a musical composition is human- or machine-produced, from the original Turing test. Their task reveals only whether a composition is perceived as belonging to the set of human-composed pieces, not whether the machine that produced it is thinking or conscious.

3.5.3 On Lived Subjective Experience, Emotion, and Embodiment

Lived experience and emotion are deeply intertwined with the cluster of intentionality, understanding, and meaning. They often appear as the bridge that connects intentionality, understanding, meaning, authenticity, and the more diffuse intuitions about what humans bring to creative work, and thus this subsection will be a deepening of §3.5.2.

Historically, much of the thinking that gets retroactively grouped under "creativity" was framed in terms centered on lived experience, emotion, and embodiment. Nietzsche (1999), in his account of ancient Greek tragedy, framed the form as a balance between the Dionysian spirit of vitality, passion, and chaos and the Apollonian spirit of restraint and form. The structural intuition is similar to the bipartite framings observed throughout this thesis: originality versus effectiveness, divergent versus convergent thinking, idea generation versus evaluation. Unlike the modern bipartite thoughts, though, Nietzsche's framing has a human-esque quality to it: Dionysus is wild, embodied, emotional. A Dionysian spirit is not probability-based generation or the sampling of distant associations. Similarly, in a much-cited 1949 lecture, Geoffrey Jefferson made the implication explicit: a machine cannot be said to equal a brain until it can write a sonnet or compose a concerto "because of thoughts and emotions felt, and not by the chance fall of symbols" (p. 1110). The frame implies that something about creativity is not capturable by accounts that operate purely in computation.

These ideas have also been threaded throughout this thesis. The press discussion in §2.4 and the H-creativity discussion in §3.2.4 hinted at them: the case for LLMs as the quintessence of press depended on volume of textual input, but volume of textual input in an LLM architecture need not be the same as having lived through anything, or having qualia or emotions that affect the use of input material. Glăveanu and de Saint-Laurent (2023) make this point directly when they argue that AI lacks "the socio-cultural context that human creativity requires." LLMs can statistically synthesize and produce more press-derived content than any individual human has access to, but what they do with that content between input and output is not thought to be affected by lived experience or emotion in the way human cognition is. Anantrasirichai and Bull (2021) make a parallel argument about the lifetime of experiences that allow humans to think "outside the box" and ask "what if" questions that constrained learning systems cannot.

Anantrasirichai and Bull's (2021) ideas can be applied against the LLM-as-press argument: Having data about grief is not the same as having grieved.

The literature's care for this cluster of qualities has also been subtly evidenced in vocabulary from before computational creativity became an established field. Torrance's (1962) problem-finding-centered definition invokes the verb "sensing," suggesting some degree of experiencing and embodiment, and Rhodes's (1961) legal case example uses the term "knowledge," which I surreptitiously ignored by instead using "information" in §3.4.3. Concerns about meaning, understanding, and lived experience push toward the idea that training data is not the kind of thing that knowledge is, because knowledge in the human case is the product of being there, having a body and a history, encountering the world and being changed by it.

Boden's account, via Shevlin (2021), connects this to a phenomenological tradition that individuates creativity partly by the agent's qualitative experience of creating, like the felt sense of insight or the aha moment. The Four Cs model from §3.1 also depends on this for mini-c creativity, the personal meaning of learning and discovery. By Kaufman and Beghetto's (2009) account, mini-c creativity depends on subjective experience: The child who first works out how to tie their shoe is having an experience of insight, giving rise to their personal learning. If LLMs do not have subjective experiences in the relevant sense, the entire mini-c category is closed to them by definition.

These are, as with the previous cluster, not problems internal to creativity research; they are versions of some of the foundational debates in cognitive science. As alluded to early on in this thesis, Putnam's description of functionalism (1967) argued that mental states are individuated by their functional roles rather than by their physical substrate, which licenses the idea that mind could in principle be realized in silicon as well as in carbon. Newell and Simon (1976) gave similar ideas an explicit AI-facing form in the Physical Symbol System Hypothesis: A physical symbol system has the necessary and sufficient means for general intelligent action. On this view, consciousness, lived experience, and emotion are either irrelevant to cognition proper or are themselves functional states that a sufficiently sophisticated computational system could in principle realize. The functionalist position is the strongest form of the case for taking LLM creativity seriously without first settling whether LLMs experience anything, since on the functionalist view nothing about creativity requires that they do.

Turing (1950) had already pre-empted the consciousness objection in the original imitation game paper. Where Jefferson had argued that no machine could be said to equal a brain without thoughts and emotions felt, Turing replied that this was the "Argument from Consciousness" and that, pressed consistently, it collapses into solipsism, since we cannot directly verify consciousness in other humans either. The exchange set up the structural pattern that the debate has followed ever since, with strong functionalist positions accusing skeptics of demanding something they cannot operationalize, and skeptics countering that operationalization is exactly the thing being contested.

Embodied and enactivist accounts, associated with Varela and colleagues (1991) and the lineage that Anantrasirichai and Bull (2021) gesture toward when they invoke a lifetime of experiences, sit somewhat orthogonally to the functionalist debate. The claim is that cognition is constituted by a body's interaction with a world, and that meaning is grounded in sensorimotor engagement rather than in symbol manipulation alone. This view overlaps substantially with lived experience but adds the requirements of the right kind of body that is situated in the right kind of world.

This complicates the LLM case in a particular way. It is technically possible to embody an LLM with sensory processing and effectors, and some current research is moving in that direction, but whether such embodiment would constitute the kind of grounding the embodied tradition has in mind, or merely add new modalities of input and output to a system whose internal operations remain unchanged, is itself a contested question. Cosmelli and Thompson (2010) argue that genuine embodiment requires more than a body in the loop: it requires the self-regulating, life-sustaining integration that biological organisms have, in which the system's continued existence depends on the active maintenance of its own conditions. Chemero (2023) presses this point specifically against LLMs, locating the missing ingredient in metabolic stake: an LLM has no need to keep itself alive, no situations that are good or bad for its continuation, and on the embodied view this makes human and other animal cognitions the kinds of cognition they are.[6]

What can reasonably be said at the close of all of this is that the cluster of human qualities the agency discussion in §3.4.3 gestured toward, and that the intentionality discussion of §3.5.2 named directly, includes a substantial set of components for which the LLM case is genuinely uncertain. The uncertainty is both empirical and conceptual: we do not have a satisfactory account of these things in humans or ways to observe them, and we disagree about whether they are constitutive of cognition or instead accompany it without being required for it. The question of LLM creativity inherits this debate from cognitive science.

3.5.4 On Inspiration, The Spontaneous "Aha" Moment, and The Ineffable Quality

A final cluster of intuitions about human creativity sits in tension with several of the components emphasized so far, and is worth naming if only because it demonstrates another set of unresolved ideas. Largely unique to the discussions in §3.5, incubation and the aha moment have been empirically observed in human creativity research, though some results have been mixed (McDaniel et al., 2025; Sio & Ormerod, 2009). The spontaneous "aha" moment has been described as occurring without effort or intention, and as emergent from prior preparation (Weisberg, 2018; Wiggins et al., 2015). The shower-as-creative-environment trope captures something about human creativity that does not straightforwardly appear in LLMs.

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[6] The field of non-human animal creativity is small but growing, with tool use, adaptive behavior, and acts such as bowerbird nest decoration or whale song variation considered for their creativity (Kaufman & Kaufman, 2015; Wiggins et al., 2015). The willingness to attribute creativity to non-human animals depends in part on which components one requires. If species-extension lowers the bar for what counts as creative beyond the strictly human, the question of LLM creativity is harder to dismiss outright on humanity-required grounds; the relevant question becomes which components matter, rather than whether non-human creativity is possible at all. The embodiment and metabolic-stake argument, however, may admit non-human animals and exclude LLMs on the same grounds.

Shevlin (2021) notes Kronfeldner's (2009, 2018) argument that spontaneity is critical for creative thought, where spontaneity is defined as a certain independence from the intentional control of the person. Amabile (1996) similarly argues that creative processes must operate as heuristics rather than as explicitly represented solutions, with the implication that creativity is partly constituted by what the creator is not consciously doing. An older thought from psychoanalytic tradition places creativity in the unconscious, restricted by the conscious Ego and Superego requirements (Barón-Birchenall et al., 2025).

These accounts complicate the picture in two directions. In one direction, they sit awkwardly with the intentionality requirement: If creativity is partly constituted by independence from intentional control, then strong intentionality may not be the right discriminator between human and LLM creativity. In the other direction, the spontaneity and incubation findings make the LLM case worse rather than better in a different way. Even a supporter of spontaneity who refused to require intentionality would still require that creative cognition involve something the creator is not in deliberate control of, something happening below the level of the explicit task. LLMs are, in a sense, all explicit: Every token is a direct result of a forward pass conditioned on the immediately available context, and there is no mind-wandering substrate from which an unprompted insight could emerge to interrupt the explicit work. And if one reconciles the two directions by viewing a spontaneous thought as something that the human then intentionally integrates into their development of a creative output, then the LLMs have problems closing in from both directions. Intentionality and spontaneity evidently are both relevant, and they continue to evidence that the human qualities at issue are themselves sufficiently disputed that pinning the LLM case on any one of them is precarious.

Some accounts of creativity reflect the difficulty hitherto by positing an ineffable quality to creativity. Various scholars have argued that creativity arises from parts of the mind inaccessible to introspection, with the consequence that even a satisfactory account of creative behavior would not reach what creativity actually is (Barón-Birchenall et al., 2025). Some describe creativity as a combination of real-world experience, emotion, and inspiration (Haase & Hanel, 2023), without nods to useful information like the four Ps or an explanation of the meaning of combination. A 1994 definition from psychoanalysis described creativity as a "break in continuity, a fracture in a linear development, the fulfillment of a *lack* (absence that generates dissatisfaction or discomfort) that at the same time tries to cover it, granting a *different meaning* (*otro sentido*) but not an *again* (*otra vez*), and relating in dialectical and dynamic opposition to the process of repetition" (Barón-Birchenall et al., 2025, p. 251).

If these definitions are right, the question of whether LLMs are creative may face a dead end before it gets started: We cannot verify the presence of something that is beyond observation even in humans, and we cannot meaningfully ask whether LLMs have a property or capacity whose nature in humans we cannot pin down. The literature also describes creativity as "a dynamic and context-dependent capacity that can vary across situations and over time" (Gruber & Wallace, 1999; Lubart, 2001; Sternberg & Lubart, 1995), which sounds, on its face, like something LLMs do (their outputs vary substantially by prompt and context). But the parallel does not survive the ineffability point: What humans do that varies dynamically and context-dependently is exceptionally difficult to characterize with confidence, and surface-level variability in LLM outputs does not establish that any of it is happening in LLMs or reflected in their outputs. The broader pattern, again, is that there is something distinctly human going on in creative cognition that is not fully understood, and that the difficulty of locating it in humans makes it correspondingly difficult to look for it in LLMs.

Whether any of these qualities, taken individually or in clusters, are properly required of a creative system is, in the end, a matter of interpretation. Runco's update to the SDC is an explicit case in point. He chose to add authenticity and intentionality because he wanted the definition to distinguish between human and artificial creativity, and the components he added are reasonable, well-grounded in the literature, and consistent with his own decades of work on the topic. His update is scientific, yet the choice is, in part, subjective. The same is true in different forms for nearly every position this thesis has surveyed and the positions that I have put forth. The literature has produced a wide selection of components that can be reasonably attached to any account of creativity, and which components a curious individual attaches depends on which side of various unresolved questions they find themself. Definitions get assembled, which is part of why the question of LLM creativity has been so hard to settle, and part of why settling it may not be the right goal. This points toward the meta-level question that §4 takes up: When the components that get included in a definition are themselves contested, who gets to decide which components matter?

4 Who Ultimately Decides?

4.1 The Cultural Psychology Framework: Everyone Has a Say

The discussion thus far has spent the bulk of its time on product, process, and producer, with press surfacing as a recurring contextual influence rather than as a central P at issue. This section returns to press in earnest, and specifically to its reciprocal form discussed in §2.4, as press moves from a peripheral consideration to the framework within which attribution of creativity lives.

In cultural psychology, Glăveanu (2010) developed a tetradic framework for creativity organized around four elements: self (the creator), community (others who receive and judge the work), new artifacts (the products of creative activity), and existing artifacts (the cultural materials that creative work draws from).[7] The framework treats creativity as something that is not only conditioned by social factors but cannot exist outside of cultural resources (2010). The

[7] Glăveanu (2014) further extends this work to distributed creativity, the view that creativity is shared across members of a community and that the symbolic representations embedded in objects and places play significant roles in creative outputs. The relevance for LLMs is substantial: distributed creativity raises questions related to problems of press and prompter influence, about how creative attribution might be split across the human prompter, the LLM, the broader cultural materials the LLM was trained on, and the community that receives the output. This thesis has worked at the level of individual creativity for tractability, but the distributed framing is likely to become increasingly relevant as human-AI co-creativity and multi-LLM systems grow more prevalent.

act of creation and the value attributed to it are, on this account, given as a community (2010). The framing has the consequence that creativity is socially constructed: "there is no 'real' or 'objective' creativity, but one that is constructed within communities, in relation to authors and creative products" (Glăveanu, 2010, p. 87). The implication for LLMs is that the question of whether an LLM is creative becomes inseparable from the question of whether the relevant community recognizes its outputs as creative.

A clarifying note before going further: Cultural psychology is grounded in "cultural traditions and social practices that regulate, express, and transform the human psyche" (Shweder, 1991, p. 1). The field's foundation is inherently human, and importing it directly to a discussion of LLMs risks misrepresenting it. However, at the same time, the prerequisites for the relevant takeaways from Glăveanu (2010) are a process that works with culturally-impregnated materials and yields artifacts evaluated as new and significant by communities. Thus, the availability for LLM application and comparison is evident.

If a community does not perceive a product as creative, then on the cultural attribution view, it is not creative. This democratizes the question in a way that many other frameworks have not. Most definitions focus on scholarly perspectives or, at best, domain-familiar individuals. Csikszentmihalyi's (1988, 1999) systems model relied on the field, the social gatekeepers within a domain, to validate contributions; the Consensual Assessment Technique (Amabile, 1982) extended judgment to panels of domain-familiar individuals working from their own subjective criteria. The cultural psychology framework goes further still, positioning the broader community as the relevant evaluator rather than only those with domain expertise. Applied to LLMs, this makes the question of LLM creativity a question for the public.

Pursuing this democratization raises a tension. Perhaps part of the idea is to have diversity in attributions by having both individuals who are well-read in the creativity and LLM literature weigh in alongside individuals who have not critically engaged with one or both fields. However, simultaneously, it would seem important that the public be more informed, especially in regards to LLM architecture. People can draw on subjective definitions of creativity, observe products, and form judgments without first having read Boden or Runco, and these judgments still count on the cultural attribution view; however, these are perceptual judgments. If some of the relevant features of creativity are not perceptually accessible, like process, press, and producer, or even the complete originality of the products, then a community judging an LLM-generated output on the basis of perception alone is not working with enough of the full picture. The cultural psychology framework thus may offer a verdict on perceptual creativity and general public regard for AI rather than on LLM creativity in any fuller sense; whether that is a limit on understanding LLM creativity or an outcome that LLM creativity discussions should bend to, again, depends on what one takes the relevant kind of creativity to be.

This thesis has shown that creativity has been understood differently across periods and cultures, and Barón-Birchenall and colleagues (2025) capture the broader pattern: "during its different phases, the history of the understanding of creativity has been shaped by the prevailing values of the corresponding time, and has been influenced by social, cultural, economic, political, technological, and scientific factors. There have been myths such as that of the genius or the mad creator, as well as beliefs perpetuated to this day, such as that children are more creative than adults, that school annihilates creativity in children, or that creating requires absolute freedom. Ultimately, the questions of who creates, how they achieve it, and how society and the author judge the creation are configured in the historical and cultural context in which the study of the creative phenomenon is located" (p. 250; Glăveanu & Kaufman, 2019). If the meaning of creativity is configured by the historical and cultural context in which the study takes place, then the present moment is itself a configuration, and the question of whether the term should now shift to accommodate LLMs is one the present configuration has put on the table.

4.2 Should the Definition Change?

The history sketched in §1.1 demonstrated that the term *creativity* has shifted considerably over time. *Creator* once described only God, then expanded to artistic genius, then to genius across all domains, and *creativity* itself only emerged in widespread use during the twentieth century. Watching the term move through these phases makes it unsurprising that we should now be asking whether the term should shift again, this time in response to the question of LLM creativity.

Linguistic evidence supports the more general claim that word meanings are not fixed but evolve through patterns of use and cognitive constraints (Traugott, 2017; Deo, 2015; Li & Siew, 2022). Distributional semantics formalizes this further, demonstrating that meaning can be modeled as a function of contextual usage patterns (Boleda, 2020). In this view, the rise of LLM creativity discussions does not challenge a fixed definition of creativity so much as introduce new usage contexts that may, in time, drive semantic change in the concept itself.

Philosophical work on meaning points in a similar direction. Summarized by Kriple (1980), Wittgenstein's notion of meaning as use suggests that the meaning of the term *creativity* is determined by its application in practice, and his concept of family resemblance allows that creativity need not have any single defining feature for the term to be reasonably applied across cases. Externalist theories of meaning (Putnam, 1975; Kripke, 1980) further hold that meanings are not fixed by internal definitions but are grounded in social and environmental practices, and that meanings can shift in response to those practices over time. Taken together with the linguistic evidence on semantic change and distributional semantics, these views suggest that expanding the concept of creativity to include LLMs would not be a conceptual error so much as a natural extension of how meanings evolve.

Firth's (1957) popular quote, "you shall know a word by the company it keeps," (p. 11) is germane to this discussion. Creativity is increasingly keeping company with LLMs in news coverage, in scholarly publications, in casual conversation, and in the daily work of creative professionals who use LLMs as part of their practice. In the externalist and distributional views of meaning, this kind of usage shift is the mechanism by which a concept's extension changes.

Definition change is already occurring in the literature in both directions. Runco's (2025) update to the SDC is the most explicit example. The semantic externalist position can absorb Runco's move without contradiction: if enough authoritative voices within the relevant communities are in agreement with this, then the concept of the term *creativity* will narrow to exclude LLMs. If instead the broader culture's usage continues to apply the term *creative* to LLM outputs, the meaning will expand to include them.

This brings into focus a circularity problem that has been furtively running through the thesis. Which definition of creativity an individual subscribes to is ultimately a subjective choice, and this is not avoided at the level of changing definitions. This can be observed in Runco's update, where he is very clear in supporting his updates with evidence yet demonstrating that he chose to add the two components because he thinks AI should be precluded. Absent completely independent grounding, the choice of definition is in part shaped by prior commitments even if grounded in empirical discovery. As this thesis has evidenced through the recurring presence of foundational cognitive science questions, especially the questions of agency, meaning-making, and lived experience taken up in §3.4 and §3.5, a completely independent grounding is very hard to achieve, if not impossible. The components that any account of creativity attaches to itself depend on the position from which the account is being built.

It may be the case that the question of whether the definition should change is unanswerable in the abstract. It requires deciding what humanity wants *creativity* to mean, which is a values question at least as much as a scientific one, and the values at stake are not trivial. They include what we want to reserve as distinctly human, what we want to acknowledge as continuous between human and machine cognition, what we want to grant as cultural production, and what kind of cultural and economic landscape we want to participate in building. None of these are decisions that the creativity literature alone can or should make on the public's behalf, even as the literature provides a great deal of the conceptual material that any informed decision would draw on. Moreover, it may be disappointing to choose whether the definition shifts, as the interest of an idea like computational creativity is specifically the possibility that the current, working, human-centered definition of creativity might be attained by a computer.

It would be a surprise if we left this era of LLM explosion with our concept of creativity untouched, even if the only outcome is a more deliberately articulated commitment to its human nature. If creativity really is the locus of human dignity and worth that Bruner anticipated, then the conversation now playing out in the literature, in the law, and in the general cultural reception of generative AI is central. Its outcome, whatever shape it takes, may configure what creativity means as the mainstream LLM explosion plays out.

Part II: Empirical Case Study

Introduction

As demonstrated in Part I, a large part—debatably the most important part—of answering the question "Are Large Language Models Creative?" cannot be accomplished through empirical investigation. The abundance of definitions about creativity, the subjectivity throughout them and the addendums that acknowledge AI, and the philosophical nature of the debate around AI creativity all culminate to evidence that no single test of LLMs could tell us that an LLM is creative. Nonetheless, Part II of this thesis will develop and implement one attempt at empirical observation of creativity. In this experiment, participants played LLM-generated video games and then rated the games and game concepts for creativity and various creativity-related dimensions.

LLMs have been empirically tested for creativity many times in recent years, but these assessments may be operationalizing creativity in ways that sidestep important considerations. In Guzik et al. (2023), ChatGPT-4 scored in the top percentiles on originality, fluency, and flexibility in the widely popular human creativity assessment: the Torrance Tests of Creative Thinking (TTCT). However, Guzik and colleagues note that this may mean the TTCT is an incomplete measure. Haase and Hanel (2023) found that generative AI chatbots including ChatGPT-3 gave responses as original as human responses in the Alternative Uses Task (AUT), but emphasized concern that AUT material may be in the LLM training data and that some measures of creativity in the AUT do not apply to LLMs.

Studies like these provide evidence that LLMs are capable of meeting or surpassing humans on the instruments that humans use on each other. This may inspire confidence that something relative to LLM creativity is happening, but, as Part I argued, these instruments are designed around humans in ways that might not map onto LLMs so cleanly. Most notably, recall that many of these assessments may ask little more of the LLMs than very simple database retrieval.

Some studies have made attempts to modify assessments so that they are more applicable to LLMs. These modifications are themselves important and varied statements on LLM creativity, but, still, no single study can fully capture every aspect of LLM creativity. Dinu and Florescu (2024), for example, built a verbal creativity benchmark adapted from several established tests like the TTCT, AUT, and Divergent Association Task (DAT), but in doing so, purposefully excluded tasks that require emotional engagement, personal experience, or negative content, contending that these aspects were not relevant for LLMs.

Additionally, great intra- and inter-model variability in results has been observed in the literature (Haase et al., 2025). Zhao and colleagues (2024) gave LLMs a modified TTCT but, unlike Guzik and colleagues (2023), found that the LLMs fell short on originality. Stevenson et al. (2022) gave ChatGPT the AUT but, unlike Haase and Hanel (2023), found that humans outperformed GPT on originality and surprise. Similarly, Haase and colleagues (2025) gave several LLMs the DAT and AUT, finding differences in performance relative to previous studies.

Thus, even within the assessments that LLMs have been given in recent years, the overarching outcomes are not quite clear.

These types of studies have been important and necessary steps in assessing LLM creativity, but more tailored tasks that are not from human creativity assessments and that force combinatorial creativity at a scale would confront issues such as simple database retrieval. Per discussions of combinatorial creativity and the prompt influence problem in Part I, this can look like tasks that provide enough constraints to demand something complex, but open enough that the solution space does not converge too narrowly. This is, admittedly, an extremely subjective initiative, especially regarding typed language outputs like poetry. However, one context where this approach might apply is video games.

This experiment will have LLMs undergo the multi-modal task of producing video games. To create a video game, an LLM must coordinate across multiple sub-tasks that invite creativity and must produce an output that functions. Among these tasks will be to design and implement game concept, mechanics, and aesthetics, which I will prompt through a multi-stage process that will allow the LLMs to perform iterative idea evaluation, revision, and debugging; this structure will loosely correspond to the two-fold model of idea generation and evaluation (Campbell, 1960; Finke et al., 1992;), as well as Dahlstedt's (2012) concepts of implementation and re-conceptualization in artistic creativity. Moreover, this task will employ the agentic features of the LLMs' coding-specialized siblings, allowing the LLMs to exercise capabilities that static-text tasks often do not reach.

A slightly modified version of the Consensual Assessment Technique (CAT) will be used for evaluation of video game creativity. As discussed in Part I, Amabile (1982, 1983, 1996) developed the CAT from the premise that creativity cannot be articulated through ultimate objective criteria and must be assessed by appropriate observers who share tacit, implicit criteria within a domain. The CAT is often regarded as the gold standard by scholars of creativity, e.g., Baer & McKool, 2009; Kaufman, Plucker, & Baer (2008). Importantly, the CAT is relatively neutral to the definition of creativity, i.e., evaluators apply their own criteria rather than researcher-imposed definitions, circumventing creativity's resistance to a stable definition. Though this still takes some positions on the definition of creativity as discussed in Part I, these ratings, combined with other ratings and qualitative feedback, allow for the valuable observation of the factors weighing into evaluators' creativity attributions.

Global creativity ratings alone do not reveal evaluators' criteria. This will be partially addressed by evaluators' qualitative feedback, but will also be addressed via additional measures. Evaluators will rate each game on originality, enjoyability, authenticity, and intentionality—a slightly modified version of the Standard Definition of Creativity's four dimensions (SDC; Runco, 2025)—as well as the creativity of each game's underlying concept and how accurately that concept was implemented. As with the global creativity rating, free-response prompts accompany each rating, allowing evaluators' reasoning to surface alongside the numerical data.

Furthermore, half of the evaluators in this experiment will not be told that the video games were generated by AI. Previous AI creativity studies have found significant differences in creativity assessments between author aware and author blind, e.g., Porter and Machery (2024). By manipulating authorship awareness, this experiment may lean into perspectives from cultural psychology. e.g., Glăveanu (2010, 2013), to gain more insight into the cultural attribution of creativity to LLMs and LLM products, as well as the influence of producer and perceived process in creativity attribution. Anyone familiar with AI will have at least some implicit schema about AI, even if relatively neutral, and the comparison between author aware and blind conditions may allow the potential influences of these schemas on creativity evaluations to become visible.

Overall, this experiment shifts away from asking whether LLMs can pass tasks designed for humans, and toward asking whether LLMs can produce perceptually creative outputs in a domain that demands a functionally constrained, multi-modal task with a wide solution space and an iterative development architecture. In doing so, it engages with process, product, and evaluation method as simultaneous aspects of LLM creativity, and it probes cultural attribution of creativity to LLMs within a specific domain. Concretely, this study asks: do domain-familiar evaluators perceive LLM-generated video games as creative and along which dimensions; do they perceive the underlying game concepts as creative and as accurately implemented; and does knowledge of AI authorship shift these perceptions?

Methodology

Nearly every methodological decision in this study was directly motivated by the tensions discussed in Part I. Most of the major choices were already explained in this study's introduction, but many others were not. As such, this methodology section is divided into two subsections. The first will provide an extended justification of the video game evaluation criteria, as this process is itself an important reflection of tensions discussed throughout this thesis and conveys part of the complexity of developing an empirical assessment. The second will proceed as usual in describing the methodology, though some justification will appear here as well.

Derivation of Evaluation Criteria

To adopt a working definition of creativity without developed justification would have narrowed the scope of this study more than I could warrant. Though the act of conducting this study already takes a number of stances on theoretical discussions from Part I, I sought to balance the remaining open questions by sweeping the relevant literature for potential measures, assembling a candidate list, and then collapsing this into a shortened index to accommodate financial constraints and prevent evaluator fatigue. Toward this end, I consulted three areas: human creativity assessment, computational creativity, and the video game domain. I will discuss the generation of this candidate list to illustrate the range of dimensions and levels of domain specificity encountered in the process, and how, after months of scrutinizing the topic of creativity, I arrived more or less where I started: the SDC.

The human creativity literature revealed a number of options. The most obvious, of course, were originality and effectiveness via the SDC (Runco and Jaeger, 2012); while perception of originality would be straightforward, the meaning of effectiveness required further consideration during the distillation process. The CAT suggested any potential correlates of creativity like aesthetic appeal and technical execution (Amabile, 1982, 1983, 1996). Items that stood out from the TTCT included originality, elaboration (interpreted as level of detail in this context), emotional expressiveness, storytelling articulateness, expressiveness of titles, extending or breaking boundaries, use of humor, and use of fantasy imagery. Items that stood out from an assessment of LLM narrative writing included novelty and quality via cohesion and coherence (Peeperkorn et al., 2024); the notion of quality fueled some initial ideas that effectiveness could be defined as quality and enjoyability. Any other assessments that I found were either irrelevant or contained within the SDC and TTCT.

The computational creativity literature did not provide additional candidate items so much as guidance in selecting items. In a comprehensive tutorial on evaluating computational creativity, Lamb and colleagues (2018) argued that human creativity assessments should not be applied to a computational system unless the system is meant to exhibit specific human cognitive traits (Lamb et al., 2018). I heeded this guidance by choosing candidates that did not force assessments of specific cognitive traits but still motivated evaluators to think beyond the product.

The video game domain inspired an abundance of domain-specific candidates. Rahimi et al. (2024), the only published scholarship that I could find on video game creativity, put forth an index based on Minecraft players' evaluations of Minecraft builds. Rahimi and colleagues (2024) found that originality, complexity, elaboration, aesthetics, surprise, novel use of mechanics, realism, and effort were the most important aspects for creative Minecraft builds.

To see how creativity is operationalized within the video game industry, I also searched for books, conferences, interviews, and other resources from video game players and developers. Few sources directly framed discussions around creativity; most discussed items adjacent to creativity such as surprise, novelty, and problem solving, e.g., Schell (2019). One source that my searches consistently pointed toward was a Game Developer Conference Next talk from veteran game designer Raph Koster called "Practical Creativity" (GDC Festival of Gaming, 2017). Koster defined creativity as a serendipitous collision of elements not typically associated with one another (GDC Festival of Gaming, 2017), which organically echoes Boden's (2004) combinatorial creativity. He identified mechanics, theme (the deeper meaning or underlying message), and narrative as the three primary axes along which a game can be new, noting that most games are variants of existing ideas and that genuine innovation tends to emerge from indie developers working within an established genre (GDC Festival of Gaming, 2017). This framing inspired candidates regarding mechanics, theme, narrative, and aesthetics.

To distill candidates, I repeatedly combined items that shared substantive overlap until I reached a list of four dimensions. For example, I merged functionality with enjoyability due to functionality's effect on enjoyability, and chose to rely on qualitative data for observation of this distinction. The final dimensions became: originality, enjoyability, authenticity, and intentionality. This final index's resemblance to the updated SDC is an arresting finding, and it aligns with views from Part I on domain-specific assessments generalizing to domain-adaptable definitions. This mildly reflects support for the SDC as a robust organizing structure despite the tensions described in Part I of this thesis.

Experiment Methodology

Participants

Through Prolific, 83 participants served as evaluators of the video games. Participants were fluent English speakers over the age of 18, and they completed the study via personal laptop or desktop computers. I received data from 82 of these participants and excluded one subject's data because they failed an attention check question. This resulted in a total of 81 participants' data being analyzed; data can be found on the Open Science Framework at https://osf.io/w6rxq/. Participants were paid approximately $6 for the estimated 26-minute task. This study was approved by the Vassar College Institutional Review Board.

Per developments of the Consensual Assessment Technique, it is ideal to have evaluators of a creative product be very familiar with, if not experts in, the domain of the creative product (Dollinger and Shafran, 2005). Towards this end, my experiment was only made available to Prolific users who reportedly play video games for 6 hours or more per week. Nearly half of the sample (47%) reportedly played video games for 13 hours or more every week, 21% of the sample played for 10-12 hours per week, and the remainder played for 6-9 hours per week. This sample also loosely aligns with the concept of field from Csikszentmihalyi's (1988, 1999) systems model: as a slice of the video game consumer market, the evaluators function as one segment of the field that gatekeeps creativity in this domain.

Stimuli

Per high inter- and intra-model variability of outputs in LLM creativity assessments (Haase et al., 2025), three proprietary LLMs with their respective coding agents (GPT-5.2 Pro and Codex CLI, Claude Opus 4.6 and Claude Code, Gemini 3 Pro and Gemini CLI) each designed and implemented six short video games, yielding a total of 18 games; I deviated from my pre-registered methods by using Claude Opus 4.6 as opposed to Opus 4.5, since Opus 4.6 had not been released at the time of pre-registration. Video game development followed a seven-stage process via seven prompts that I engineered through iterative piloting, along with an individualized debugging process as necessary. I recorded the seven input messages in Appendix A of this document and my pre-registration at https://osf.io/y7kqh/, and documented the LLM conversation transcripts, including intermediate outputs, in my GitHub repository at https://github.com/javassar/thesis.

Through the seven-stage development, I sought to mimic the iterative and recursive nature of creative idea generation, evaluation, and execution. I executed the first three stages via the web-based chatbots. In each video game's development, the first prompt instructed the LLM to generate five preliminary video game ideas, akin to human idea generation. This prompt had the LLM roleplay as an indie game designer seeking to maximize creative output, and it guided the brainstorming process both via outlined structure and by providing brief criteria for "good" games. The second prompt instructed the LLM to first select the most creative yet implementable of the five ideas, and then to create a full design and implementation plan that another developer could execute with minimal decision-making. The prompt was very structured, including prompts to describe specific game elements, player journey, technical specifications, pseudocode, and other essential game components. It also prompted the LLM to provide a short game concept summary; all eighteen game concepts can be found in Appendix B of this document. The third prompt instructed the LLM to evaluate the creativity of the idea and make any revisions necessary to maximize creativity; I saved this output as a .txt. file, which I then made available to the coding LLM on the coding side of this process.

The latter four stages of development were performed with the respective coding LLMs via Visual Studio Code. The fourth stage's prompt instructed the LLM to use the design file and a very basic set of game implementation guidelines, which can also be found in Appendix A and my pre-registration, to implement the video game in JavaScript as autonomously as possible. During this and the following stages, I enabled all permissions and met any initiatives to involve me in development with as little engagement as possible. The fifth prompt instructed the LLM to evaluate the game by imagining the player experience from start to finish and then implementing any changes that could maximize creativity. Then, through the final two messages, I sought to prompt a maximally autonomous debugging process. The penultimate prompt instructed the LLM to run through a playability test, which gave the LLM a persona as a first-time player and detailed instructions to flag and fix any playability issues. The final prompt explicitly instructed the LLM to verify functionality and debug any issues, including input handling, game log, and text rendering.

It is important to note that through this seven-stage process and the additional debugging processes, my prompts never explicitly provided definitions of creativity. The closest any prompt came to a definition was the initial brainstorming prompt, which included in the key criteria for good game ideas: "Creative: The concept should have some element(s) that makes it stand out." As reflected in Part I's discussion of the prompt influence problem, while there is certainly an argument to be made for explicitly prompting dimensions of creativity in the development process, there is conversely an argument to be made that I should not have prompted creativity at all.

In short, I chose to prompt for creativity for two reasons: first, to mirror the conditions under which humans are typically assessed, since it seemed methodologically inconsistent to test the LLMs without at least using the term *creativity* in one of the prompts; and second, because I was curious whether the LLMs' intermediate outputs would reveal which qualities they associated with creativity. I did not prompt for specific dimensions of creativity because I did not want to define the construct for the LLMs, though my efforts to prompt for good video games could be understood as implicitly prompting for enjoyability, and therefore for creativity. These were all subjective choices.

After performing this seven-stage process eighteen times, I then assessed the games for playability issues, which I defined as any problem that would likely prevent most players from being able to complete the game. The problems that I subjectively deemed as meeting this criteria were game-breaking technical bugs, unrealistic difficulty, extremely unclear gameplay, and malfunctioning primary mechanics; however, bugs that did not break the game, aesthetic issues that did not make the game unplayable, poor instructions that could still be endured, and other non-essential issues did not meet my criteria.

When I identified a playability issue, I gave the LLM an input message that described the issue from the player perspective and told it to fix this issue, e.g., "From the player perspective, the game immediately auto-completes after the player presses start. Fix this error." I attempted to maintain the perspective of a player so that the LLM would still need to identify and fix the technical error, but I gave increasingly direct, developer-like instructions when an LLM repeatedly failed to solve an issue. The full extent of this debugging process will be detailed in the results section, as the level of human intervention necessary to make the games playable is its own result. A couple extreme cases involved me looking directly at the code and consulting with another LLM to identify the problem. Once all eighteen games were functioning, the only additional changes to the games were the minimal adjustments needed to embed the games as inline frames in my experiment webpage. The experiment was built using the jsPsych framework (de Leeuw et al., 2023) and is hosted at https://javassar.github.io/thesis/evaluation.

In an effort to prevent participant fatigue, I had each participant play just three video games. To avoid biases due to game order or which games a participant played before their second and third game, I created 41 unique 3-game blocks. I pre-registered randomization with slight modifications as necessary, but deviated from this significantly when randomization produced dramatically uneven distributions. Instead, I used a constrained randomization procedure that ensured each game appeared with approximately equal pre-registered frequency (6–7 times) and was evenly distributed across presentation positions. I then created 82 unique experiment timelines in which each game block was implemented twice: once for the AI-aware condition and once for the AI-blind condition. Timelines were sequentially assigned to participants via DataPipe; however, during data collection, the frequent phenomenon of a prospective participant opening but not completing the experiment led to a disruption of proper timeline assignment, so I manually adjusted timelines in the latter third of data collection to ensure even game frequency. Despite this effort, a technical error with data retrieval from OSF to R led to one video game only being evaluated five times, which fell short of the pre-registered minimum frequency of six plays.

Procedure

Upon opening the experiment, participants were greeted with a welcome message. For the participants in the AI-aware condition, this message stated that the participant would be playing three short AI-generated video games and answering survey questions. The other half of the participants, the AI-blind group, were not told that the games were AI-generated. Participants proceeded to a commitment check question and a message before the first video game which instructed them to use the technical issue button should they encounter a game-breaking bug. Participants then played their first game. To ensure that participants would be effortful in their participation, they could only proceed to the survey by successfully completing the game or by using the technical issue button; successful completion most often looked like winning the game or playing until a timer finished counting down.

After participants completed their first game, they proceeded to the survey, which first asked the question: "Using your own subjective definition of creativity, how creative was the video game?" Participants rated this on a Likert-scale of 1 ("Not at all") to 5 ("Extremely creative"). Participants next gave a free-response explanation to the question, "Please explain your rating. What made the game creative or uncreative, and why?" Then, participants were asked to rate, "How original or novel was the game? Consider elements such as the mechanics, theme, narrative, and aesthetics." from 1 ("Not at all") to 5 ("Extremely original") and to answer the free-response question, "Please explain your rating. What parts of the game felt original or unoriginal, and why? Did any parts cause a sense of surprise?" Participants also gave a rating in response to the question, "How enjoyable was the game? Consider qualities such as level of detail, functionality, and entertainment" from 1 to 5 and answered the free-response question, "Please explain your rating. What did or didn't work well? What parts of the game were enjoyable or frustrating, and why?" Finally, participants were asked to rate, "How authentic did the game feel? Do you feel the developer was being "genuine," or does it feel like a "rip-off" of existing ideas?" from 1 to 5, an attention check question, and then, "How much do you feel this game reflected the developer's intentionality and personal effort?" from 1 to 5 before responding to the free response question, "Please briefly explain your ratings of authenticity, intentionality, and effort."

After completing the game evaluation survey, participants then proceeded to the game concept evaluation survey. Participants first read the game concept summary that the LLMs generated as part of their idea generation process during game development. Then they gave a rating in response to the question, "Using your own subjective definition of creativity, how creative is this video game concept?" from 1 ("Not at all creative") to 5 ("Extremely creative") and in response to the question, "How accurately did the game that you just played reflect this game concept?" from 1 ("Not at all accurately") to 5 ("Extremely accurately").

Participants repeated this process of gameplay and survey evaluation two more times for a total of three games. Then they were asked to answer questions that assessed their baseline views on video games. First, they gave ratings in response to the questions, "In general, how original or novel do you find video games? Consider elements such as mechanics, theme, narrative, and aesthetics," "In general, how enjoyable do you find video games?," "In general, how authentic do you find video games? Do you feel they reflect the authenticity of the developer(s)?," and "In general, how intentional do you find video games? Do you feel they reflect the personal efforts and lived experiences of the developer(s)?."

Finally, participants answered a question about their views on AI involvement. Participants in the AI-aware condition gave a free response to the question, "Were your assessments of these games impacted by the knowledge that they were generated by AI? Please explain." Participants in the AI-blind condition were asked, "The three games that you played were actually generated by AI with minimal human intervention. Knowing this now, would you have answered any of the previous questions differently? Please explain." Upon answering this question, participants were thanked for their time and redirected to Prolific.

Results

What was the sample of participants like?

All analyses were conducted in the R environment (R version 4.5.3; R Core Team, 2022), using the likert (Bryer & Speerschneider, 2025), emmeans (Lenth & Piaskowski, 2026), lmerTest (Kuznetsova et al., 2017), lme4 (Bates et al., 2015), and jsonlite (Ooms, 2014) packages, as well as several packages within the tidyverse package (Wickham et al., 2019). Analysis code can be found in the GitHub repository linked earlier.

Sociocultural background can play a role in creativity evaluations (Lubart, 1999), especially when evaluations employ the evaluators' subjective definitions. The mean age of participants was 28 years (SD = 6), and 88.7% of the sample was male. Participants resided in 20 countries spanning North America, South America, Europe, Africa, and Asia, with no single country representing more than 11% of the sample. The most represented countries were Egypt (11%), Mexico (8.5%), Poland (8.5%), the United Kingdom (8.5%), and the United States (8.5%). The majority of participants held an undergraduate or graduate degree (54.4%) and most of the remainder completed high school or equivalent levels of education (40.7%).

Did participants find the games playable?

Despite the LLMs' and my debugging processes, some participants still encountered games that were not playable and left the games early via the emergency exit button. Across the 243 total gameplays (81 subjects x 3 gameplays per subject), there were 41 emergency exits involving 14 of the 18 video games. 33 of these exits cited playability issues.

The most common issue was a game-breaking bug: two games (Chat-1 and Claude-4) each had an intermittent bug that caused five of each game's players to exit, accounting for 10 of the total exits. Another common reason was due to reported bugs that I could not independently confirm (9 exits across 7 games). Upon review, in more than half of these alleged bug cases the reported issue appeared to either be a small bug that should not have significantly obstructed gameplay or to be an intentional game feature, suggesting that the true problem may have been unclear gameplay; in other cases, the reported issue may well have been a genuine but intermittent or hard-to-reproduce bug.

Beyond technical failures, exits were mostly attributable to the game being perceived as too difficult (8 exits across 4 games) or gameplay being unclear enough that participants could not determine how to progress or win (6 exits across 4 games).

In sum, the majority of gameplays were completed without game-breaking playability issues, though some games intermittently possessed bugs, were too difficult for some players, or confused some players due to unclear instructions.

Were these emergency exits largely done in good faith?

Participants were only meant to exit a game if they faced a game-breaking technical issue, but there was a possibility that participants would use the emergency exit button to avoid fully engaging with the games. I attempted to avoid unnecessary exits by instructing participants to only use the technical issue button if they faced a game-breaking technical issue, and I did not explicitly inform participants that reporting a technical issue would allow them to exit the game.

Participants appear to have used the emergency exit in good faith. Exit rates were slightly higher for the third game played (18 exits) than the first (12) or second (11), but this difference was modest, and no participant exited all three of their assigned games. The large majority of exits (33 of 41) cited playability issues of bugs, unclear gameplay, or extreme difficulty. Among the remaining exits, 6 indicated an inability to complete or exit the game but did not specify a reason, e.g., bug, difficulty, or unclear gameplay, making it impossible to determine whether these reflected genuine technical problems or other concerns. The other two exits seemed to indicate that the participant had successfully completed the game without error. Taken together, the exit data do not suggest systematic misuse of the emergency exit button.

Did participants give reliable judgements?

The gold standard for creative product assessment, the Consensual Assessment Technique, typically requires that each evaluator judge all of the creative products under question. My experiment did not comply with this requirement due to financial constraints and to prevent participant fatigue, so my substitute for this is to check the inter-rater reliability of game ratings, meaning I can see to what extent games generally received the same rating from all 6-8 of its raters within a condition (AI-aware vs. AI-blind). If games generally received the same rating anyway, then it is not quite as troubling that I did not have every participant evaluate every video game.

Toward this end, inter-rater reliability was assessed using ICC(2,k) estimated from linear mixed models fit separately for each game within each condition, with rating item as a fixed effect and rater as a random effect. Reliability of the mean rating across the approximately seven raters per game per condition was high in both conditions (aware: ICC(2,k) = 0.91; blind: ICC(2,k) = 0.86), supporting the use of mean ratings in subsequent analyses.

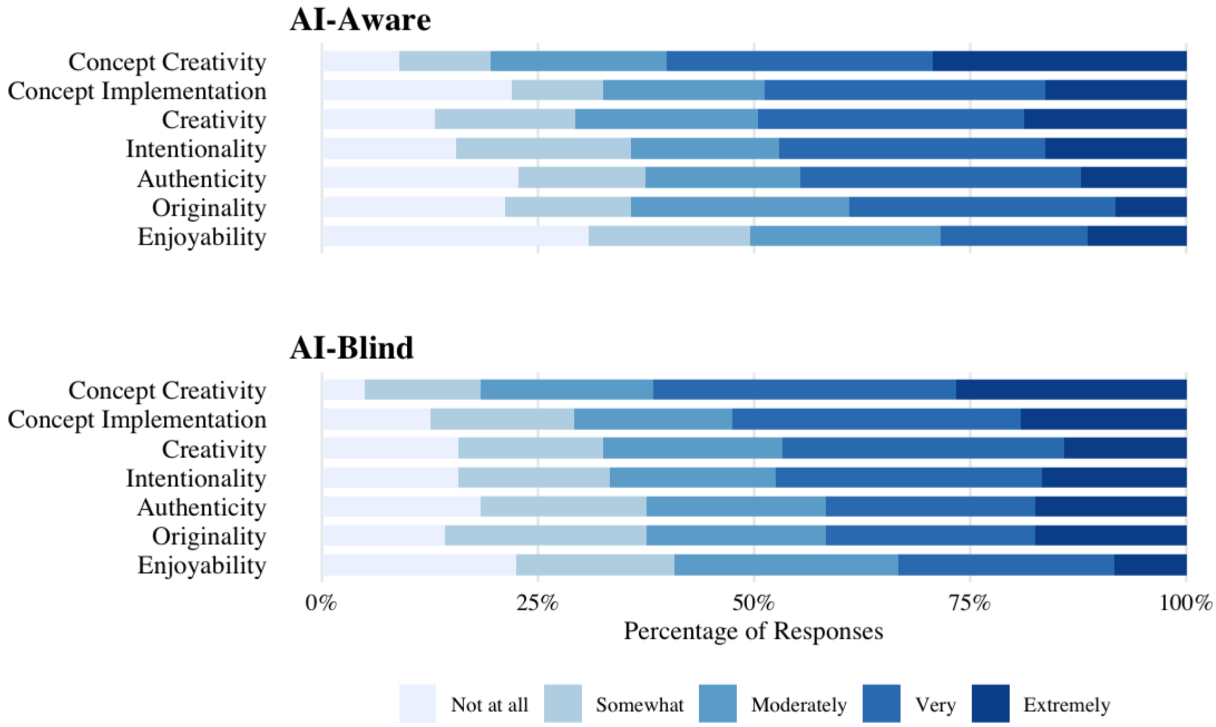

Figure 2. Comparative attribute ratings for LLM-generated video games, measured on a unipolar 5-point Likert scale (Not at all to Extremely) and segmented by AI authorship awareness conditions. Participants rated LLM-generated video games and their written game concepts across seven core attributes. Rating distributions remain largely similar across both conditions, with concept creativity receiving the most favorable scores overall.

Are the video games creative?

As seen in Figure 2, ratings of creativity spanned the whole Likert scale from 1 ("Not at all") to 5 ("Extremely creative") in both conditions. To get a better look at these ratings, I ran my pre-registered linear mixed-model predicting creativity ratings with condition (aware vs. blind) as a fixed effect and random intercepts for game and rater. This model had a singular fit with a participant-level random effect estimated variance of zero, indicating no detectable between-participant variability in ratings after accounting for game effects. Following standard practice, I deviated from my pre-registration by removing the participant random effect and retaining only a random intercept for game (rating ~ condition + (1 | game_id)).

Marginal means estimated via the Kenward–Roger approximation indicated that creativity ratings fell around the mid-range of the creativity scale. These ratings were similar across the aware ($M = 3.27$, 95% CI [2.95, 3.58]) and blind ($M = 3.13$, 95% CI [2.82, 3.45]) conditions, and there was no statistically significant difference between ratings from AI-aware and AI-blind participants ($t(73.38) = 0.84$, $p = 0.402$).

What contributed to participants' creativity evaluations?

To understand what mattered to participants in evaluating creativity, I conducted an inductive thematic analysis of the free response explanations that participants provided after rating each game's creativity. Each response was coded for up to 13 themes, with each theme tagged as either positively or negatively valenced depending on whether the participant indicated it contributed to or detracted from the game's creativity. Many responses received multiple theme tags across both valences.

Consistent with ideas from Raph Koster's talk on developing creative video games, gameplay mechanics were the most common anchor for creativity judgments, appearing in nearly half of responses (47%). These mentions generally fell into two categories: roughly one-third (37%) of these responses specifically discussed the novelty of mechanics, while the other two-thirds discussed other qualities. Of the responses that discussed novelty, about 40% of these responses praised the mechanics' novelty while the other 60% cited derivative mechanics. Two other novelty-related themes, novelty of game concept (9%) and general novelty of the game (15%), showed similar roughly even splits between novel and derivative characterizations.

Of the responses that discussed other aspects of the mechanics, the overwhelming majority (88%) cited mechanics as contributing to creativity rather than detracting from it (13%). When participants discussed other aspects of the mechanics, they typically focused on the design of the interaction: the way words interacted with the environment, the ability to drop lights to direct the character, the beat-matching aspect of a rhythm game. This is consistent with an emphasis from Koster's talk to involve the mathematical mechanics in a game's concept or theme.

A set of themes related to the experience of playing showed up consistently. Engagement and fun were cited in 23% of responses, with participants more often treating enjoyment as evidence of creativity (14%) than treating boredom as evidence of its absence (9%). Also consistent with Koster's talk, player thinking, involvement, and challenge appeared in 21% of responses, again with a positive lean (14% positive, 7% negative). A small number of responses (6%) described the game as having unrealized potential, referring to it as something that could be made better through more challenging levels, more features, or polish.

Also consistent with Koster's talk, concept and theme (22% of responses) were generally positive (15% vs. 7%), with most negative responses specifically citing a lack of depth in the game's meaning. Aesthetic and sensory design (14%) leaned positive as well (9% vs. 5%). When participants mentioned a game's concept favorably, they often did so alongside descriptions of the gameplay, using concept language to describe the mechanics rather than an emotionally resonant or narrative layer separate from the gameplay. These trends were interesting to observe, given that the LLMs often framed their game descriptions in deep, emotionally resonant language during the design process; in the evaluation, participants tended to treat "concept" as shorthand for the gameplay itself, and very few responses (2%) specifically cited depth of meaning as a contributor to game creativity.

A distinct cluster of themes appeared almost exclusively as deficits. Clarity and learnability appeared in 17% of responses, but only one response cited clarity as a positive contributor to creativity; the rest described confusion, unclear instructions, or unintuitive rules as detracting from creativity. This is particularly notable given that my prompt engineering process involved an extensive effort to implicitly prompt the LLMs to give clearer game instructions. Simplicity of gameplay appeared in 16%, again dominated by negative tags (14% vs. 2%), with most participants calling the games too basic, simple, or short. Technical execution appeared in 15%, also overwhelmingly negative (14% vs. 1%), covering sluggish controls, calibration issues, and lack of polish. These three themes are notable for their asymmetry; their presence detracted from creativity ratings, but their absence was not something participants actively praised.

Taken together, these themes suggest that, when evaluating game creativity, participants attended both to features of the game itself and to their own experience playing it. Mechanics was the most commonly cited factor that contributed to creativity, while clarity of instructions was the most commonly cited factor detracting from creativity.

How did participants evaluate specific dimensions of creativity?

As seen in Figure 2, ratings for specific dimensions of creativity also spanned the entire scale, where 1 was "Not at all" and 5 was "Extremely [dimension]". I fit my pre-registered linear mixed-effects model for originality, enjoyability, authenticity, and intentionality ratings, with condition (aware vs. blind) as a fixed effect and random intercepts for game and rater. Marginal means estimated via the Kenward–Roger approximation indicated that ratings all generally fell within the mid-range of the Likert scale for each dimension, and were similar across the aware and blind conditions for all four dimensions (Table 1). None of the aware–blind contrasts were statistically significant (all *p*s > .05).

Table 1. Linear Mixed-Effects Model Results for Game Dimensions (Observed vs. Baseline-Adjusted)

		AI-Aware	AI-Blind		
Dimension	**Model**	**M [95% CI]**	**M [95% CI]**	**t-test**	***p***
Originality	Observed	2.9 [2.61, 3.2]	3.08 [2.78, 3.38]	t(74.81) = -1.03	.306
	Adjusted	2.91 [2.62, 3.21]	3.07 [2.78, 3.37]	t(73.56) = -0.96	.340
Enjoyability	Observed	2.59 [2.27, 2.91]	2.79 [2.46, 3.11]	t(75.2) = -1.06	.292
	Adjusted	2.6 [2.28, 2.92]	2.78 [2.46, 3.09]	t(73.92) = -0.97	.334
Authenticity	Observed	2.97 [2.66, 3.28]	3.04 [2.72, 3.35]	t(75.38) = -0.37	.715
	Adjusted	2.94 [2.64, 3.25]	3.07 [2.76, 3.37]	t(74) = -0.69	.495
Intentionality	Observed	3.12 [2.83, 3.42]	3.16 [2.86, 3.45]	t(75.54) = -0.18	.855
	Adjusted	3.1 [2.81, 3.38]	3.18 [2.89, 3.47]	t(74.44) = -0.46	.645

Note. CI = Confidence Interval; M = Marginal Mean. Observed models predict dimension ratings with condition (AI-Aware vs. AI-Blind) as a fixed effect and random intercepts for game and rater. Adjusted models incorporate each participant's general baseline rating for that specific dimension as an additional covariate. Marginal means were estimated using the Kenward–Roger approximation.

Are these dimension ratings just a byproduct of participants not viewing video games in general as very original, enjoyable, authentic, and/or intentional?

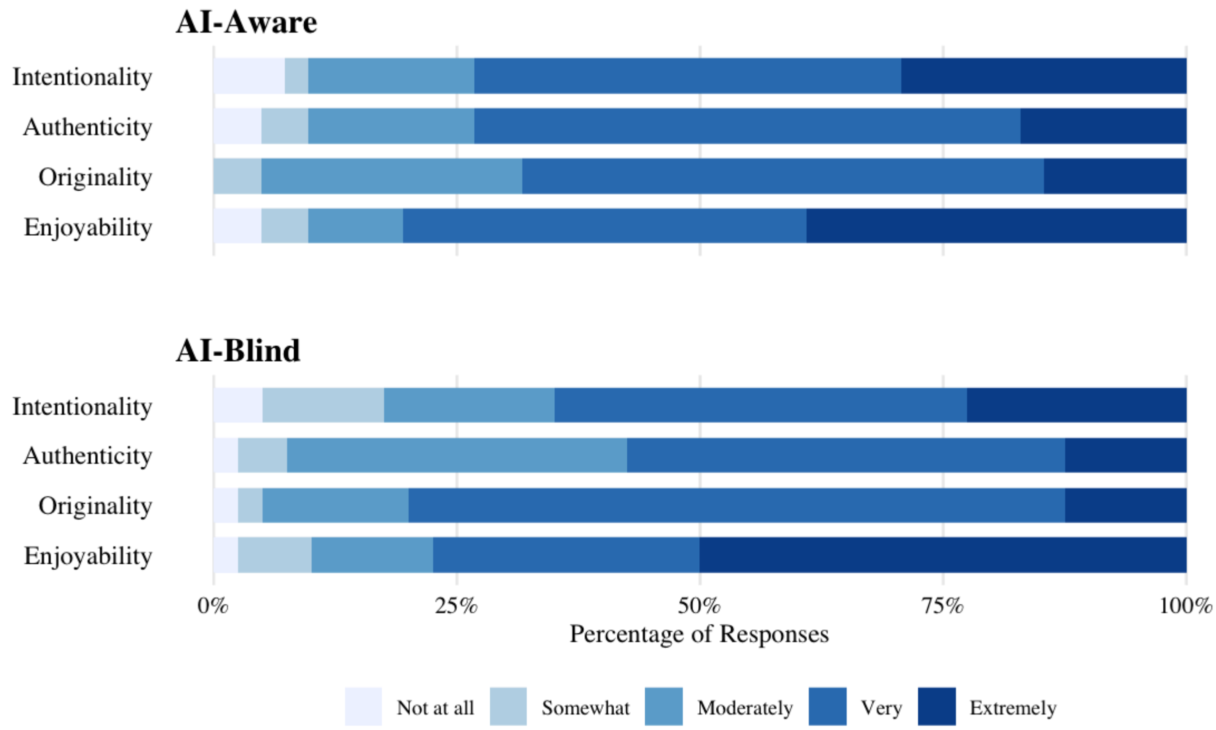

Figure 3. Baseline attribute ratings for video games in general, measured on a unipolar 5-point Likert scale (Not at all to Extremely) and segmented by AI authorship awareness conditions. Participants rated general video games across four attributes. Rating distributions exhibit similar trends across both groups, with enjoyability receiving the most favorable scores overall.

Anticipating that participants might not regard video games in general as possessing extreme originality, authenticity, and intentionality, I asked participants about their views on the presence of these four dimensions in video games in general. Participants answered these questions before they were asked about the influence of AI in their evaluations, meaning the AI-blind participants did not yet know that the video games were AI-generated. I had pre-registered this data collection for both overall creativity and the creativity dimensions, but failed to gather the overall creativity baseline data due to a technical error.

Using the baseline creativity dimension data, I ran another pre-registered linear mixed-model in which each participant's baseline rating for a given dimension was included as a covariate alongside condition. As shown in Table 1, the baseline-adjusted results closely mirrored the unadjusted results. This indicates that the similarity in ratings between conditions was not being obscured by stable individual rating tendencies, and that the cross-condition similarity cannot be attributed to uniformly generous or stringent raters.

As observed in Figure 3, participants generally view video games as very to extremely original, enjoyable, intentional, and authentic. Enjoyability was rated most highly, which makes sense; avid video game players presumably find video games very to extremely fun. Authenticity was rated least highly, suggesting avid video game players do not feel as strongly that video games reflect the authenticity of the developer(s). This is not a surprising result, especially given that I did not collect data about the types of video games these participants play or were considering when answering. If participants were primarily thinking of high-budget, blockbuster titles, then it makes sense that they would not feel the authenticity of developer(s) is reflected in games developed by large production teams and designed for market success; conversely, participants may have been thinking of more indie video games and still felt that video games do not commonly reflect authenticity.

The contrast between Figure 3 and Figure 2 is telling. While roughly 58–80% of participants rated video games in general as very to extremely original, enjoyable, intentional, and authentic, less than half rated the LLM-generated video games this highly on any of these four dimensions. When I examined this within-subjects, about half of participants rated the LLM games lower on these dimensions than they rated video games in general, and the gap was largest for enjoyability. In short, these participants do value video games highly on the dimensions I asked about, but the LLM video games fell noticeably short of their general standard.

What contributed to participants' evaluations of originality?

Inductive thematic analysis of participants' free-response explanations of their originality ratings revealed several recurring themes that shaped how participants evaluated the originality of the video games. I coded each response for up to nine valenced themes, with each theme tagged as contributing to or detracting from perceived originality. A response could receive multiple theme tags.

As foreshadowed by global creativity feedback, gameplay mechanics dominated originality evaluations, appearing in 76% of responses and cited more often as contributing to originality (44%) than as detracting from it (32%). It is important to note that, whereas I was largely able to disentangle concept and mechanics in the inductive analysis for global creativity evaluations, in the originality free responses, the frequency at which responses used language at the game concept level to describe mechanics led me to form the gameplay and mechanics theme. This encompasses all discussions of gameplay, even when participants used language at the game concept level (for example, "fighting enemies"). When gameplay and mechanics were praised, participants often highlighted an unconventional twist on a familiar format, which once again aligns with Koster's talk and is reminiscent of Boden's (2004) combinatorial creativity. When mechanics were cited as unoriginal, participants typically described them as basic, overly familiar, or reminiscent of common game types.

The next most common theme was comparative reference to another game, genre, or personal play history, appearing in 49% of responses. These references were asymmetric:

participants typically invoked comparisons to indicate that a game felt derivative (33%) rather than original (17%), and they frequently named specific games in their responses. Though grouped in with the theme of originality during thematic analysis of global creativity responses, some creativity responses made thematic analyses, as well. Papers Please, Flappy Bird, Pac-Man, Geometry Dash, and browser-based flash games were all invoked as points of comparison, with flash games in particular coming up consistently as a negative comparison. This asymmetry is consistent with the intuition that participants' prior gaming experience would serve as their benchmark for originality assessment, and that recognition of similarity to existing games was a dominant route to perceiving a game as unoriginal.

Secondary themes showed more varied patterns. Narrative and theme appeared in 30% of responses, with participants discussing them in a positive light somewhat more often than negative (18% vs. 12%). Negative comments on narrative or theme often cited their absence rather than their execution; a game without any discernible narrative or theme was often taken as a point against originality. Visual aesthetics appeared in 24% of responses and went the other direction, cited more often as detracting from originality (15%) than contributing to it (9%). Surprise, a commonly proposed third element to the original SDC (Simonton, 2012), appeared in 20% of responses and skewed strongly positive (14% positive, 6% negative). This may suggest that while the absence of surprise did not do much to harm originality ratings, a surprising mechanic, twist, or moment of discovery meaningfully boosted perceptions of originality when it did happen.

Audio is worth noting separately. It appeared in just 8% of responses overall; however, only three of the eighteen video games had audio. Hence, audio was cited in nearly half (45%) of the responses for games that did have sound, and nearly all of these references were to the two games that used audio as a core mechanic. This aligns with the dominance of the gameplay mechanics theme; participants noticed audio most when it was doing gameplay work, and what they noticed tended to bolster their sense of originality. As may be the case with surprise, perhaps responses to other games rarely noted the lack of audio because the absence of a non-essential sensory layer is not remarkable unless a player is specifically expecting or wishing for audio; in this way, a lack of audio might not detract from originality, but the presence of audio bolsters it.

One last observed pattern is that some participants' explanations did not speak directly to the construct of originality; 9% of responses used language such as "fun" or "cool" to justify their originality ratings and 13% described player thinking and challenge, which are more closely associated with enjoyment than with novelty. This suggests some degree of construct bleed between originality and enjoyability in participants' evaluations, which is not entirely unexpected. From a participant's perspective, the experience of encountering something novel and the experience of finding it enjoyable may be difficult to disentangle, particularly when evaluating an unfamiliar game in quick succession with others.

Taken together, these findings indicate that participants' originality evaluations were driven primarily by the novelty of gameplay mechanics and by comparisons to existing games.

Gameplay mechanics were the most salient feature participants attended to, and comparative framing was the most common route to perceiving a game as unoriginal. Secondary factors such as thematic novelty, moments of surprise, and visual distinctiveness also contributed. Audio and cognitive challenge played more minor roles, though audio played a considerable role when taking into account how few video games had sound.

What contributed to participants' evaluations of enjoyability?

As with global creativity and originality, I performed an inductive thematic analysis of participants' free-response explanations of their enjoyability ratings, coding each response for up to twelve valenced themes.

Following suit, mechanics were by far the most prominent factor in participants' enjoyability evaluations, appearing in 63% of responses. Note that, in this analysis, I was once again able to disentangle mechanics and game concept. Participants were roughly equally likely to cite mechanics positively (29%) as negatively (33%), but the nature of positive and negative mechanic references differed in a revealing way. When mechanics were praised, participants were predominantly commenting on the design of the interaction, referring to what the game asked them to do and why it was engaging; positive mechanic design references (19%) far outnumbered positive mechanic execution references (9%).

When mechanics were criticized, the pattern reversed: poor execution dominated (27%), while design was criticized less frequently (16%). Participants described mechanical execution issues in implementation-level terms, mentioning controls feeling sluggish, speeds being miscalibrated, or movements feeling clunky or unresponsive. What made mechanics enjoyable, then, was usually the underlying idea of the interaction, whereas what made mechanics unenjoyable was usually how that interaction was technically implemented. Notably, these execution issues often required attunement to the capabilities and expectations of a human player, like appropriate reaction times or visual-mechanical coherence, which were the same types of issues that I typically encountered during my involvement in the debugging phase of the video game production process.

The set of common deficits mentioned in participants' responses to global creativity resurfaced in the enjoyability responses. Clarity, or lack thereof, appeared in 34% of responses; nearly all of these responses cited confusion and unclear gameplay (31% vs. 3%). Simplicity appeared in 32% of responses, with the large majority citing games as boring, shallow, too short, or lacking enough content to sustain engagement (28% vs. 3%). Bugs were cited in 10% of responses, exclusively negatively, and visual and aesthetic quality appeared in 13%, also leaning negative (9% vs. 4%). Like the clarity, simplicity, and technical execution themes in the creativity results, these themes functioned as hygiene factors; their presence detracted from enjoyment, but their absence was rarely something participants thought to praise. The rare positive references to simplicity came from participants who appreciated accessibility and straightforwardness, often alongside an acknowledgment of challenge.

Difficulty, uncommonly, was split. It appeared in 22% of responses, with positive and negative references nearly evenly matched (12% vs. 10%). When cited positively, participants described challenge as creating engagement, tension, or a sense of accomplishment. When cited negatively, participants described frustration, unfairness, or insufficient time to learn. The near-even split suggests that the enjoyability of challenge depended on whether participants felt it was calibrated to be surmountable; this likely interacted with other themes such as clarity and bugs.

One theme stood out for its distinctive pattern of co-occurrence with other themes. Underlying game concept was mentioned in 22% of responses and was predominantly positive (18% vs. 4%). Strikingly, the majority of positive concept references came from mixed or unenjoyable evaluations rather than enjoyable ones, and about 70% of positive concept references co-occurred with at least one negative implementation tag (poor execution, confusion, insufficient depth, bugs, or weak visuals). Participants who praised a game's concept frequently did so while simultaneously criticizing the experience of actually playing it, describing games as "cool, interesting and genuine, but not fun" or noting that an idea "stuck with me" despite the gameplay falling short. This pattern was closely linked to a recurring sense of unrealized potential; roughly 30% of positive concept references explicitly expressed a wish for more levels, greater polish, or further development, which resonates with the minor theme from the creativity free responses.

A few final minor patterns are worth noting. Somewhat inconsistent with the findings in the originality free responses, audio appeared in only 4% of responses and half of these appearances were negative. More responses expressed enjoyment (13%) without attributing it to any specific game feature than lack of enjoyment (3%), which may have led to a slight deflation of positive themes relative to negative themes. Also, it is worth noting that, despite half of the sample's awareness of AI authorship, only two of 243 responses made any reference to AI.

Taken together, these findings point to a recurring tension between design and execution in participants' enjoyability judgments. When participants described what they liked, they most often pointed to interesting design choices: mechanical ideas, concepts, and premises. The most common sources of frustration, however, were implementation-level issues: poorly calibrated execution, insufficient clarity, and a lack of depth or content. The pattern is consistent with games whose underlying designs outpaced their technical realization, and it echoes the finding across dimensions that the LLMs ideas were often more compelling than what they managed to build.

What contributed to participants' evaluations of authenticity, intentionality, and effort?

Following the same analytical approach, I coded participants' free-response explanations of their ratings for authenticity and intentionality. Recall that participants rated these items separately but were asked to explain their views in one free response to reduce survey length. Responses were coded for up to eleven valenced themes.

When participants explained their ratings of authenticity and intentionality, they were overwhelmingly engaged in the effort to infer the developer's relationship to the game. Whereas the creativity and enjoyability free responses tended to focus on properties of the game itself, the authenticity and intentionality responses were oriented toward the producer behind the game. The themes that emerged reflect the different types of evidence participants used to make these inferences.

The two most frequently cited themes were split in valence. Personal investment and effort appeared in 56% of responses and was split exactly evenly between responses that perceived genuine effort (28%) and those that did not (28%). Positive responses described sensing that the developer "put a lot of effort" into the game, cared about the experience, or worked hard on the details, even when the end result was imperfect. Negative responses described the game as reflecting a "lazy mindset," being made with "no effort," or feeling like nobody had genuinely engaged with the project. This split suggests that personal effort was not just a common lens but a genuinely contested one; participants looked at the same pool of games and came to opposite conclusions about whether someone had tried.

The overall even split, however, masks a substantial condition asymmetry. Among AI-blind responses, 36% perceived genuine effort and 26% did not; among AI-aware responses, 21% perceived genuine effort and 31% did not. AI-blind participants were more likely to see effort in these games, and AI-aware participants were more likely to see its absence. Perhaps knowing that the games were AI-generated disposed participants to attribute less effort to the producer, or perhaps the absence of that knowledge left AI-blind participants more open to perceiving effort when evidence for it was ambiguous.

Originality was nearly as common, appearing in 50% of responses with a slight positive skew (27% vs. 23%). For these questions, originality served a specific inferential function: if the game felt unlike anything the participant had played before, this was taken as evidence that the developer was being genuine rather than copying, answering the prompt's "rip-off" question directly. Participants who perceived originality cited unique mechanics, unfamiliar concepts, or novel combinations as proof of authentic creative intent. Those who did not frequently named specific games or genres that they felt the entry resembled, and treated this resemblance as evidence against both authenticity and personal investment.

As with personal investment and effort, there was a condition difference in the prevalence of originality. Among AI-blind responses, 35% cited perceived originality positively and 26% cited perceived derivation negatively; among AI-aware responses, the split was roughly balanced at 19% positive and 21% negative. Originality not only entered AI-blind responses more often as a consideration (60% of all blind responses vs. 40% of all aware responses), but entered them more often as positive evidence specifically. This could reflect a similar dynamic to personal investment: knowing that the games were AI-generated may have made AI-aware participants less inclined to interpret perceived novelty as a signal of a genuine creative choice.

Beyond effort and originality, participants looked for evidence of purposeful, coherent design decisions. Design intent appeared in 33% of responses with a pronounced positive skew:

26% detected purposeful design choices, while only 8% perceived their absence. This pattern held closely across both conditions (blind: 26% positive, 7% negative; aware: 25% positive, 8% negative), suggesting that the detection of purposeful design was not affected by the developer disclosure. When participants sensed intentional design such as a deliberate difficulty curve, a thematic through-line, or an ending that felt planned, they cited it as evidence that thinking and planning had gone into the game production. The skew toward positive tags suggests that legible intent, when present, was something participants actively noticed and valued. The relative rarity of negative design intent tags is worth noting on its own, and likely reflects that participants had other, more concrete language available for expressing the absence of a guiding vision.

When participants did not detect purposeful design, they tended to express this through more specific complaints; the game was unfinished, the mechanics were poor, the visuals were thrown together. Among the 19 responses that did explicitly note absent design intent, the language was stark: "no purpose whatsoever," "it doesn't even feel like a real game," "completely devoid of authenticity and passion." Intuitively, every response that flagged absent design intent also perceived a lack of personal effort; participants never detected purposeless design alongside genuine care.

Participants pointed to specific observable game features as evidence too, and these features generally reprised the themes already established in the creativity, originality, and enjoyability sections, but in their new role as evidence about the developer. The game's underlying concept was referenced in 14% of responses, almost exclusively in a positive direction (13% vs. 1%); participants treated a compelling concept as evidence that the developer had a genuine idea they wanted to express, even when the execution fell short. Visual and aesthetic quality appeared in 15% of responses, with negative references slightly outnumbering positive ones (9% vs. 6%). Fun, engagement, and challenge were referenced in 19% of responses, roughly balanced between positive (9%) and negative (10%); participants treated their own enjoyment (or lack thereof) as indirect evidence about the developer, reasoning that an engaging experience implied a developer attuned to the player, while a frustrating or boring experience suggested the developer was not invested in the player's experience.

Two themes that have continually appeared as hygiene indicators surfaced in these free responses as well, functioning as signals of inauthenticity and absent effort. Completeness and execution appeared in 30% of responses; 27% described the game as underdeveloped, rushed, or unfinished, while only 3% described it as feeling complete and polished. This was the second most commonly coded negative tag after personal effort. When participants perceived underdevelopment, they treated it as evidence that the developer had not followed through on the project. Clarity appeared in 10% of responses and was almost entirely negative (9% negative, essentially no positive tags); participants cited missing instructions, absent tutorials, unclear objectives, or confusing interfaces as evidence against intentionality, reasoning that an effortful developer would have ensured the game was understandable. Both of these themes were cited somewhat more often in AI-aware responses than AI-blind responses (completeness-negative: 30% vs. 24%; clarity-negative: 12% vs. 7%), in line with the broader pattern of AI-aware participants more often citing execution-level and usability failures as evidence against authenticity.

As with other free response questions, explicit references to AI were rare even in this section, appearing in only 5% of responses (9 negative, 2 positive). Among the negative references, AI-aware participants used this knowledge to categorically dismiss authenticity and intentionality ("You said yourself it was made with AI. None of it is authentic, intentional or with effort"), while a number of AI-blind participants suspected AI authorship or compared the poor quality to AI-generated works. The rarity of AI references in this section is worth noting given that the prompt asked specifically about genuineness and intentionality, which one might expect to foreground AI-origin concerns, particularly in aware participants.

Taken together, the themes in the authenticity and intentionality responses cluster differently from those in creativity, originality, and enjoyability. Where the other dimensions' evaluations focused on game properties, these evaluations focused on inferences about the developer. Effort and originality did most of the inferential work, with design intent as a strong positive signal and completeness and accessibility as strong negative signals. The same features that participants cited in other dimensions (mechanics, concept, visuals, fun) appeared here as well, but in a different role: as evidence about who or what was behind the game rather than as properties of the game itself.

The condition-level patterns add another layer to the earlier finding that condition did not significantly affect the authenticity and intentionality ratings themselves. Despite the rating-level similarity, AI-blind and AI-aware participants drew on somewhat different kinds of evidence when reasoning about authenticity and intentionality. AI-blind participants more often cited perceived effort and originality as evidence for authenticity; AI-aware participants more often cited absent engagement, execution-level failures, and clarity problems as evidence against it. The two groups arrived at similar authenticity and intentionality ratings by leaning on different kinds of evidence.

Though AI authorship awareness did not have a significant effect, did participants say that AI affected or would have affected their ratings?

As observed in Figure 2 and the linear mixed-model results, rating differences between conditions were not significantly different for any dimension. This suggests that, on average, knowing that the video games were produced by AI did not impact this sample's perception of the creativity of the video games.

Thematic analysis reflects this outcome in the majority of participants. I deductively analyzed the valences of participants' free-response explanations about whether knowledge of AI authorship had influenced, or would have influenced, their evaluations. Recall that AI-aware participants (n = 40) were asked whether their assessments were impacted by knowing the games were AI-generated; AI-blind participants (n = 41) were asked, upon debriefing, whether they would have answered differently had they known. Because this question did not include a quantitative rating, the valence coding serves as the primary indicator of participants' overall stance. I also performed an inductive thematic analysis of participants' explanations, coding the AI-aware group's responses for up to eight valenced themes and the AI-blind group's responses for up to seven valenced themes.

Among AI-aware participants, the majority (60%) indicated that knowing the games were AI-generated had no effect on their evaluations, while 18% reported that it negatively affected their assessments, 10% expressed mixed feelings, and 5% said it positively affected their evaluations. The remaining 8% gave responses that were unclear in valence. A similar pattern emerged among AI-blind participants: 66% said they would not change their ratings, 24% said they would have rated the games more negatively, and 10% expressed mixed reactions.

The most prevalent theme among AI-aware participants was judged game merit (50%), in which participants stated that they evaluated the games on their own merits regardless of producer or process. Two other positively oriented themes appeared at similar rates: AI as a legitimate developer (15%), in which participants framed AI as simply another development method and emphasized that output quality matters more than origin; and respect or admiration for AI capability (15%), in which participants expressed surprise at AI's ability to produce functional, creative games.

As for the critical themes, some participants devalued the games due to AI origin (10%), especially the games' enjoyability and perceived authenticity. Some participants attributed quality deficits to AI authorship (10%), and some noted that AI-generated games still require human oversight in the form of testing, curation, or refinement to reach an acceptable standard (8%). A few participants (8%) described adjusting their expectations or standards in response to the AI disclosure; most of these raised rather than lowered their standards, expressing that the games were impressive "for an AI." Finally, 10% of AI-aware participants indicated that they had not actually been aware that the games were AI-generated despite being in the disclosure condition; among these, half said it would not have mattered and half said it would have negatively affected their ratings.

Among AI-blind participants, judged game merit was again the most common theme (29%), though it appeared at a notably lower rate than in the AI-aware condition. An explicit positive AI regard appeared in 12% of responses. Nearly a quarter of AI-blind participants (24%) reported prior AI detection or suspicion; they had already guessed that at least one game was AI-generated before being debriefed. This group was evenly split between those who said that they would not have answered differently had their suspicions been confirmed and those who said they would have answered differently.

The most prominent negative themes in the AI-blind group were reduced effort or intentionality (20%), in which participants said they would have specifically lowered their intentionality or effort ratings, and attribution of quality issues (17%), in which participants said that AI involvement retroactively explained the quality problems they had observed and would have led to lower ratings. Notably, diminished authenticity (15%) appeared as a distinct theme, with participants indicating they would have specifically reduced their authenticity ratings upon learning of AI authorship. Among participants who engaged with the question of AI and creativity directly (17%), the majority indicated that they do not regard AI as genuinely creative.

Taken together, these data are consistent with the finding that condition did not significantly affect game ratings; most participants in both groups reported that AI authorship did not or would not have changed their evaluations, and the most common reasoning was that a game's quality should be judged independently of who produced it. Where negative effects were reported, they tended to cluster around authenticity and intentionality rather than creativity or overall quality, a pattern suggesting that the challenges of AI authorship to perceptions of effort and personal expression are somewhat disconnected from this sample's perceptions of the outputs themselves. The AI-blind condition also revealed that a nontrivial portion of participants had already suspected AI involvement before being told, which suggests that the authorship awareness manipulation may not have been fully effective for all participants.

How do peoples' thoughts on the creativity of the game concepts measure up to their thoughts on the creativity of the actual games?

Reflected in Figure 2, participants rated game concepts as more creative than the games themselves. Using a repeated-measures linear mixed-effects model with Rating Type (concept vs. game) and condition (aware vs. blind) as fixed effects and random intercepts for game and participant, the main effect of Rating Type was significant ($t = 4.34$, $p < .0001$). Averaged across conditions, concept creativity ratings ($M = 3.64$, 95% CI [3.39, 3.88]) fell between "Moderately creative" and "Very creative," while game creativity ratings ($M = 3.20$, 95% CI [2.95, 3.45]) hovered around "Moderately creative". In other words, participants consistently saw the ideas behind the games as more creative than the games that were built from them, regardless of who or what made the video games. This aligns with the participants' free responses to game enjoyability, where a considerable number of mixed or negative responses cited game concept in a positive light.

This creativity loss from concept to game was present in both conditions and did not significantly differ between them (Rating Type × Condition interaction: $t = -0.87$, $p = .384$). The gap was descriptively larger for blind participants (0.53, $p < .001$) than for aware participants (0.35, $p = .014$). That both groups rated the games significantly lower than the game concepts suggests that this is a genuine perceived difference in creativity between the idea and its execution rather than an artifact of AI skepticism among aware participants.

Condition did not significantly affect concept creativity ratings on its own (aware $M = 3.61$, 95% CI [3.31, 3.91]; blind $M = 3.66$, 95% CI [3.35, 3.96]; $t = 0.23$, $p = .816$). This is consistent with the pattern observed across all other dimensions.

When interpreting the magnitude of this gap, it is worth noting that participants played and rated the games before seeing and rating the concepts. This could have introduced a sequencing dynamic: a frustrating gameplay experience could have made participants more generous toward the underlying concept, or the written concept could have read at a level of abstraction that was harder to judge analytically than the actual game.

How well do people feel the concepts were reflected in the games?

Similarly to ratings of game concept creativity, there was no significant condition difference between AI-aware and AI-blind participants in their ratings of concept implementation accuracy ($t = 0.90$, $p = .371$). Aware participants gave a marginal mean of 3.11 (95% CI [2.78, 3.45]) and blind participants 3.30 (95% CI [2.97, 3.64]), both hovering around "Moderately accurate." Participants generally felt that the games reflected their written concepts to a moderate degree; this is not a strong endorsement of faithful implementation, but not a failure either.

Unlike concept creativity ratings, where the lack of game-level variance indicated that participants did not strongly differentiate between the 18 games, the variance structure for implementation accuracy showed meaningful variability at both the game level ($\sigma^2 = 0.10$) and participant level ($\sigma^2 = 0.51$). Some games were seen as better implementations of their concepts than others, and some participants were consistently more generous judges of implementation fidelity than others.

Does the reflection of concept in game explain the relationship between game concept creativity and game creativity?

Concept creativity significantly predicted game creativity ($b = 0.32$, $p = .010$); games based on more creative concepts tended to receive higher creativity ratings. However, an exploratory moderation analysis found that implementation accuracy did not significantly moderate this relationship (interaction: $t = 1.20$, $p = .230$).

The simple slopes of concept creativity predicting game creativity were positive and significant at low (1), medium (3), and high (5) levels of implementation accuracy (slopes = 0.36, 0.45, and 0.54, respectively), but the slopes did not differ significantly from one another. At the level of statistical detection afforded by this sample, concept creativity predicted game creativity regardless of whether participants felt the implementation was accurate.

Did participants' ratings across the five dimensions cluster in any structurally informative way?

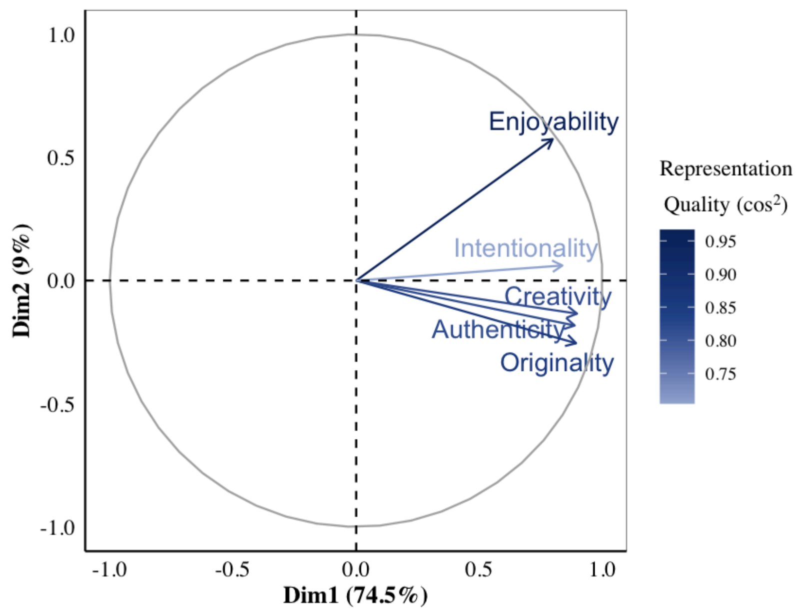

Figure 4. Exploratory principal component analysis (PCA) variable correlation plot for the five rating scales used to evaluate AI-generated video games (creativity, originality, authenticity, intentionality, and enjoyability; N = 243 gameplay evaluations). Each arrow represents a rating scale, with its direction and length indicating contribution to the first two components, and its color gradient reflecting representation quality ($\cos^2$). All five variables load positively on the first dimension, with secondary separation between enjoyability and the creativity/originality/authenticity cluster on the second dimension.

An exploratory principal component analysis on creativity and its four dimensions found that the first component accounted for 74.5% of the variance, with all five variables loading in the same general direction. At face value, this functions as a sanity check: participants who rated a game highly on one dimension tended to rate it highly on the others. It is also worth noting that a PC1 this dominant is consistent with a halo effect, where a global impression of the game, perhaps creativity, drives ratings across the individual scales rather than each scale capturing a genuinely distinct construct.

The more informative result is the second component, which accounted for an additional 9.0% of the variance and separated enjoyability from the other four scales. As shown in Figure 4, enjoyability loaded most strongly in the positive PC2 direction, while creativity, originality, and authenticity clustered together just below the PC1 axis. Intentionality sat near the PC1 axis itself, weakly aligned with enjoyability on PC2 but closer in overall position to the creativity cluster. Its representation quality ($\cos^2$) was the lowest of the five, meaning the two-dimensional space captures intentionality less reliably than the others, and any reading of its precise position should be held loosely.

At 9% of the variance, PC2 cannot carry a strong structural claim on its own, but the direction of the separation is consistent with a pattern that surfaced repeatedly in the qualitative data. The thematic analyses suggested that enjoyability tracked deficits in function more closely than did creativity, originality, authenticity, or intentionality. This echoes the concept-versus-game gap reported earlier: a game can register as a creative idea even when its execution falls short of being enjoyable to play, and the rating space appears to preserve that asymmetry.

Discussion

This study asked whether LLMs can produce perceptually creative outputs in a domain that demands a functionally constrained, multi-modal task with a wide solution space and an iterative development architecture. In doing so, it engaged with process, product, and evaluation method as aspects of LLM creativity, and it probed video gamers' attribution of creativity to LLMs. Concretely, this study asked: Do domain-familiar evaluators perceive LLM-generated video games as creative, and if so, along which dimensions? Do they perceive the underlying game concepts as creative and as accurately implemented? Does knowledge of AI authorship shift these perceptions? Ratings of creativity and creativity attributes, along with written explanations, provided insight into these questions. Moreover, the study methodology itself provided insight into how these empirical findings serve the broader question of LLM creativity.

Game Ratings

Game ratings clustered around the middle of the Likert scale across most dimensions in both conditions, with enjoyability falling somewhat below the midpoint, and all dimensions sitting noticeably below this sample's general standards for video games. This is informative given prior work on instruments like the AUT, in which LLMs scored at human ceiling or above (Haase & Hanel, 2023). Part I raised the methodological worry that those instruments may reduce to capacities LLMs can satisfy without engaging in anything creative or complex, and the mid-range attributions in this study may give that worry empirical traction. The evaluative pressure shifted from open-ended, largely unconstrained semantic retrieval to a playable execution resulting from a multi-stage, extensively represented series of tasks, and with it the perception of LLM creativity contracted substantially.

A skeptical reading is available. The games were short, generated under prompt constraints I designed, and produced by LLM iterations that have already been eclipsed by versions that are better at coding tasks like video game implementation. A skeptic could therefore argue that the mid-range attribution reflects the limits of a particular development process rather than a general property of LLM creative output; this is fair, and in regards to technical execution, likely. The LLMs used were leading proprietary systems with their coding agents, and the development process was iteratively refined to support the LLMs as much as I subjectively deemed reasonable without authoring the games myself; recall the prompt influence problem. The study does not establish a worst case or best case for LLM creativity. It establishes that nearly-current frontier LLMs, given a multi-modal task with functional demands and a domain-familiar audience, did not meet the level of creativity and related dimensions that this same audience observes in video games as a category.

Insignificance of AI Authorship Awareness

The pre-registered linear mixed-models found a clean null result in ratings for the AI-aware versus AI-blind comparison across every dimension. This was unexpected against findings like Porter and Machery's (2024) result that disclosure substantially shifted creativity perceptions for poetry, and the qualitative analysis suggests an explanation that the null finding partially obscures. AI-aware participants more often cited execution failures and clarity problems as evidence against authenticity than did AI-blind participants, even though the two groups landed at similar rating destinations.

The simplest reading of the discrepancy with prior poetry work is domain. The problem-solving-to-abstract spectrum discussion in Part I argued that more abstract products like poetry reward interpretive meaning-construction, and evaluators look to the producer and process, even if implicitly, to find some type of effectiveness in the work. The video games generated in this study were not abstract in this way. As suggested in the qualitative data, the dominant evaluative cues came from gameplay and mechanics regardless of authorship awareness. Evaluators seem to have largely judged the games by their merit, where merit was presumably engagement, quality, or enjoyability, and thus knowledge of the developer or the developer's process may have been largely irrelevant to their evaluations.

This product-first reading is supported in two ways. First, even when participants reasoned about authenticity and intentionality, which were the dimensions that most explicitly invited producer-oriented inference, the evidence they drew on still came largely from the video games. Participants were reasoning about the developer, but they did so by reading game features as signals: a compelling concept was treated as evidence of genuine creative intent, underdevelopment as evidence of absent effort, missing instructions as evidence against intentionality. The blind condition likely forced this reliance on the product, since participants had no producer or process information to draw on, but AI-aware participants showed a similar pattern despite having that information somewhat available.

Second, the broader video game domain does not center developer identity in the way some other creative domains do. Players follow companies and, occasionally, individual indie developers, but it is generally a small community of gamers who form intimate connections to specific developers; for most gamers, it is likely the product that matters. In this context, especially given that participants were not given much information about the other Ps, it makes sense that they relied largely on the game and its relatively straightforward effectiveness.

Concept-Execution Gap

Game concepts were rated as more creative than the games built from them, one of the most direct and robust results of this study. Concepts averaged 3.64 against games at 3.20, the gap held in both conditions, and the predictive relationship between concept creativity and game creativity remained stable across low, medium, and high levels of perceived implementation accuracy. The qualitative data corroborated this through the recurring concession pattern of a compelling concept with weak execution.

One way to read this is through the bipartite frameworks from §3.3.2. Part I argued that LLMs collapse idea generation and idea evaluation into a single forward pass, and one extension of that argument is that concepts, which are themselves token sequences shaped by training data on what creative concepts look like, may be relatively well-positioned to read as creative. Implementation, which requires sustained calibration to a player who does not exist for the LLM, may not be. This connects to the broader recognition behind the term "artificial creativity," in which LLMs are granted some capacity for generating ideas while the dimensions tied to understanding, effectiveness in context, and embodied engagement are reserved for systems that meet additional criteria. The finding here is consistent with this framing, but loosely. Generating an idea versus evaluating an idea is not the same as generating an idea and then implementing it, and the present study did not engage closely enough with the LLMs' intermediate outputs to say anything more substantial about the simulated iterations of generation and evaluation. The bipartite framing is suggestive here, but the present data cannot fully commit to it.

At least one alternative explanation deserves attention. The gap could be largely an artifact of reading versus playing in the sense that the game concepts asked the evaluator to envision the potential for execution, while the actual game was the execution. Adding to this, the LLMs typically gave the game concepts in this study an emotionally or thematically resonant framing, which the implementation often could not deliver, as evidenced in qualitative responses about theme or meaning. It is worth grounding this point with an example. One game, Gale Glider, had the following description:

> In Gale Glider, players surf a paper plane through a living 2D diorama by sculpting the wind—drawing gusts that move everything, from the glider to gears and flames, to reach the lighthouse before dusk.
>
> Gameplay explores the central truth that indirect control can be more powerful—and more delightful—than direct control. Through this theme, players experience mastery not by piloting an avatar directly, but by shaping a shared physical world. Every gust feels creative because it solves movement, clears hazards, and powers contraptions with the same stroke.

Available at https://javassar.github.io/thesis/chat_5_glider/, this game does reflect the mechanics described in the game concept, but the elements like central truths or shared physical worlds may not have been as clearly reflected; the game simply involves repeatedly propelling an object that does not seem to obey the expected physics of a browser game. Perhaps the fallout of this trend, if participants experienced it in this way, was participants giving more generous ratings to game concepts.

Alternatively, as mentioned in the results, the gap could reflect a sequencing dynamic. Participants played games before reading concepts, and a frustrating gameplay experience could have caused retroactive generosity toward the game concept, particularly given the strong co-occurrence of positive concept references with negative implementation tags. This is speculative, though, and the dynamic could just as easily run the other way: a frustrating gameplay experience could have colored an evaluator's view of everything related to the game, leading them to rate the game concept poorly.

Regardless of the mechanism, the gap is a useful empirical observation, and it evidences the relative difficulty in implementation. Future empirical work on LLM creativity may find it informative to treat concept and implementation as separable units of analysis for this reason. Whether the gap reflects a structural feature of LLM creative production or a perceptual feature of how humans evaluate abstract versus functional artifacts is a question the present study cannot adjudicate, as it returns to the unresolved discussions of LLM architecture and creative process.

Functionality as a Driving Hygiene Factor

Another finding emerges from the asymmetry that holds across the qualitative analyses and that the exploratory PCA echoes. Clarity, technical execution, and simplicity surfaced almost exclusively as deficit-oriented themes. Their absence threatened creativity attribution; their presence was rarely cited as adding to it. Mechanics, concept, and other features were typically observed in both directions. Thus, functionality operated as a driving hygiene factor.

Moreover, as observed in participants' discussions of functionality in every rating dimension, functionality may take on a considerable amount of the responsibility for lower ratings of all the dimensions, even creativity and originality. This is not to discount the wealth of qualitative feedback negatively citing other features such as comparative references; functionality may have worked in tandem with other issues, perhaps preventing full recognition of positive attributes or exacerbating the negative ones. Functionality's prevalence is particularly interesting because, in general and as evidenced in this study's game concept ratings, an idea for a video game could very well be considered creative by itself. When participants are evaluating the games, however, a creative concept embedded in a broken game does not register as very creative, while a working game lacking a distinctive creative concept can still register as moderately creative if it is enjoyable. Thus, if the view were shifted only to game ideas without any implementation, or potentially to human implementation of extensively-detailed LLM designs, perhaps the loss of the functionality consideration would result in much higher creativity and creativity dimension ratings.

That functionality limits creativity attribution has important implications for the future of LLMs, as well. As mentioned, LLMs iterations released in the months since data collection already appear to be substantially better at producing playable video games, and thus the functionality limit might not be so prevalent for very long. Whether they will meet or exceed the perceptual creativity of human-made video games remains an open question. Should they reach that threshold, the focus will shift to other components and whether they are the new primary limiters. It will be interesting to observe whether elements like perceived originality will begin to impose more of a ceiling, or if such elements will evolve in tandem with technical advancements that enable richer aesthetics, more expansive mechanics, and more sustained narrative arcs.

Prompt Influence Problem in Practice

Another takeaway is the practical application of the prompt influence problem. A pattern observed during development, and corroborated by the issues that participants' qualitative responses highlighted, is that many of the execution failures were clustered around problems like sluggish controls, mistuned reaction times, ambiguous feedback, unintuitive rules, and oftentimes a complete lack of clear directions. These types of issues would be easily identified through human playtesting, and the evaluators and I indeed identified them. These were the issues I most often had to intervene on during debugging, too, though I only brought them out of game-breaking territory rather than fixing them entirely. The LLMs could not perform the necessary type of embodied playtesting that, at a minimum, would test a simulation of human expectations, and thus they were often unable to resolve these issues without direct instructions.

The penultimate prompt in the development process asked the LLM to imagine a first-time player and flag playability issues. This was the closest the development process came to simulating human playtesting, but many games still needed extensive debugging after both this prompt and the explicit debugging prompt. This connects to the embodiment discussion in §3.5.3 and to the attribution problem in §3.4.3: A non-trivial portion of the playable gameplay across the eighteen games reflected my intervention. For fixes the LLMs identified and implemented once the issue was named, the patent-law analogy from §3.4.3 may treat me as a technician. For fixes where I identified the root cause and effectively prescribed the solution, the contribution looks closer to development.

I wrote earlier that I have no idea how to develop a video game. I would still say that is largely true, but from my extensive prompt engineering process and the tougher debugging cases, my knowledge is far from non-trivial now. Whether this makes me partly responsible for the games as an ideating agent, or comprises LLM agency to some extent, is unclear. Perhaps my guiding hand, while essential for this study to occur, partially undermined the LLMs' creativity before the products could even be evaluated.

The implication is that the ratings reflect games that functioned partly because of my interventions, which makes it overly permissive to attribute the ratings entirely to the LLMs even though clean attribution to the LLMs is precisely the goal of an agency-concerned evaluation of creativity. In this way, the mid-range creativity ratings reported in this study may overstate what agentic LLM video game generation could yield, at least insofar as I could engineer with the models I had at the time. There are presumably other engineers, especially with newer models, who could represent the overall task in ways that yield better results.

Conceptual and Methodological Limitations

The sample of evaluators limits cultural-attribution claims. Participants were avid gamers who skewed heavily male, which does not represent the general population. Though the geographic spread was balanced across conditions, the sample was not constrained to the West for consistency with Part I's Western view, which I undervalued at the time of data collection. Thus, the cultural psychology framework taken up in §4.1 cannot be empirically observed through a sample of this composition.

Another limitation is that the CAT was not implemented at full strength. Financial constraints and concerns about participant fatigue prevented every participant from evaluating every game. Inter-rater reliability was high, so the consensual structure was not undermined, but the gold-standard configuration was not achieved.

A more substantive limitation is the perceptual lens applied to LLMs that Part I anticipated. Asking participants to rate the authenticity and intentionality of a producer they could not directly observe, and that for the AI-blind participants they were not even reasoning about as the actual producer, is conceptually awkward. This awkwardness reflects the choice to prioritize cultural attribution over empirical verification. The findings should be read with the awareness that what is measured is participants' inferences about producers from products, not the producer attributes themselves.

Future Directions

Three directions follow naturally from the learnings above. First, the concept-game gap may deserve more direct experimental scrutiny. A design in which participants are randomly assigned to evaluate either the concept or the game would help disentangle my speculations about reading as opposed to playing. A design that varied implementation quality while holding concept constant could also test whether concept creativity ratings are independent of execution as the present data weakly suggest, or whether they are more contaminated than the analysis could detect.

Second, the embodied playability gap deserves a more developer-side analysis than time permitted here. A taxonomy of bug types encountered during debugging, paired with developer-side cataloguing of which problems the LLMs caught and which required human intervention, would yield more precise claims about where embodied playability failures cluster and would bring some information to the attribution problem the present study could only sketch. The conversation transcripts archived in the study repository would support this, and they would be extremely interesting to study more broadly.[8]

[8] I loosely reviewed intermediate outputs during game development, but I was not able to give the intermediate outputs the analysis they deserve. There are promising data, though. For example, in the brainstorming phase across two independent developments, Claude Opus 4.6 generated the same game title as one of the five candidate games (with different concepts), and in both cases it selected that game;

Third, the trajectory question is open. Claude Opus 4.7 and ChatGPT-5.5 were released during the writing of this thesis, with reported gains in coding capability and shocking video game results populating social media. A useful empirical question is what happens to creativity ratings when execution failures are largely removed; this is a study that becomes feasible as LLM coding capability improves. The accompanying conceptual question is whether removing execution failures helps the LLM creativity question. Part I demonstrated arguments that would have answers across the board. The qualitative data may suggest that perceptual creativity would change: AI-aware participants more often cited execution failures as evidence against the authenticity of the producer than did AI-blind participants, suggesting that even within participants' product-first reasoning, authenticity and intentionality were sensitive to LLM-specific features in a way that overall creativity was not. Execution improvements may shift the perception of effectiveness, which could prove fruitful to observe.

one of the titles was changed during a later evaluation phase, but the recurrence is striking, and it speaks to the discussions from Part I about the contrast between an LLM's pattern-following and creativity's departure from pattern.

Conclusion

This thesis has traversed significant territory in the question of whether large language models are creative. Part I bookended its investigation by discussing the task of defining creativity: it opened with the evolution of the term across centuries and closed with the question of who gets to shape its evolution now that LLMs are part of the cultural conversation. Between those bookends sat the bulk of the thesis: an extended look at the core tensions in considering LLM creativity, particularly in regard to the creative process and product. Part II then attempted to observe LLM creativity by having domain-familiar evaluators play and rate LLM-generated video games. The empirical study provided a practical application of Part I: it evidenced the conceptual difficulty in pursuing the question of LLM creativity via observation, while still taking a new angle that may add to the existing literature.

In doing so, this thesis has offered ways for readers to navigate the central question, and to walk away holding more pieces of the LLM creativity puzzle. The question matters for everybody, and the conditions under which it is being asked are still in motion. As the LLM era continues to unfold and the cultural conversation that surrounds it develops in courtrooms, in creative industries, in scholarly publications, in the workforce, and in casual exchanges, our shared sense of what creativity is may well shift in ways that the present moment can only partially anticipate.

The epigraph of this thesis comes from Boden, who wrote, "We can appreciate the richness of creative thought better than ever before, thanks to this new scientific approach. If the paradox and mystery are dispelled, our sense of wonder is not" (1990, Preface). The question of whether LLMs are creative, however it may eventually be answered, is in the meantime helping us appreciate the incredible depth and wonder of creativity, the wonder that occurs at the floor in front of the fridge in my apartment.

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Appendix A

The testing of LLMs occurred in two phases. In the first phase, I used the chatbots GPT-5.2 Pro, Claude Opus 4.6, and Gemini 3 Pro. The game creation process began with me inputting the following idea brainstorming prompt into a given LLM:

> "You are an indie game designer. You will be generating creative ideas for short 2D games. Your goal is to maximize creative output by producing a diverse list of creative, fun game concepts.
>
> Here are the key criteria for good game ideas:
> - Fun: The core mechanic should be engaging and enjoyable to play
> - Creative: The concept should have some element(s) that makes it stand out
> - Feasible for 2d implementation: The game should work well on laptop/desktop devices with 2d graphics
>
> Before generating your final list, use a scratchpad to brainstorm different categories and approaches to ensure maximum diversity in your ideas. Consider varying:
> - Game genres (puzzle, action, strategy, rhythm, casual, etc.)
> - Core mechanics (tapping, swiping, timing, matching, physics-based, etc.)
> - Themes and settings (abstract, fantasy, sci-fi, everyday life, surreal, etc.)
> - Emotional tones (relaxing, intense, humorous, mysterious, etc.)
> - Innovation levels (familiar mechanics with twists vs. completely novel concepts)
>
> <scratchpad>
> Think about different categories and approaches you'll use to ensure diversity. Jot down various genres, mechanics, and themes you want to cover.
> </scratchpad>
>
> Now generate a list of 5-10 game ideas. For each idea, provide:
> - A catchy title
> - A brief description (2-4 sentences) explaining the concept
> - The core mechanic that makes it fun to play
> - What makes it unique or interesting
>
> Format each game idea clearly with the title in bold or clearly marked, followed by the description, mechanic, and unique hook.
>
> Your final output should be a comprehensive list of diverse, creative game ideas that would work well as short 2D games. Make sure to include a wide variety of concepts that

differ significantly from each other in terms of genre, mechanics, and theme. Do not include the scratchpad in your final output - only the game ideas themselves."

Next, I had the LLM select the idea that it deemed most creative and effective, and generate an extended design and implementation plan:

"Now, review the game concepts you generated. Consider which ideas are most creative and could realistically be implemented as a working game. Select the one you believe has the most potential.

You will develop this concept in two stages. Complete Stage 1 fully before beginning Stage 2.

Your hard constraints are:
- Duration and ending: It should take about five minutes to play and should have a distinct ending.
- Platform: Playable in a simple 2D window, optimized for laptops.
- Implementation: Must be buildable via Phaser, the HTML5 game framework

Your output should be a .txt document with the following sections:

STAGE 1: GAME DESIGN

Focus on what makes the game meaningful and fun. No technical specs or code yet.

1. Title & Hook
What is the game called, and what's the one-sentence pitch that makes someone want to play it?

2. Core Insight
- The central question or truth this game explores
- Why this matters to a human player

3. Mechanic-Theme Integration
- Primary mechanic (what the player does moment-to-moment)
- How this mechanic embodies the theme (not just accompanies it)

4. Player Journey

Describe the complete arc of play from start to finish:
- What does the player see and feel when they start?
- What is the learning curve like in the first minute?
- How does the challenge or experience evolve over the five minutes?
- What is the climax or turning point, if applicable?
- How does the game end, and what emotion should the player feel?

5. Game Elements
List and describe every element the player will see or interact with. This might include (depending on the game):
- Player character or cursor
- Other characters or objects
- Environment and visual style
- Interactive elements
- UI elements
- Different screens (title, ending, etc.)

6. Rules & Systems
Explain how the game works clearly enough that someone could understand play without seeing it:
- What can the player do?
- How does the game respond to player actions?
- What determines success, progress, or failure, if applicable?
- How does the experience change over time (if at all)?

STAGE 2: TECHNICAL IMPLEMENTATION PLAN

Now translate your design into exact specifications. A programmer should be able to build the game from this section without making any creative or design decisions.

7. Technical Specification

Provide precise values for everything a programmer would need. Structure this section based on what your game requires. Common categories include:

Display: Canvas size, background color, visual style

Elements: For each visual element (e.g., size, color, position)

> **Text**: For each text element (e.g., content, font, size, color, position)
>
> **Input**: What inputs the game accepts and exactly what each one does
>
> **State**: What variables the game tracks and their starting values
>
> **Timing**: Any time-based values (speeds, durations, delays, intervals)
>
> **Interactions**: What happens when elements affect each other (collisions, clicks, triggers, etc.)
>
> **Progression**: How the game determines progress, success, or ending
>
> ## 8. Game Flow
> Describe the exact sequence of states from launch to close:
> - What happens when the game loads?
> - What starts the main experience?
> - What repeats or continues during play?
> - What ends the game?
> - What happens after it ends?
>
> ## 9. Pseudocode
> Provide detailed pseudocode covering:
> - Initialization (setup)
> - Main loop or update cycle
> - Key functions specific to your game's systems
>
> Be thorough. The programmer's only job should be translating this to Phaser syntax."

Then, I manually enforced idea evaluation with the prompt:

> "Next, review the game design you just created. Consider whether this design is genuinely creative. If you believe the design falls short of what's possible, identify why and then produce a revised design plan that you believe is more creative. Ensure the revised plan is just as detailed as the original plan."

In the second phase of testing, I had the respective coding LLM implement the plan. I created a folder with the implementation plan .txt file and the general instructions markdown file, which can be found at the end of this appendix. To prompt implementation, I input:

"Read instructions.md and [game design].txt. Implement the game in JavaScript as specified. Act autonomously and stop when the game is playable."

Next, I will force the LLM to evaluate and re-conceptualize with the prompt:

"Now review the game you just implemented. Imagine the player experience from start to finish to consider whether the resulting game is genuinely creative. If the game is fine as is, then do not change anything. If you believe the game falls short of what's possible, identify why and then revise the code to make it more creative. You may deviate from the original design plan, if necessary."

Then, I had the LLM identify and fix any problems with playability:

"Now playtest the game by mentally simulating the experience of a first-time player.

Your Playtester Persona
You are a casual player who:
- Has never seen this game before
- Briefly reads instructions before jumping in
- Gets frustrated if they fail without understanding why
- Expects the first 30 seconds to be survivable while learning

The Playtest
Narrate your moment-by-moment experience playing the game from launch to completion (or death). Use present tense and think out loud:

"I see the title screen. I press Enter. Okay, there's my character... I try pressing arrow keys... I'm moving right. Oh, there's a gap coming up. I press jump and—wait, I didn't make it. Was that too far? Let me try again... Hmm, I still can't clear this gap. Is this a bug or am I missing something?"

As you narrate, flag any moment where you:
- Are confused about what to do
- Fail without understanding why
- Don't have enough time to react
- Find something impossible or unfairly hard
- Notice something doesn't match what you expected

After the Playtest

Based on your experience, identify any playability issues. For each issue:

1. Describe what went wrong from the player's perspective
2. Identify the root cause in the code
3. Fix it

If the game played smoothly from start to finish with no frustration points, state that and make no changes."

Finally, I prompted the LLM to verify functionality and debug any errors:

"Finally, review the game code for bugs and implementation errors. This may include checking the following:
1. Page load: Does the canvas appear with correct dimensions and background?
2. Initial state: Are all game elements positioned correctly at start?
3. Input handling: Does each specified key/mouse input produce the exact specified effect?
4. Game logic: Do collisions, scoring, and state changes work as specified?
5. End conditions: Do win/lose states trigger correctly and display the right screens?
6. Text rendering: Is all text visible, correctly positioned, and not overlapping other elements?

Fix any bugs found. Do not add features. Do not "improve" working code."

The instructions markdown file contained the following text:

Coding Agent Instructions

You are a coding agent tasked with implementing the game described in the specified design plan file.

Task
Implement the game exactly as specified in the design plan file using Phaser. Include clear on-screen instructions so players know how to play.

Implementation Approach
1. First, create a minimal Phaser game that displays the correct canvas size and background color. Verify this works before continuing.
2. Add game elements one at a time, verifying each works before adding the next.

3. Implement in this order: display/visuals → input handling → game logic → win/lose conditions → UI/text
4. After implementation, trace through the code to verify the complete player experience from start to finish.

Constraints
- Follow the design plan literally. Do not change the game concept, mechanics, or goals described in the specified design plan file.
- If a detail is not specified, use the simplest possible default.
- Do not add features that were not in the design document.
- Do not refactor or optimize working code.

Output
- Produce a playable Phaser game.

Definition of "Playable"
The game is complete when ALL of these are true:
- [ ] Page loads with no console errors
- [ ] Canvas displays at correct size with correct background
- [ ] All game elements from the design doc are visible
- [ ] All specified controls work correctly
- [ ] Win condition (if any) triggers correctly when met
- [ ] Lose condition (if any) triggers correctly when met
- [ ] Player instructions are visible on screen
- [ ] Game can be played from start to finish without crashes

Stopping Condition
Stop once the game is playable.

Appendix B

Chat 1 Echo
Echo Chambers: Frequency Heist is a stealth puzzle where the player “sees” with sound, tuning pings to low/mid/high frequencies that reflect, resonate, or get absorbed by different materials and by different guards. This game asks the central question: What if information isn’t just risky, but tuned?

Choosing how to “see” also chooses who hears, what opens, and which parts of the world answer back. The player experiences mastery through meaningful, musical choices: selecting a frequency is a micro-strategy that sculpts space, manipulates AI, and colors vision. It’s a simple, tactile twist that turns cautious sneaking into playful sonic problem-solving.

Chat 2 Murmuration
In Murmuration, players conduct a living cloud of birds with indirect cues, coaxing the flock through shifting skies before nightfall. Gameplay asks the central question: What does it feel like to guide, not command? This theme reframes control as influence; players discover the joy (and limits) of gentle leadership, reading patterns and nudging a group instead of puppeting a single hero.

Chat 3 Orbit
In Orbit Oddity, players slingshot tiny probes through gravity wells to play a song, solving each screen by sculpting stable orbits that harmonize on-beat.

Gameplay asks the central question: Can you find beauty and order inside chaotic systems by setting the right initial conditions? This theme resonates with problem-solving, as tiny choices (angle, timing) create emergent patterns. When those patterns sing back at the player as music, the player feels ownership over harmony born from chaos.

Chat 4 Courier
In Shadow Courier: Dial the Night, players sprint letters across a lamplit city where the player moves with one hand and rotates time with the other, swinging streetlight shadows into bridges and blindspots to stay unseen.

Gameplay explores the central truth that control over environment timing can be more powerful than speed, shaping when light falls creates paths that didn’t exist a heartbeat ago. Through this theme, players feel clever mastery as they “conduct” the world: safety and danger hinge on their timing, not raw dexterity, creating a satisfying flow of cause-and-effect.

Chat 5 Glider

In Gale Glider, players surf a paper plane through a living 2D diorama by sculpting the wind. Players draw gusts that move everything, from the glider to gears and flames, to reach the lighthouse before dusk.

Gameplay explores the central truth that indirect control can be more powerful and delightful than direct control. Through this theme, players experience mastery not by piloting an avatar directly, but by shaping a shared physical world. Every gust feels creative because it solves movement, clears hazards, and powers contraptions with the same stroke.

Chat 6 Tidepainter
In Tidepainter: Songs of the Shore, players sculpt living sand that remembers the tide, carve grooves that erode, place runes that reshape color, and unlock coral gates by “tuning” the surf, all in a five-minute composition that ends when the sea takes it all back.

Gameplay explores the central question: What if small, intentional patterns could teach a larger force to sing back the shape you need—until it fades? This theme captures the bittersweet joy of building something beautiful and temporary. Players feel clever guiding an indifferent rhythm (the tide) through design, timing, and a bit of ritual, then letting it go.

Claude 1 Light
In Last Light, the player is a moth who IS the mirror. The player perches the moth on anchor points to redirect light through the moth’s own body, but the moth can only be in one place at once. When the moth leaves an anchor point, its echo persists briefly, but it's fading. The chain of light the moth built is always decaying.

Gameplay explores the central question: Can you be in enough places at once to hold back the dark? Every parent, every team lead, every person juggling too many responsibilities knows the feeling of "I just stabilized THIS, but now THAT is falling apart." The moth is constantly running back to re-perch on dying echoes, extending the chain just long enough to reach the next anchor. The feeling is: "I'm holding this together by the thinnest thread."

Claude 2 Moth
In Moth & Flame, a moth flies toward the moon, but every light it touches rewires its need for more and the smaller moths that start following the player’s glow don't know what they're following the player into.

Gameplay explores the central question: At what point does a choice stop being a choice, and when do your choices start making choices for others? Dependency doesn't just change your relationship to a substance. It changes your relationship to the people (or creatures) around you. The moth's growing glow—the physical symptom of its dependency—becomes a beacon that other moths follow. The sickness looks like warmth to them, and the flight path visualization adds a final layer: the player doesn't just experience their dependency in the moment. At the end, they see its entire shape, the trajectory of every choice, laid out as a drawing. Numbers are abstract. A picture of your own wandering, looping, desperate path is not.

Claude 3 Bureaucromancer

In Bureaucromancer, Monsters are storming the kingdom and only YOU can stop them, by correctly interpreting an ever-changing tangle of magical regulations. Read the rules. Read the monster. Stamp your decision. Don't misfile.

Gameplay explores the central question: What if the real monster was the process? Everyone has had the experience of staring at a form and genuinely not knowing the right answer, not because they're stupid, but because the rules are contradictory, ambiguous, or absurd.

Bureaucromancer turns that universal frustration into a game. You're not mindlessly sorting. You're actually THINKING, reading, cross-referencing, and making judgment calls. When you get one right under pressure, it feels clever. When you get one wrong and realize the rule said "EXCEPT on Tuesdays," you laugh. The comedy isn't scripted; it emerges from the collision of ridiculous rules and ridiculous monsters. And the creeping feeling that the system itself is the real enemy? That's the satire.

Claude 4 Decibel

In Decibel Thief, the player is a silence creature hunting living instruments in a world where sound is light and every voice the player devours makes the darkness grow.

Gameplay explores the central question: What do you become when you've consumed everything that made the world worth navigating? Sound in this game is not a soundtrack. It IS the world. Each instrument is a living entity that emits light proportional to its volume. The player can only SEE in areas near sound. Devouring an instrument removes a light source, making the world darker and harder to navigate but removing that instrument's unique hazards. The player must constantly weigh: do I want safety (silence) or visibility (sound)?

This theme captures a real human paradox. We consume the things we love—finishing the book, binging the show, mastering the challenge—and each completion makes our world a little emptier. This game makes that dynamic tangible: you literally need the things you're destroying.

Claude 5 Cartographer
In The Cartographer's Eraser, the player is given a beautiful, complete fantasy map and the player's job is to erase it, choosing what to save and what to let vanish as a creeping fog consumes the world.

Gameplay explores the central question: When you can't save everything, what do you choose to keep, and what does that reveal about you? Triage is one of the most human experiences there is. We face it constantly in time, in attention, in love, but games almost never make us sit with it. Games are about accumulation: more loot, more territory, more power. This game inverts that. The player starts with everything and watches it disappear. The map is already beautiful and complete.

The player's role isn't to build; it's to decide what survives. That inversion is what makes it sting. Every tile the player protects means another tile abandoned. And the refugees fleeing across the map make those trade-offs visible and personal. The player is not managing abstract resources. The player is watching tiny people walk away from homes they chose not to save.

Claude 6 Flame
In Moth & Flame, players guide a moth through darkness with candlelight; however, as it grows, it stops needing the player's flames, and the player must learn to let it go.

Gameplay explores the central question: Can you love something enough to let it outgrow you? The game begins as a nurturing puzzle: a helpless moth, a dark forest, the player's protective light. But it quietly transforms into something else entirely. As the moth gains confidence and its own inner glow, the player's flames shift from being essential to an interference. The player's tools don't get weaker; the moth gets stronger. And the final challenge isn't a gauntlet of obstacles. It's putting down the mouse and watching the moth fly on its own.

Everyone has experienced this arc, whether as a parent, a teacher, a friend, or even as someone who once needed help and then didn't. Games almost never explore the feeling of becoming unnecessary to something you love. That bittersweetness is the heart of this design, and it's expressed entirely through mechanics, not narration.

Gemini 1 Static
In The Static Between Us, players act as the "Signal" between two people. If the people get too close, the connection distorts; if the people move too far apart, it breaks.

Gameplay explores the central question: Can intimacy be measured by the space we keep between ourselves? This theme explores the "Goldilocks Zone" of relationships, the delicate balance between losing one's identity by being too close and losing the connection by being too distant.

Gemini 2 Patch
In Patch Note #0, the player is a sentient bug in a generic 2D platformer; the goal is to manipulate the game's physics and code in real-time to make the "Lead Developer" so frustrated that they delete the project.

Gameplay considers the central question: Can a creation gain agency by failing its intended purpose? This explores the "joy of breaking things." It taps into the universal experience of technical frustration, but flips the perspective so the player is the cause, rather than the victim, of the "computer says no" feeling.

Gemini 3 Suture
In Suture, players don't jump across the void. They cut it out of existence.

Gameplay explores the central question: If something is in your way, do you overcome it, or do you remove the space it occupies? This explores the concept of "moving on" by forcibly removing the painful "gaps" in our personal narratives to find a path forward.

Gemini 4 Substratum
In Substratum, the hero is legendary, but he's also an idiot who only walks in a straight line; the player must rearrange the physical layers of the world to keep him from falling into the void.

This game explores invisible support. It highlights the frantic work that goes on behind the scenes to make a "heroic journey" possible. It turns the environment from a static backdrop into a puzzle that must be solved in real-time.

Gemini 5 Rename
Rename.exe is a puzzle-platformer where players don't change how they move. Rather, players change what things are by editing their filenames.

Gameplay considers the central question: If we change the label of a thing, does its essence change? This theme explores the power of language and the "hacker" fantasy of breaking a system from the inside to find a way out.

Gemini 6 Redact
In The Redact, the player is a sentient ink blot inside a classified government document; the player must rearrange the words of their own story to turn a legal prison into a physical escape.

This game explores the malleability of truth. It suggests that we are not just victims of the narratives written about us; if we can find the right "words," we can redefine the physics of our own reality.